Towards ecosystem-based techniques for tipping point detection

Deevesh A. Hemraj* and Jacob Carstensen

Aarhus University, Department of Ecoscience, Frederiksborgvej 399, DK-4000, Roskilde, Denmark

*Author for correspondence (E-mail: adhemraj@ecos.au.dk; Tel.: +4550651268).

ABSTRACT

An ecosystem shifts to an alternative stable state when a threshold of accumulated pressure (i.e. direct impact of environmental change or human activities) is exceeded. Detecting this threshold in empirical data remains a challenge because ecosystems are governed by complex interlinkages and feedback loops between their components and pressures. In addition, multiple feedback mechanisms exist that can make an ecosystem resilient to state shifts. Therefore, unless a broad ecological perspective is used to detect state shifts, it remains questionable to what extent current detection methods really capture ecosystem state shifts and whether inferences made from smaller scale analyses can be implemented into ecosystem management. We reviewed the techniques currently used for retrospective detection of state shifts detection from empirical data. We show that most techniques are not suitable for taking a broad ecosystem perspective because approximately 85% do not combine intervariable non-linear relationships and high-dimensional data from multiple ecosystem variables, but rather tend to focus on one subsystem of the ecosystem. Thus, our perception of state shifts may be limited by methods that are often used on smaller data sets, unrepresentative of whole ecosystems. By reviewing the characteristics, advantages, and limitations of the current techniques, we identify methods that provide the potential to incorporate a broad ecosystem-based approach. We therefore provide perspectives into developing techniques better suited for detecting ecosystem state shifts that incorporate intervariable interactions and high-dimensionality data.

Key words: threshold, regime shift, ecosystem state, bifurcation, non-linear, ecosystem resilience, dimensionality, alternative stable state.

CONTENTS

IX. Supporting information

I. INTRODUCTION

An ecosystem tipping point involves a threshold beyond which an ecosystem transitions from one stable state to another. The change in stable state is generally expected to follow a non-linear bifurcation driven by accumulation of pressure on the ecosystem over time. This accumulation of pressure gradually erodes resilience until a small additional increase in pressure is sufficient to cause a large shift in ecosystem state (Scheffer *et al*., 2001; Folke *et al*., 2004). Once an ecosystem has shifted to a different state, recovery to the previous reference state tends to be very complex and lengthy (Moreno-Mateos *et al*., 2020; Hemraj *et al*., 2022).

Our understanding of tipping points (Table 1) and ecological state shifts is primarily rooted in non-linear mathematical models that exhibit such behaviour in response to changes in a single driver. However, natural biological systems are subjected to multi-layered stressors that interlink in different ways (e.g. synergistic, antagonistic, or additive) and drive different patterns of ecosystem-level response (Harley *et al*., 2017). As opposed to a single pressure, detecting the responses of multiple ecosystem components under various combinations of interlinking pressures remains particularly difficult because of the complexity of interactions between response variables and pressures that can cause, or even prevent, a shift in stable state (e.g. linear and non-linear relationships, cumulative effects, or feedback loops; Fig. 1). Multiple perspectives showcase the potential intricacy of interlinkages in ecosystems that need to be accounted for when inferring state shifts (e.g. Pilosof *et al*., 2017; Kefi *et al*., 2022), but appropriate statistical techniques to detect the presence and causes of these shifts remain limited.

In the Anthropocene, cumulative pressures are being exerted on marine and terrestrial ecosystems (Halpern *et al*., 2008; Watson & Venter, 2019; O'Bryan *et al*., 2020). The overlap and interlinkages among these pressures influence the response of the ecosystem, including significant feedback responses (Fig. 1). For example, the feedback loops within a smaller subsystem in an ecosystem may be shifted by multiple pressures. This shift in one subsystem, in turn, can drive a cascading impact on other subsystems within that ecosystem (Fig. 1). Such interconnections between different subsystems that respond to multiple stressors drive changes in ecosystem resilience and can cause an accelerated or delayed shift in ecosystem stable state (Fig. 2). Therefore, based on the pressures and response variables being investigated, it may be difficult to detect exact ecosystem tipping points from empirical data, especially if data are inconsistent (Carstensen, Telford & Birks, 2013). Recent challenges to the concept that thresholds leading to tipping points can be identified from empirical data instead suggest that there are more gradual shifts in response magnitude and variance in relation to increasing pressure, and that bi-modality or multi-modality, even if they exist, are too difficult to identify because of the noisiness of ecological data (Hillebrand *et al*., 2020, 2023). However, most studies tend to be based primarily on a few environmental drivers and ecosystem components (i.e. a subsystem of the whole ecosystem, see Fig. 1), thus raising the question of which ecological dimensions (e.g. ecosystem type, species composition, keystone species or functional groups) exhibit threshold responses and to what extent detecting tipping points from one pressure at a time is reliable (Dudney & Suding, 2020).

There remains the problem that different combinations of pressures can drive or prevent shifts in the ecosystem. For example, a synergistic interaction between increased nutrient and CO_2 concentrations accelerated the expansion of filamentous turfs at the expense of calcifying algae in an Australian rocky reef while nutrients or CO_2 alone did not have such a drastic effect (Russell *et al*., 2009). On the other hand, the antagonistic effects of multiple stressors may cause a delay in tipping point. In a mesocosm experiment, the introduction of zebra mussels, an invasive species, delayed a shift in state due to increased nutrient input by controlling algal biomass (Dzialowski & Jessie, 2009). These examples show that detecting tipping points in ecosystems from multiple stressors remains complex as it requires some level of incorporation of the interlinkages between the stressors in a multivariate manner (Fig. 1), a consideration that few methods currently incorporate. The other problem is that bifurcation at tipping points may occur in different ways (e.g. fold bifurcation, pitchfork bifurcation, Hopf bifurcation, or transcritical bifurcation), i.e. the shift of an ecosystem from one stable state to another can follow different pathways, including more gradual shifts rather than sharp sudden shifts (Randsalu-Wendrup *et al.*, 2016). In addition, an ecosystem state shift may be delayed in relation to the peaking of combined environmental stressors (Fig. 2; Hillebrand *et al.*, 2023), or the ecosystem may evade tipping altogether (Rietkerk *et al.*, 2021).

From an ecological perspective, these different pathways and lags can be driven by variation in responses, such as through different levels of community interaction (Connell & Ghedini, 2015), species-specific responses (Folke *et al.*, 2004), the resilience of the ecosystem (Dakos *et al.*, 2019), functional redundancy within the ecosystem (Céréghino *et al.*, 2022), changes in the system's feedback loops that regulate dynamics (Biggs, Peterson & Rocha, 2018), evolutionary tipping points (Botero *et al.*, 2015), or lags driven by extinction debt (Tilman *et al.*, 1994). Additionally, spatial fluxes between systems, such as metapopulation and metacommunity interactions, biogeographical range shifts, and spatial distribution of specific cumulative pressures can drive feedback mechanisms that either push ecosystems towards tipping points or help them evade them (see Conversi *et al.*, 2015; Fisher *et al.*, 2015; Schneider & Kéfi, 2016; Rietkerk *et al.*, 2021; Kéfi *et al.*, 2022). Altogether, these ecological mechanisms render the response of ecosystems complex, meaning that using only a subsystem within the ecosystem to identify tipping points is likely to lead to inaccurate and unreliable outcomes, which altogether will not provide good guidance for ecosystem management.

Currently, there are two primary research pathways towards identifying tipping points. One involves using early warning signals (EWS) to forecast how far an ecosystem is from a tipping point: i.e. a 'predictive' approach. Ecosystems reach a tipping point when increasing pressures drive a reduction in resilience and stability to a point where a small additional increase in pressure will cause a shift in ecosystem state. As resilience and stability decrease, the recovery time to the original equilibrium increases, that is a 'critical slowing down' occurs (Van Nes & Scheffer, 2004; Dakos *et al.*, 2012). Simultaneously, the closer the ecosystem is to its tipping point, the greater is its tendency to show early warning signals (EWS) including higher variance, autocorrelation, and skewness in response (Scheffer *et al.*, 2009; Lenton, 2011). Consequently, predicting potential tipping points provides a key tool for prevention of ecosystem state shifts. While increasing variance, autocorrelation and skewness have formed the basis of forecasting tipping points, recent advancements in statistical and computer science have provided new approaches to forecasting tipping points, including from multivariate datasets (e.g. Dakos *et al.*, 2012; Grziwotz *et al.*, 2023; O'Brien *et al.*, 2023). As an alternative to the 'predictive' approach, the 'detection' approach involves uncovering past shifts in ecosystem state from empirical data and using that information to restore or manage ecosystems. Using retrospective detection of ecosystem-specific variables that cause state shifts, or are susceptible to change, is extremely useful for setting restoration targets and protecting ecosystems against reoccurrence of state shifts.

The focus of the current work is on moving towards an ecosystem-based technique for the 'detection' approach, primarily on a temporal scale. While prediction techniques and spatial-scale feedback are also important aspects to be considered, incorporating these here will create an overly exhaustive review. Some of the key methods for prediction have been described previously by O'Brien *et al.* (2023). Current advances in detecting major shifts in ecosystem state from empirical data highlight that a broader ecosystem approach is essential (Dakos *et al*., 2019; Dudney & Suding, 2020; Kéfi *et al*., 2022; Hillebrand *et al*., 2023). There are numerous techniques available for detecting tipping points from univariate or multivariate ecological data. Importantly, however, few of these methods are suitable for detecting tipping points in ecosystems by evaluating the combination of non-linear interactions between multiple response and pressure variables. Considering the challenges presented by interlinkages between ecosystem variables, and the ever-growing intricacy caused by inclusion of more response and pressure variables (e.g. increasing dimensionality increases difficulty of data processing and can decrease predictive power), it is difficult to devise an ideal method that comprehensively and accurately identifies tipping points from ecological data. Nonetheless, here, we review the literature to identify the major techniques applied for detecting non-linear shifts in ecosystem state, including univariate and multivariate methods, and how they fail properly to assess shifts driven by combined stressor effects on combined response variables. We also put forward some techniques that come closer to the inclusion of multiple variables for identifying changes in ecosystem state that may or may not lead to drastic state shifts. We aim to delve into the advantages and limitations of such techniques and discuss the future perspective of what would constitute a broad ecosystem-based approach to detect the presence or absence of a shift in ecosystem state.

II. TECHNIQUES FOR DETECTING STATE SHIFTS

Several techniques currently exist for detecting non-linear shifts in empirical data that represent thresholds. We performed a systematic search of peer-reviewed papers that identify tipping points following the PRISMA guidelines (which aim to standardise and increase reporting quality for systematic reviews and meta-analyses by using structured, transparent, and reproducible methods to collate and summarise evidence; O'Dea *et al.*, 2021). We explicitly sought studies reporting non-linear interactions that caused shifts in ecological data by using the search terms "(nonlinear OR non-linear) AND (response OR impact OR stressor OR pressure) AND (ecosystem OR ecological) AND (threshold OR tipping point)" on the *Web of Science* published between 1st January 1990 and 15th April 2023 (see online Supporting Information, Table S1 and Appendix S1). Our search resulted in 658 papers. We then used the following criteria to target papers that were based only on original research using methods to identify tipping points: (1) the study attempted to identify a shift in their data that would signify a tipping point, and (2) the study used or tested one or more techniques that are suitable to identify a non-linear shift in the data. We identified 507 peer-reviewed papers that met our criteria (Table S1). Study identification and screening were performed manually and without the assistance of software. Among these papers, 36% sought to determine the effect of one predictor on two or more response variables (Fig. 3A), 11% determined the response of one variable in relation to one predictor (Fig. 3B), 30% combined multiple response variables and multiple predictors (Fig. 3C), and 6% used two or more predictors to identify shifts in one response variable (Fig. 3D). The remaining 17% involved syntheses (including reviews, perspectives, or meta-analyses) on specific habitats, methods, or environmental management involving tipping points. Among the most applied techniques were segmented type regressions (ST in Fig. 3), curve derivative-based models (NL) (e.g. Gaussian model, General additive model, continuous interaction models, threshold non-additive formulation), multivariate analysis-based methods [MV, e.g. principal components analysis (PCA)], machine learning (ML), e.g. 'significant zero crossings' (SiZer), regression tree-based methods and other clustering type analysis, network analysis (NB), or logistic/sigmoid (LS) models. These methods were used in 55% of all studies, however, the usage of each method within each category differed (Fig. 3). Curvilinear model-based analysis (NL in Fig. 3) was the method used most often in all categories. Overall, about 55% of studies were from terrestrial ecosystems, 17% from freshwater ecosystems, and 28% from marine ecosystems. In marine and terrestrial studies, most detections of thresholds were performed using curvilinear model fitting (NL in Fig. 3) or segmented-type (ST in Fig. 3) regressions. In freshwater studies the major methods used were curvilinear model fitting (NL) and PCA-based (MV) methods. Machine learning-based (ML) methods were generally more common in terrestrial and freshwater studies than marine studies.

While many of the methods above are focused on detecting non-linear shifts in data (either they are fully non-linear or they utilise some type of sequential analysis to identify non-linear breakpoints), they tend to be more applicable for low-dimensionality data where fewer variables can be managed more easily. To utilise data of high dimensionality better, techniques such as combined multivariate analysis (Vasilakopoulos & Marshall, 2015), threshold indicator taxa analysis (TITAN; Baker & King, 2010), or gradient forest (Ellis, Smith & Pitcher, 2012) have been developed. However, a common drawback of these methods remains their inability to integrate the synergistic, antagonistic, or additive interactions between variables (whether ecosystem components or stressors). Thus, even if multiple variables are used to infer shifts in stable state (e.g. community shifts), variables often are either analysed in a pairwise manner (i.e. different combinations of one predictor against one response variable) or using linear multivariate techniques that reduce dimensionality (e.g. PCA-based techniques: MV in Fig. 3). That said, some machine learning techniques (e.g. regression tree-based) and network techniques do enable analysis of the interactive effects of multiple predictor or response variables.

Although most methods and studies reduce dimensionality before inferring changes in stable state (Fig. 3; green arrows), some studies seek to integrate interactive effects between variables. For example, 30% of studies that used two or more predictors to determine the effect on one response variable sought to integrate interactive predictor effects. Similarly, 8% of studies looking at the effect of one predictor on multiple response variables and 21% of studies looking at the effect of multiple predictors on multiple response variables assessed interactive predictor effects. These studies tried to incorporate intervariable interactions to infer a better understanding of how variables interlink to cause a shift in the stable state of ecosystems. Nonetheless, the number of studies that integrated interactive effects remains low and in multiple instances the exact non-linear relationship between two or more response or predictor variables was unclear (e.g. in machine learning-based methods). For example, nitrogen fertilisation of semiarid shrubland can drive both linear and quadratic relationships in community data (annual plants, microorganisms, biocrusts, edaphic fauna) depending on bottom-up effects, top-down effects, or reciprocal effects (Ochoa-Hueso, 2016). However, the interactive effects of other stressors might change the nature of these relationships (e.g. quadratic to logistic). As a more complex example, freshwater flow in streams can drive changes in environmental parameters such as a linear change in temperature. This temperature change can cause non-linear shifts in community driven by taxa with steep temperature sensitivities (logistic-type shift) while other taxa increase more smoothly but are simultaneously affected by other parameters such as oxygen availability (e.g. Gaussian-type response to temperature and oxygen combinations) (see Rosenfeld, 2017). Currently, no techniques have been applied to detect thresholds from a broader ecosystem-based perspective by the combination of non-linear interactions between variables for high-dimensionality data. Below we describe some of the most common methods for identifying state shift thresholds to determine their advantages but also limitations regarding incorporation of non-linear interactions between variables and the use of broad data sets.

III. METHODS BASED ON FEW VARIABLES AND LOW DIMENSIONALITY

(1) Segmented or changepoint regression

Segmented regression is a technique that identifies discontinuities in data based on changing slopes. This type of regression can involve non-linear regression with a 'hinge' function where the trajectory of data transitions smoothly (Toms & Lesperance, 2003; Francesco Ficetola & Denoël, 2009; Carstensen & Weydmann, 2012). It also has the advantage of identifying the specific value at which the data trajectory changes. The non-linear method is based on models that allow a smooth transition between two segments of different slopes (Fig. 4A). Several smooth transition models can be applied for this purpose, such as hyperbolic-tangent, locally estimated scatterplot smoothing (LOESS), and others (see Toms & Lesperance, 2003; Carstensen & Weydmann, 2012), and the fit of the model is evaluated by calculating the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Changepoint regression works in a similar manner as segmented regression but does not identify a change in slope. Rather, it focuses on separating segments of response variable Y where the mean and variance change along predictor X using sequential F tests (Fig. 4B). Both segmented and changepoint regressions are commonly used to determine shifts in univariate or multivariate data sets, including multiple ecosystem components or stressors. However, they can only infer shifts in one ecosystem component against one stressor at a time. In the case of multiple variables, one can either alternate between different pairs of ecosystem components and stressors (e.g. Berdugo *et al.*, 2020) or utilise multivariate analysis of ecosystem components to identify shifts of multiple components against time or a stressor. For example, they are commonly used together with PCA or multidimensional scaling (MDS) of ecological communities, to determine where along a time series or stressor gradient the community shifts (e.g. Xuan *et al*., 2019). In either of these alternatives, segmented and changepoint regressions are unable to capture and use interlinkages between different variables to infer shifts in the state of the variables. In addition, segmented or changepoint regressions may not always identify the correct breakpoints in the data (e.g. with noisy data) and small variations or transformation of the data may cause large differences in threshold identification (Francesco Ficetola & Denoël, 2009; Matthews *et al*., 2014; Gkioulekas & Papageorgiou, 2019).

(2) Curve-based methods

An alternative, purely non-linear, method for threshold detection involves fitting a curvilinear model to the data and identifying the points at which major changes in trajectory occur. Curve fitting is commonly used for detecting thresholds in different disciplines because of the multitude of possibilities available for fitting different types of data sets. For example, quadratic, Gaussian, or Weibull functions are often applied in thermal biology to estimate temperature thresholds of metabolic activity or thermal performance (Angilletta, 2006; Sinclair *et al*., 2016). On the other hand, log-logistic type functions are often more appropriate in dose–response types of analysis such as in ecotoxicology or biochemistry (Ritz *et al*., 2015). Finally, generalised additive models (GAMs), or modified versions such as threshold-GAM, are widely used in many ecological contexts for identification of thresholds from large data sets that may have different types of predictor variables (continuous, categorical, or ordinal), data sets that contain missing values, and data sets with one or more potential turning points (Benedetti *et al*., 2009; Samhouri *et al*., 2017; Aspin *et al*., 2019). Thresholds from fitted curves can be detected from changes in the first and second derivatives of the curves which represent the slope and inflection, respectively. The derivatives can be calculated directly for any curvilinear model individually and used to identify a threshold. Alternatively, multiple derivatives from different bandwidths of a curvilinear model can be calculated by SiZer (Fig. 5; Sonderegger *et al.*, 2009) and plotted on a SiZer map to identify a threshold based on the best fitting model bandwidth (tuning parameter that controls the smoothness of the resulting curve).

SiZer is a method that uses a non-parametric method to approximate the response function and derivatives, and then uses these to identify thresholds (Sonderegger *et al.*, 2009). The advantage of SiZer is that it makes fewer assumptions than conventional threshold models (e.g. segmented-type regressions) and uses a range of smoothing functions in the process of determining a threshold in the predictor variable. When fitting a curvilinear model to data (e.g. LOESS, locally weighted polynomial), the first and second derivatives inform the user on the change in trajectory of the data. However, one major complication comes with the choice of the smoothing function and its derivatives. Since smoothing algorithms are dependent on a bandwidth (tuning parameter; h) that defines how smooth the resulting curve is, the choice of bandwidth will determine whether the curve overfits or underfits the data. For example, three lines are fitted to the data based on three different bandwidths (Fig. 5A; SiZer uses locally weighted polynomial as a default). At $h = 7$ (blue) the curve underfits the data whereas at $h = 0.7$ (yellow), the curve overfits the data. At $h = 2$ (green), the line seems less likely to be underfitted or overfitted. SiZer uses a range of bandwidth values (ranging between 0.5 and 10) to fit the model to the data and then produces a SiZer map (Fig. 5B) that plots the first and second derivative values along the fitted curve for each bandwidth. The white lines give a visual representation of the size of the bandwidth. The horizontal distance between the lines is drawn to be $2h$, indicating the effective width of the locally weighted polynomial (Chaudhuri & Marron, 1999; Sonderegger *et al.*, 2009). In the SiZer maps, blue represents a positive derivative value (i.e. 95% CI only contains positive values), red represents a negative derivative value (i.e. 95% CI only contains negative values), and purple represents possibly zero (i.e. 95% CI contains some zero values). To interpret the sizer map and identify a tipping point, one identifies when the derivative crosses zero to become either negative or positive, i.e. there is a turning point in the line fit. This turning point represents when the response of the ecosystem component changes along the gradient of a pressure or along a temporal scale. In Fig. 5, the first derivative shows a clear change at around 2005 and the second derivative shows that two changes occurred (~2005 and ~2012–2013) in the response variable when the model at bandwidth $h = 2$ is fitted. The maps also show that at $h = 7$, the model fails to identify any major shifts, i.e. only the broad pattern in the data is represented. On the other hand, at $h = 0.7$, the model overfits the data and identifies many shifts, i.e. more detailed patterns in the data come into focus. Therefore, from Fig. 5 it can be inferred there is a large change in the response of the ecosystem component in year 2005, and somewhere between 2012 and 2013. SiZer thus provides a means of assessing the data set at different scales to gain a clear understanding of the trends in the data (see Appendix S2).

Using curve-based models for threshold detection allows for the application of purely non-linear techniques that can capture different types of relationships between variables and effectively identify a point where the predictor causes a non-linear change in the response variable. Nonetheless, some models may perform better than others depending on the spread of the data (e.g. more gradual patterns or abruptly changing patterns). For example, piecewise regressions and GAMs allow for the distinction between linear and non-linear dynamics and tend correctly to identify breakpoints in data while logistic regressions do not perform well with more gradual patterns in data [see Francesco-Ficetola & Denoël (2009) and Rocha-Santos *et al*. (2016)]. In addition, depending on the amount of data available, false detection of a threshold may be a problem. For example, for a data set with less than 50 observations, SiZer would generally minimise false detections compared to piecewise regressions or changepoint regressions although care must be taken when selecting the bandwidth (see Daily *et al*., 2012). Importantly, however, the problem regarding integration of multiple variables and their combined interlinkages remains unresolved. This is because in their current use, curve-based models have primarily (although not always) been applied for one-on-one regressions. Nonetheless, they do offer the possibility of extending analysis to more variables. For example, GAMs can include multiple covariates and interactive effects. In addition, GAMs such as threshold-GAM or continuous interaction GAM provide the ability to estimate interlinking effects between more than two variables (Ciannelli *et al*., 2004). Similarly, log-logistic (dose–response) models commonly used in ecotoxicology studies also offer the possibility of deriving the influence of a predictor variable on the effect that another predictor variable has on a response variable, that is, implementing antagonistic, synergistic, or additive effects (Bjergager *et al*., 2017). While the possibility of incorporating multiple variables exists, the appropriate application of GAMs or dose–response models to inferring thresholds from multiple interacting predictor and response variables is yet to be implemented.

(3) Multivariate analysis-based method

The application of multivariate analysis for the detection of thresholds allows the use of large data sets with high dimensionality. Therefore, this provides an opportunity to infer shifts in stable state of a whole community or ecosystem over time or in response to a stressor. However, the first step in this analysis of large data sets with high dimensionality is to use a dimensionality-reduction technique, such as MDS or PCA, to compress the data by linear combinations while seeking to maximise explanation of the variation in the data with the fewest variables (Petersen *et al*., 2008; Andersen *et al*., 2009; Van Der Maaten, Postma & van den Herik, 2009). In PCA, for example, each sample is allocated a PC score which denotes its position from the first PC line (PC1), or the second PC line (PC2) and so on, that reflect the variation in the data set (Abdi & Williams, 2010). The PC1 or PC2 scores of the samples can then be plotted against time or a stressor to identify whether there is a large change in PC scores over time or along a stressor gradient. The point at which there is a large change in PC scores denotes a shift in response of the whole community along the timescale or pressure gradient. A segmented or changepoint regression is commonly used to identify more clearly the time or stressor value where the shift in PC scores occurs and infer statistical significance (e.g. Xuan *et al*., 2019; Tonno *et al*., 2021). The majority of ecological studies that aim at identifying thresholds from multivariate data tend to be based on PCA or MDS for dimensionality reduction. While these methods deliver good estimations of when an ecosystem or community may shift, there are several issues that make them inadequate for estimating interactive and non-linear effects. Firstly, PCA or MDS are strictly linear-based dimensionality-reduction techniques, meaning that any non-linear relationships between the predictor variables or between the response variables are disregarded. To include non-linear relationships, PCA or MDS may be replaced by other techniques such as non-linear canonical correlation analysis, non-metric MDS (with non-linear function), or stochastic neighbour embedding. Secondly, they focus on the variables that produce the most variation in the data set, that is, they disregard the effects of some variables and their potential interlinking effects with other variables (problems related to proportion of variance represented; McCune & Grace, 2002). As such, they do not represent the full variation in the set of variables since generally only the PC1 or PC2 scores are utilised, thus they can have issues with consistency and accuracy (McCune & Grace, 2002). Thirdly, PC1 scores are generally plotted against time or one stressor at a time, meaning that interactive effects of multiple stressors are not accounted for. A possible alternative to a PCA-based method would be to use multivariate adaptive regression splines (MARS). MARS, most commonly used for species distribution models, can typically allow examination of the relationship between one response variable and multiple interacting pressure variables. Additionally, it can be used with multiple response variables by averaging the residual squared errors across all response variables (species) and using this average to investigate the effect of multiple interacting pressures, through piecewise regressions, in driving these changes in species composition. However, by averaging residual squared errors, it does not include non-linear interactions between species. Finally, unless paired with a segmented or changepoint regression, this PCA-based method remains primarily graphical. Alternatively, one may use a chronological clustering technique with a distance-based matrix (typically Euclidean), which is able to determine the timepoint of change in community structure through Ward's linkage method or other similar linkage methods (Legendre, Dallot & Legendre, 1985; Andersen *et al*., 2009).

IV. METHODS THAT INCORPORATE BOTH MULTIPLE PREDICTORS AND MULTIPLE RESPONSE VARIABLES

(1) Importance of variable selection

Prior to discussing the methods that incorporate multiple variables to identify tipping points, it is imperative to stress that properly considering relevant variables for analysis should be a priority before formal analysis. Firstly, the quality and quantity of data required to undertake any specific test properly needs to be considered carefully as some analyses require large and very high-quality data sets, while others may be more forgiving. Secondly, the selection of variables should emphasise relevant community–environment relationships. Common problems of overfitting or overinterpretation otherwise may arise. For example, ecological modelling and the increasing use of machine learning techniques have seen the usage of numerous predictor variables incorporated into models without clear consensus on their importance or relevance to the response variables. Such practice can drive model overfitting and unreliable predictions through autocorrelation (Meyer *et al*., 2019), imbalance between accuracy and stability (Fox *et al*., 2017), inability to detect whether a variable indeed has an effect or not (Galipaud *et al*., 2014), and increased Type 1 error and overestimation of explained variance (Blanchet, Legendre & Borcard, 2008), among others. Therefore, it is important for the researchers to familiarise themselves with the response and predictor variables they use, and cautiously select the variables that promote relevant signal (through ecological relevance, for example) rather than those that just add noise.

(2) Machine learning methods

The past decade or so has seen an exponential rise in the application of machine learning techniques for ecological methods. Naturally, the development of machine learning techniques such as regression trees, boosted regression trees, and random forest have benefitted methods for detecting tipping points (e.g. Lintz *et al*., 2011; Ellis *et al*., 2012). Among the most versatile methods incorporating multiple predictor and response variables in the detection of thresholds and ecosystem state change is the use of gradient forest (Ellis *et al*., 2012) – an extension of random forest aimed at identifying overall ecosystem component compositional changes based on pressure gradients (Fig. 6; see detailed description in Appendix S2). The method identifies splits in each response variable data set based on the predictor gradient and synthesises cross-validated and dimensionless R^2 values and accuracy importance measures from random forests for each response variable. Consequently, it results in a ranking of predictors based on their importance in driving compositional turnover in response variables and indicates where along the pressure gradient such compositional turnover occurs. The levels of the highest ranked predictors at which most response variables split are considered thresholds that will likely cause the compositions to tip into an alternative stable state. Secondly, the method uses the cumulative importance of splits in the data along the stressor gradient to identify where along the stressor a threshold may be. While the method includes multiple response and predictor variables and identifies the predictors that are most influential on driving tipping points, a major drawback is the incapacity of implementing interconnections between variables, that is, non-linear interactions between multiple response or predictor variables. In addition, the method is based on one-to-one inference of predictor impact on response variables, although the conditional importance (in relation to other predictors) of predictors is considered. In that sense, in its current form, this method provides a useful avenue for including multiple variables, but still fails at implementing non-linear interactions, unlike boosted regression trees, for example, which can determine interactive effects but at the cost of the ability to determine which predictor variables are most influential on the response variables. In addition, gradient forest does not maintain the overall dimensionality of data, especially by analysing predictor variables individually.

More recently, a multiple factor model was utilised to incorporate the combined effects of multiple pressures on driving tipping points in phytoplankton productivity and species composition globally (Ban, Hu & Li, 2022). The model included ensemble learning and a stacking method for stressors that combines several individual models to obtain better performance by training the first-level regressors with the training set and then the second-level regressors as a meta-regressor using predictions from the first-level regressors as input (Breiman, 1996; Ganaie *et al.*, 2022). Ban *et al.* (2022) also applied multilayer perceptron (MLP) algorithms to determine the effect of multiple stressors on phytoplankton production and species composition. MLP models are neural network-based models with the capability of learning non-linear functions between an input and the output layer, including one or more non-linear layers in between, meaning that the non-linear interactions of multiple variables can be accounted for (see Taud & Mas, 2018). Following these models incorporating the effects of multiple pressures, Ban *et al*. (2022) then analysed changes in phytoplankton productivity and species richness trends as a function of multi-stressors over time. The use of MLP or stacking methods opens the way towards better integration of interlinking multi-stressor impacts on response variables, even if phytoplankton interspecies interactions are not accounted for. Unfortunately, however, another issue remaining is a lack of transparency on how exactly multiple stressors are interacting with each other, a common problem with machine learning techniques.

(3) Combined multivariate method

A commonly applied method for detecting shifts in ecosystem state is one that utilises multivariate plotting (e.g. PCA or MDS) of ecosystem components and subsequent plots of the multivariate scores (e.g. PC1 scores) against time or a pressure to detect where along the timescale or pressure intensity a major change in PC scores occurs (e.g. Hare & Mantua, 2000; Möllmann *et al*., 2009: Xuan *et al*., 2019). This shows whether there is a major shift in the overall composition of the ecosystem components and where along the timescale or pressure intensity gradient this shift occurs. The common problems with this method, as presented above, are (1) its inability to capture non-linear relationships between ecosystem components, (2) distortion effects by reduction of dimensionality, (3) the inability to infer the effect of covarying pressures (two or more pressures) on ecosystem components, and (4) that it is mainly graphical, unless it is paired with segmented-type regressions. It is also important to consider the values of ecosystem components used in the analyses as when some ecosystem components have large values (e.g. high abundances) and others low values, an 'arched' distribution in the two-dimensional representation may be obtained which may bias the interpretation of shifts in the composition of ecosystem components.

Nonetheless, assuming a system where (1) and (2) are of low concern, multivariate-based methods have seen major development in terms of incorporation of the effects of multiple pressures in determining tipping points (Fig. 7; Vasilakopoulos & Marshall, 2015; see Appendix S4). In brief, this is achieved by using PC scores obtained from PCAs of response variables (RV PC) and predictors (Pred PC), independently. Shifts in PC scores, either RV PC or Pred PC along a timeline are detected by sequential regime shift detection. The RV PC, which represents the shift between alternate states, and the predictor PCs are incorporated into a GAM and threshold-GAM model to determine the effects of combined predictors on combined response variables [hence, the term 'combined multivariate method']. The RV PC is then plotted against the Pred PC, and different models are fitted (fit of null, linear, quadratic, cubic) at the alternate states to determine the relationship between RV PC and Pred PC at each alternate state. The estimated position of a tipping point (e.g. in which year) can be visualised by adding a line to the best fitted models and associating each data point on the graph to its categorical value (e.g. the year). Consequently, the year in which the fitted model shows the largest change (e.g. change in slope or a breakpoint) denotes the year when the combined predictor variables caused a large shift in the combined response variables. Subsequent additions to this method include the analysis of one- or two-year lags in pressure components as explanatory variables in the GAM and threshold-GAM models to identify whether delays in ecosystem responses to pressures are involved (Vasilakopoulos *et al.*, 2017). While this multivariate approach integrates multiple pressures in determining tipping points using multiple ecosystem components, the inherent issues regarding its inability to integrate non-linear relationships between either ecosystem components or pressure remains a limitation. In addition, the multivariability is reduced to univariability because it reduces the variation of multiple response variables and predictor variables to one response PC and one predictor PC which are then typically used in a one-to-one regression. Finally, here, the time lags between response and predictor variables are being examined from the PCA scores. Alternatively, one may also analyse lags between the variables (e.g. between responses A, B, and predictor C) and adjusting their data prior to performing the PCAs. While this may potentially increase the accuracy of implementing lags and response/predictor relationships within this method, it may also complicate analyses.

(4) Dose–response-based method

Dose–response methods are commonly used in biochemistry, ecotoxicology, and microbiology. Their application involves identifying well-defined changes in the response of a substance based on the concentration (dose) of another (Ruberg, 1995; Goutelle *et al*., 2008). Dose–response analyses are mostly based on log-logistic-type regressions, which can identify the dose of a substance required to cause a large non-linear shift in the response of another substance. Responses can be continuous, binary, or discrete (Ritz *et al*., 2015). The application of this method in identifying tipping points using pressure and response variable empirical data can be derived from the determination of half-maximal effective concentration (EC_{50}). EC_{50} represents the concentration of a substance that causes 50% of the maximal response of another. In ecological terms, it would represent the pressure intensity resulting in 50% of the maximal response of an ecosystem component (Fig. 8). EC_{50} on the log-logistic model is situated where the gradient of the line is the steepest, meaning that a small additional increase in pressure can drive a large change in ecosystem component response, similar to tipping points.

Dose–response models can be applied using multiple simultaneously occurring pressures to identify tipping points in response variables (Yadav *et al*., 2015). To visualise this, one may plot the log-logistic response of a variable against a gradient of pressure A, given different intensities of pressure B (e.g. 0%, 20%, 40%, 60%) and determine the new EC_{50} of pressure

A based on the intensity of pressure B (Bjergager *et al.*, 2017). Fig. 8 shows different interactive effects that pressures can have on response variables and that can be reflected through dose–response curves. For example, additive effects (Fig. 8A) of pressures A and B would cause the EC_{50} of the response variable to decrease through a combination of the individual effects of A and B. On the other hand, in antagonistic interaction (Fig. 8B), the pressures counteract each other's effects and therefore the EC_{50} of the response variable increases when both pressures A and B are present. Finally, in a synergistic effect (Fig. 8C), pressures A and B interact to reduce the EC_{50} of the response variable, but not in a simply additive manner. For example, in Fig. 8C, although the EC_{50} of pressures A and B are similar, their interactive effect decreases the response variable EC_{50} (green line). A threshold for synergy can be considered as the intensity of pressure B that decreases the EC_{50} of pressure A by another 50% (Bjergager *et al.*, 2017). The advantage of using dose–response models to identify tipping points is that, theoretically, one can implement a large suite of pressures in a stepwise manner and determine their cumulative impacts at different intensities. In addition, dose–response models allow one to take a more conservative approach by defining a tipping point as the EC_{25}, for example, rather than EC_{50}. While dose–response models can work well with multiple pressures, simultaneously determining the effects of multiple response variables can remain difficult as it would require some type of index that combines multiple response variables (e.g. multifunctionality index, diversity index) or the use of multivariate dose–response modelling (multivariate analysis of variance-based model; Brown *et al.*, 2012). These may still not allow incorporating non-linear relationships between the response variables. Incorporation of multiple response and predictor variables and their non-linear interlinkages may require adaptation of methods such as multivariate non-linear regression for use with dose–response curves.

V. FUTURE PERSPECTIVES

The need for a broad ecosystem-based approach for detecting shifts in ecosystem stable state has become more evident as we increasingly understand the complex relationships between multiple ecosystem variables, and the interplay between ecological mechanisms driving ecosystem responses to multiple interlinked stressors (Botero *et al*., 2015; Dakos *et al*., 2019; Hillebrand *et al*., 2023). While examining how one or few ecological processes change over time is useful for understanding parts of an ecosystem, it is insufficient to guide broad-scale ecosystem management. For example, detailed understanding of specific processes that drive components such as fisheries, biodiversity, or ecosystem functions are important. However, in terms of broad ecosystem management, focusing on these components individually will limit the extent to which ecosystem management is possible. In addition, the continued 'layering' of multiple stressors (either driven by human activity or climate change) creates a novel environment to which multiple species must continually acclimate or adapt, and in which they co-exist (Thompson *et al*., 2021; Ward *et al*., 2023). Managing such changing ecosystems requires an understanding of the multitude of processes and interactions that are changing and which of these are likely to cause a major shift in the ecosystem state. Several frameworks have been developed to estimate such risks and to develop possible management strategies related to multiple environmental pressures and their possible interaction with ecosystem components [e.g. Resist-Accept-Direct (RAD) (Thompson *et al*., 2021), or Spatial Cumulative Assessment of Impact Risk for Management (SCAIRM) (Piet *et al*., 2023)]. However, while these frameworks provide an overview of management possibilities for ecosystems that are changing, including possible state shifts, they are unable to quantify exactly how these ecosystems are changing, and which pressure combinations may be leading to tipping points.

Reflecting the spatiotemporal complexity of ecosystem dynamics, including variation in intensity of multiple stressors and their non-linear effects on metacommunity interactions remains statistically challenging. Nonetheless, several methods have attempted to combine multiple ecosystem variables in a multivariate manner and to use that combination to detect shifts in ecosystem stable state. These methods have generally been successfully applied to detect shifts in empirical data that may signify thresholds leading to tipping points. Overall, these methods (including machine learning or dimension reduction-based methods) are robust in terms of statistical performance, however, care should be taken in interpreting their outcomes in relation to ecological relevance. It is important to select the predictor and response variables carefully based on ecological relevance and the hypothesis being tested rather than compiling all possible variables. Importantly, the major limitation of most of these approaches is the lack of inclusion of non-linear relationships between variables as many of these techniques are based on linear relationships (Table 2). Secondly, even when data from numerous variables are utilised, the methods reduce dimensionality by using one-to-one regressions, or PCA followed by one-to-one regression of PC scores against time or a predictor. This means that, although these methods tend to capture some broadly shifting patterns in data that can signify thresholds, they fail to capture the complex non-linear interactions that cause, or in some cases prevent, ecosystems to shift between stable states. Therefore, to provide a more holistic, ecosystem-based approach to determining ecosystem state shifts from ecological data, these two major limitations need to be addressed.

High-dimensionality data is a common feature of disciplines such as information technology, astronomy, or bioinformatics. Its use and application in ecology remains relatively low, although in fields such as evolutionary biology it has become a necessity. One of the biggest advantages of using high-dimensionality data in ecology is its ability to capture complex relationships within the data, and it can be used for detailed analysis/modelling where subtle changes in data may inform about important processes (such as in genomics/proteomics/transcriptomics). Hence, high-dimensionality data can allow for comprehensive and precise representation of both complex ecological processes and more nuanced ecological processes which may be missed when low dimensionality data are used. Working with high-dimensionality data, however, has its own problems (Clarke *et al.*, 2008). Firstly, high-dimensionality data have the ‘curse of dimensionality’, i.e. data are noisy, which can lead to estimation instability, model overfitting and local convergence. Inferring true patterns in the data therefore may be difficult (which is why some methods reduce dimensionality before analysis). In addition, high-dimensionality data are inherently quite large data sets requiring high computing power and appropriate statistics (e.g. Bühlmann & Van De Geer, 2011; Bühlmann, Kalisch & Meier, 2014). Finally, visualisation of data and patterns becomes more complex with increasing dimensionality, since we tend to be limited to three-dimensional visualisation (x,y,z). However, multiple tools now are available for processing and visualising high-dimensionality data which can be applied in an ecological context (see Liu *et al.*, 2016). Considering that computing power is increasing and multiple high-dimensionality data-handling techniques can be transferred from other fields (e.g. bioinformatics, information technology) and adapted to ecology, the prospect of using high-dimensionality data in ecology for detecting changes in overall ecosystem stable states is high, but remains to be adequately implemented. While retaining dimensionality and taking a broad ecosystem-based approach to identifying shifts in stable state is important, caution must be used in the selection of variables, especially predictor variables. In this sense, high dimensionality may mean the selection of different numbers of parameters for different ecosystems and that the complexity of the ecosystem (i.e. the number of interacting parameters that drive the ecosystem) will determine the number of parameters included.

Although current methods are not yet suitable for fully integrating non-linear intervariable interactions and high-dimensionality ecological data, some methods seem to have the potential for further extension to allow analysis of complex ecosystem non-linear interactions and changes in stable state. For example, network analysis can comprehensively incorporate non-linear relationships between multiple predictor and response variables to examine ecosystem state (e.g. Ban *et al*., 2022; Hemraj *et al*., 2017). Deep learning neural networks offer great potential to integrate multiple ecosystem variables as they were designed to incorporate complex interlinkages systematically and are thus appropriate to predict changes in behaviour of ecosystems in relation to increasing environmental stressors (Özesmi & Özesmi, 1999; Millie *et al*., 2012). However, a major drawback is that they provide little explanatory insight into the relative influence of each variable and how predictions are made, although there may be potential to overcome such drawbacks (Arrieta *et al*., 2020). In addition, without multiple preventive adjustments (e.g. Ying, 2019), overfitting of models becomes a common problem. To provide better clarity on intervariable relationships, network-based methods can be complemented with non-linear models which can then be incorporated within the network. Causal networks (or interaction networks) are such an example whereby the relationship between variables is predefined and then implemented into an interaction network to determine the state of an ecosystem and its drivers (e.g. Chang *et al*., 2022; Zhao *et al*., 2023). This would allow for the determination of multiple ecosystem states based on the interactions between all the variables that drive that ecosystem. An alternative to network techniques that potentially can maintain high dimensionality and integrate non-linear relationships would be the curve-based dose–response model. Dose–response models are based on the non-linear relationship between a response variable and a predictor. The idea would be to integrate the interactions of multiple response variables and multiple predictors, simultaneously. As described above, the integration of multiple predictors can be done by predicting a response to a predictor as a function of other predictors (i.e. interactive effects). Interaction between multiple response variables is generally integrated through multivariate dose–response models, but these disregard non-linear relationships between response variables (Brown *et al*., 2012) and a non-linear adaptation of this method will be required. Similarly, GAMs or MARS offer the ability to identify interactive effects of multiple predictors on a response variable, however, integrating the interactive responses of multiple response variables remains to be implemented.

VI. CONCLUSIONS

(1) Multiple methods exist to detect thresholds and tipping points in empirical data, but they inherently lack the capacity to evaluate comprehensively the state of the ecosystem because they either do not account for non-linear interactions between multiple ecosystem variables or they reduce dimensionality in some way.

(2) Hillebrand *et al*. (2020) argued that tipping points cannot be detected in empirical data and even if a threshold is identified in the data, its real-life application in terms of broader ecosystem management may be limited. We identified that current methods used for detecting shifts in stable state are not comprehensive enough and the number of predictors or response variables used are often low meaning that ecosystem shifts are being inferred from changes in only part of the ecosystem.

(3) Nevertheless, there are techniques that have the potential to assess the state of an ecosystem more comprehensively and detect shifts between stable states, if these are present. These methods require large amounts of high-dimensionality data, which may not always be available.

(4) Inferring ecosystem state shifts from restricted data sets or from only a few components of an ecosystem may capture important changes in subsystems or processes but not overall

shift in ecosystem stable state because they do not fully capture the overall response of the ecosystem in relation to all its combined stressors.

(5) In terms of analysing ecosystem state and detecting shifts that may occur, the underlying question remains whether high-dimensionality data should become a rule, rather than an exception, for inferring broad ecosystem state patterns.

(6) Our perception of tipping points may have been driven by partial data sets and the methods available for analysing them. However, the new challenges in detecting tipping points from a broader ecosystem context suggest that the development of more holistic analytics is required to understand better how and to what extent ecosystems shift stable states.

VII. ACKNOWLEDGEMENTS

This manuscript is a result of the GES4SEAS (Achieving good environmental status for maintaining ecosystem services, by assessing integrated impacts of cumulative pressures) project, funded by the European Union under the Horizon Europe program (grant agreement no. 101059877), www.ges4seas.eu.

VIII. REFERENCES

References identified with an asterisk (*) are cited only within the online Supporting Information.

ABDI, H. & WILLIAMS., L.J. (2010). Principal Component Analysis. *Wiley Interdisciplinary Reviews: Computational Statistics*.

ANAND, A., PANDEY, M. K., SRIVASTAVA, P. K., GUPTA, A. & KHAN, M. L. (2021). Integrating multi-sensors data for species distribution mapping using deep learning and envelope models. *Remote Sensing* **13**(16), 3284.

ANDERSEN, T., CARSTENSEN, J., HERNANDEZ-GARCIA, E. & DUARTE, C.M. (2009). Ecological thresholds and regime shifts: approaches to identification. *Trends in Ecology & Evolution* **24**, 49–57.

ANGILLETTA JR, M.J. (2006). Estimating and comparing thermal performance curves. *Journal of Thermal Biology* **31**, 541–545.

AROGUNDADE, S., MDUDUZI, B. & HASSAN, A. S. (2022). Spatial impact of foreign direct investment on ecological footprint in Africa. *Environmental Science and Pollution Research* **29**(34), 51589-51608.

ARRIETA, A.B., DÍAZ-RODRÍGUEZ, N., DEL SER, J., BENNETOT, A., TABIK, S., BARBADO, A., GARCÍA, S., GIL-LÓPEZ, S., MOLINA, D., BENJAMINS, R. & CHATILA, R. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion* ***58***, 82–115.

ASPIN, T.W., KHAMIS, K., MATTHEWS, T.J., MILNER, A.M., O'CALLAGHAN, M.J., TRIMMER, M., WOODWARD, G. & LEDGER, M.E. (2019). Extreme drought pushes stream invertebrate communities over functional thresholds. *Global Change Biology* **25**, 230-244.

BAKER, M.E. & KING, R.S. (2010). A new method for detecting and interpreting biodiversity and ecological community thresholds. *Methods in Ecology and Evolution* **1**, 25–37.

BAN, Z., HU, X. & LI, J. (2022). Tipping points of marine phytoplankton to multiple environmental stressors. *Nature Climate Change* **12**, 1045–1051.

BENEDETTI, A., ABRAHAMOWICZ, M., LEFFONDRÉ, K., GOLDBERG, M.S. & TAMBLYN, R. (2009). Using generalized additive models to detect and estimate threshold associations. *The International Journal of Biostatistics* **5**.

BERDUGO, M., DELGADO-BAQUERIZO, M., SOLIVERES, S., HERNÁNDEZ-CLEMENTE, R., ZHAO, Y., GAITÁN, J.J., GROSS, N., SAIZ, H., MAIRE, V., LEHMANN, A. & RILLIG, M.C. (2020). Global ecosystem thresholds driven by aridity. *Science* **367**, 787–790.

BIGGS, R., PETERSON, G.D. & ROCHA, J.C. (2018). The Regime Shifts Database. *Ecology and Society* **23**.

BJERGAGER, M.B.A., DALHOFF, K., KRETSCHMANN, A., NØRGAARD, K.B., MAYER, P. & CEDERGREEN, N. (2017). Determining lower threshold concentrations for synergistic effects. *Aquatic Toxicology* **182**, 79–90.

BJERGAGER, M.B.A., DALHOFF, K., KRETSCHMANN, A., NØRGAARD, K.B., MAYER, P. & CEDERGREEN, N. (2017). Determining lower threshold concentrations for synergistic effects. *Aquatic Toxicology* **182**, 79–90.

BLANCHET, F. G., LEGENDRE, P., & BORCARD, D. (2008). Forward selection of explanatory variables. *Ecology* **89**, 2623–2632.

BOTERO, C.A., WEISSING, F.J., WRIGHT, J. & RUBENSTEIN, D.R. (2015). Evolutionary tipping points in the capacity to adapt to environmental change. *Proceedings of the National Academy of Sciences* **112**, 184–189.

BREIMAN, L. (1996). Stacked regressions. *Machine Learning* **24**, 49–64.

BROWN, C.C., HAVENER, T.M., MEDINA, M.W., KRAUSS, R.M., MCLEOD, H.L. & MOTSINGER-REIF, A.A. (2012). Multivariate methods and software for association mapping in dose-response genome-wide association studies. *BioData Mining* **5**, 1–15.

BÜHLMANN, P. & VAN DE GEER, S. (2011). *Statistics for high-dimensional data: methods, theory and applications*. Springer Science & Business Media.

BÜHLMANN, P., KALISCH, M. & MEIER, L. (2014). High-dimensional statistics with a view toward applications in biology. *Annual Review of Statistics and Its Application* **1**, 255–278.

CARSTENSEN, J., & WEYDMANN, A. (2012). Tipping points in the Arctic: eyeballing or statistical significance? *Ambio* **41**, 34–43.

CARSTENSEN, J., TELFORD, R.J. & BIRKS, H.J.B. (2013). Diatom flickering prior to regime shift. *Nature* **498**, E11–E12.

CÉRÉGHINO, R., TRZCINSKI, M.K., MACDONALD, A.A.M., MARINO, N.A., ACOSTA MERCADO, D., LEROY, C., CORBARA, B., ROMERO, G.Q., FARJALLA, V.F., BARBERIS, I.M. & DÉZERALD, O. (2022). Functional redundancy dampens precipitation change impacts on species-rich invertebrate communities across the Neotropics. *Functional Ecology* **36**, 1559–1572.

CHANG, C.W., MIKI, T., YE, H., SOUISSI, S., ADRIAN, R., ANNEVILLE, O., AGASILD, H., BAN, S., BE'ERI-SHLEVIN, Y., CHIANG, Y.R. & FEUCHTMAYR, H. (2022). Causal networks of phytoplankton diversity and biomass are modulated by environmental context. *Nature Communications* **13**, p.1140.

CHAUDHURI, P., & MARRON, J. S. (1999). SiZer for exploration of structures in curves. *Journal of the American Statistical Association* **94**, 807–823.

CIANNELLI, L., CHAN, K.S., BAILEY, K.M. & STENSETH, N.C., (2004). Nonadditive effects of the environment on the survival of a large marine fish population. *Ecology* **85**, 3418–3427.

CLARKE, R., RESSOM, H.W., WANG, A., XUAN, J., LIU, M.C., GEHAN, E.A. & WANG, Y. (2008). The properties of high-dimensional data spaces: implications for exploring gene and protein expression data. *Nature Reviews Cancer* **8**, 37–49.

CONNELL, S.D. & GHEDINI, G. (2015). Resisting regime-shifts: the stabilising effect of compensatory processes. *Trends in ecology & evolution* **30**, 513–515.

CONVERSI, A., DAKOS, V., GÅRDMARK, A., LING, S., FOLKE, C., MUMBY, P.J., GREENE, C., EDWARDS, M., BLENCKNER, T., CASINI, M. & PERSHING A. (2015) A holistic view of marine regime shifts. *Philosophical Transactions of the Royal Society B: Biological Sciences* **370**. 20130279.

DAILY, J.P., HITT, N.P., SMITH, D.R. AND SNYDER, C.D. (2012). Experimental and environmental factors affect spurious detection of ecological thresholds. *Ecology* **93**, 17–23.

DAKOS, V., CARPENTER, S.R., BROCK, W.A., ELLISON, A.M., GUTTAL, V., IVES, A.R., KÉFI, S., LIVINA, V., SEEKELL, D.A., VAN NES, E.H. AND SCHEFFER, M. (2012). Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PloS one* 7, p.e41010.

DAKOS, V., MATTHEWS, B., HENDRY, A.P., LEVINE, J., LOEUILLE, N., NORBERG, J., NOSIL, P., SCHEFFER, M. & DE MEESTER, L. (2019). Ecosystem tipping points in an evolving world. *Nature Ecology & Evolution* **3**, 355–362.

DUDNEY, J. & SUDING, K.N., (2020). The elusive search for tipping points. *Nature Ecology & Evolution* **4**, 1449–1450.

DZIALOWSKI, A.R. & JESSIE, W. (2009). Zebra mussels negate or mask the increasing effects of nutrient enrichment on algal biomass: a preliminary mesocosm study. *Journal of Plankton Research* **31**, 1437–1440.

ELLIS, N., SMITH, S.J. & PITCHER, C.R. (2012). Gradient forests: calculating importance gradients on physical predictors. *Ecology* **93**, 156–168.

FISHER, J. A. D., CASINI, M., FRANK, K. T., MÖLLMANN, C., LEGGETT, W. C., & DASKALOV, G. (2015). The importance of within-system spatial variation in drivers of marine ecosystem regime shifts. *Philosophical Transactions of the Royal Society B: Biological Sciences* 370, 20130271.

FOLKE, C., CARPENTER, S., WALKER, B., SCHEFFER, M., ELMQVIST, T., GUNDERSON, L. & HOLLING, C.S. (2004). Regime shifts, resilience, and biodiversity in ecosystem management. *Annual Review in Ecology, Evolution and Systematics* **35**, 557–581.

FOX, E.W., HILL, R.A., LEIBOWITZ, S.G., OLSEN, A.R., THORNBRUGH, D.J. & WEBER, M.H. (2017). Assessing the accuracy and stability of variable selection methods for random forest modelling in ecology. *Environmental Monitoring and Assessment* **189**, 1–20.

FRANCESCO FICETOLA, G. & DENOËL, M. (2009). Ecological thresholds: an assessment of methods to identify abrupt changes in species–habitat relationships. *Ecography* **32**, 1075–1084.

GALIPAUD, M., GILLINGHAM, M. A., DAVID, M., & DECHAUME-MONCHARMONT, F. X. (2014). Ecologists overestimate the importance of predictor variables in model averaging: a plea for cautious interpretations. *Methods in Ecology and Evolution* **5**, 983–991.

GANAIE, M.A., HU, M., MALIK, A.K., TANVEER, M. & SUGANTHAN, P.N. (2022). Ensemble deep learning: A review. *Engineering Applications of Artificial Intelligence* **115**, p.105151.

GKIOULEKAS, I. & PAPAGEORGIOU, L.G. (2019). Piecewise regression analysis through information criteria using mathematical programming. *Expert Systems with Applications* **121**, 362–372.

GOUTELLE, S., MAURIN, M., ROUGIER, F., BARBAUT, X., BOURGUIGNON, L., DUCHER, M. AND MAIRE, P. (2008). The Hill equation: a review of its capabilities in pharmacological modelling. *Fundamental & Clinical Pharmacology* **22**, 633–648.

GRZIWOTZ, F., CHANG, C.W., DAKOS, V., VAN NES, E.H., SCHWARZLÄNDER, M., KAMPS, O., HEẞLER, M., TOKUDA, I.T., TELSCHOW, A. & HSIEH, C.H. (2023). Anticipating the occurrence and type of critical transitions. *Science Advances* 9, p.eabq4558.

HALPERN, B.S., WALBRIDGE, S., SELKOE, K.A., KAPPEL, C.V., MICHELI, F., D'AGROSA, C., BRUNO, J.F., CASEY, K.S., EBERT, C., FOX, H.E. & FUJITA, R. (2008). A global map of human impact on marine ecosystems. *Science* **319**, 948–952.

HARE, S.R. & MANTUA, N.J. (2000). Empirical evidence for North Pacific regime shifts in 1977 and 1989. *Progress in Oceanography* **47**, 103–145.

HARLEY, C.D., CONNELL, S.D., DOUBLEDAY, Z.A., KELAHER, B., RUSSELL, B.D., SARÀ, G. & HELMUTH, B. (2017). Conceptualizing ecosystem tipping points within a physiological framework. *Ecology and Evolution* **7**, 6035–6045.

HEMRAJ, D. A., HOSSAIN, A., YE, Q., QIN, J. G., & LETERME, S. C. (2017). Anthropogenic shift of planktonic food web structure in a coastal lagoon by freshwater flow regulation. *Scientific Reports* **7**, 44441.

HEMRAJ, D.A., BISHOP, M.J., HANCOCK, B., MINUTI, J.J., THURSTAN, R.H., ZU ERMGASSEN, P.S. & RUSSELL, B.D. (2022). Oyster reef restoration fails to recoup global historic ecosystem losses despite substantial biodiversity gain. *Science Advances* **8**, p.eabp8747.

HILLEBRAND, H., DONOHUE, I., HARPOLE, W.S., HODAPP, D., KUCERA, M., LEWANDOWSKA, A.M., MERDER, J., MONTOYA, J.M. & FREUND, J.A. (2020). Thresholds for ecological responses to global change do not emerge from empirical data. *Nature Ecology & Evolution* **4**, 1502–1509.

HILLEBRAND, H., KUCZYNSKI, L., KUNZE, C., RILLO, M.C. & DAJKA, J.C. (2023). Thresholds and tipping points are tempting but not necessarily suitable concepts to address anthropogenic biodiversity change—an intervention. *Marine Biodiversity* **53**, 1–8.

KÉFI, S., SAADE, C., BERLOW, E.L., CABRAL, J.S. & FRONHOFER, E.A. (2022). Scaling up our understanding of tipping points. *Philosophical Transactions of the Royal Society B* **377**, p.20210386.

LEGENDRE, P., DALLOT, S., & LEGENDRE, L. (1985). Succession of species within a community: chronological clustering, with applications to marine and freshwater zooplankton. *The American Naturalist* **125**, 257–288.

LENTON, T.M., 2011. Early warning of climate tipping points. *Nature Climate Change* **1**, 201–209.

LINTZ, H.E., MCCUNE, B., GRAY, A.N. & MCCULLOH, K.A. (2011). Quantifying ecological thresholds from response surfaces. *Ecological Modelling* **222**, 427–436.

LIU, S., MALJOVEC, D., WANG, B., BREMER, P.T. & PASCUCCI, V. (2016). Visualizing high-dimensional data: Advances in the past decade. *IEEE transactions on visualization and computer graphics* **23**, 1249–1268.

MATTHEWS, T.J., STEINBAUER, M.J., TZIRKALLI, E., TRIANTIS, K.A. & WHITTAKER, R.J. (2014). Thresholds and the species–area relationship: a synthetic analysis of habitat island datasets. *Journal of Biogeography*, **41**, 1018–1028.

MCCUNE, B. & GRACE, J.B. (2002). Analysis of ecological communities. *Gleneden Beach, OR: MjM software design* **28**, 1–300.

MEYER, H., REUDENBACH, C., WÖLLAUER, S. & NAUSS, T. (2019). Importance of spatial predictor variable selection in machine learning applications–Moving from data reproduction to spatial prediction. *Ecological Modelling* **411**, p.108815.

MILLIE, D.F., WECKMAN, G.R., YOUNG II, W.A., IVEY, J.E., CARRICK, H.J. & FAHNENSTIEL, G.L. (2012). Modeling microalgal abundance with artificial neural networks: Demonstration of a heuristic 'Grey-Box'to deconvolve and quantify environmental influences. *Environmental Modelling & Software* **38**, 27–39.

MÖLLMANN, C., DIEKMANN, R., MÜLLER-KARULIS, B., KORNILOVS, G., PLIKSHS, M. & AXE, P. (2009). Reorganization of a large marine ecosystem due to atmospheric and anthropogenic pressure: a discontinuous regime shift in the Central Baltic Sea. *Global Change Biology* **15**, 1377–1393.

MORENO-MATEOS, D., ALBERDI, A., MORRIËN, E., VAN DER PUTTEN, W.H., RODRÍGUEZ-UÑA, A. & MONTOYA, D. (2020). The long-term restoration of ecosystem complexity. *Nature Ecology & Evolution* **4**, 676–685.

O'BRYAN, C.J., ALLAN, J.R., HOLDEN, M., SANDERSON, C., VENTER, O., DI MARCO, M., MCDONALD-MADDEN, E. & WATSON, J.E. (2020). Intense human pressure is widespread across terrestrial vertebrate ranges. *Global Ecology and Conservation* **21**, p.e00882.

O'BRIEN, D.A., DEB, S., SIDHEEKH, S., KRISHNAN, N.C., SHARATHI DUTTA, P. & CLEMENTS, C.F. (2023). EWSmethods: an R package to forecast tipping points at the community level using early warning signals, resilience measures, and machine learning models. *Ecography* 2023, p.e06674.

OCHOA-HUESO, R. (2016). Nonlinear disruption of ecological interactions in response to nitrogen deposition. *Ecology* **97**, 2802–2814.

O'DEA, R.E., LAGISZ, M., JENNIONS, M.D., KORICHEVA, J., NOBLE, D.W., PARKER, T.H., GUREVITCH, J., PAGE, M.J., STEWART, G., MOHER, D. AND NAKAGAWA, S. (2021). Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: a PRISMA extension. *Biological Reviews* **96**, 1695–1722.

ÖZESMI, S.L. & ÖZESMI, U. (1999). An artificial neural network approach to spatial habitat modelling with interspecific interaction. *Ecological Modelling* **116**, 15–31.

PETERSEN, J.K., HANSEN, J.W., LAURSEN, M.B., CLAUSEN, P., CARSTENSEN, J. & CONLEY, D.J. (2008). Regime shift in a coastal marine ecosystem. *Ecological Applications* **18**, 497–510.

PIET, G., GRUNDLEHNER, A., JONGBLOED, R., TAMIS, J. AND DE VRIES, P. (2023). SCAIRM: A spatial cumulative assessment of impact risk for management. *Ecological Indicators* **157**, p.111157.

PILOSOF, S., PORTER, M. A., PASCUAL, M., & KÉFI, S. (2017). The multilayer nature of ecological networks. *Nature Ecology & Evolution* 1, 0101.

RANDSALU-WENDRUP, L., CONLEY, D.J., CARSTENSEN, J. & FRITZ, S.C. (2016). Paleolimnological records of regime shifts in lakes in response to climate change and anthropogenic activities. *Journal of Paleolimnology* **56**, 1–14

RIETKERK, M., BASTIAANSEN, R., BANERJEE, S., VAN DE KOPPEL, J., BAUDENA, M. & DOELMAN, A. (2021). Evasion of tipping in complex systems through spatial pattern formation. *Science* **374**, p.eabj0359.

RITZ, C., BATY, F., STREIBIG, J.C. & GERHARD, D. (2015). Dose-response analysis using R. *PloS one* **10**, p.e0146021.y

ROCHA-SANTOS, L., PESSOA, M.S., CASSANO, C.R., TALORA, D.C., ORIHUELA, R.L., MARIANO-NETO, E., MORANTE-FILHO, J.C., FARIA, D. AND CAZETTA, E., (2016). The shrinkage of a forest: Landscape-scale deforestation leading to overall changes in local forest structure. *Biological Conservation* **196**, 1–9.

ROSENFELD, J.S. (2017). Developing flow–ecology relationships: Implications of nonlinear biological responses for water management. *Freshwater Biology*, **62**, 1305–1324.

RUBERG, S.J. (1995). Dose response studies I. Some design considerations. *Journal of Biopharmaceutical Statistics* **5**, 1–14.

RUSSELL, B.D., THOMPSON, J.A.I., FALKENBERG, L.J. & CONNELL, S.D. (2009). Synergistic effects of climate change and local stressors: CO_2 and nutrient-driven change in subtidal rocky habitats. *Global Change Biology* **15**, 2153–2162.

SAMHOURI, J.F., ANDREWS, K.S., FAY, G., HARVEY, C.J., HAZEN, E.L., HENNESSEY, S.M., HOLSMAN, K., HUNSICKER, M.E., LARGE, S.I., MARSHALL, K.N. & STIER, A.C. (2017). Defining ecosystem thresholds for human activities and environmental pressures in the California Current. *Ecosphere* **8**(6), e01860.

SCHEFFER, M., BASCOMPTE, J., BROCK, W.A., BROVKIN, V., CARPENTER, S.R., DAKOS, V., HELD, H., VAN NES, E.H., RIETKERK, M. AND SUGIHARA, G. (2009). Early-warning signals for critical transitions. *Nature* **461**, 53–59.

SCHEFFER, M., CARPENTER, S., FOLEY, J.A., FOLKE, C. & WALKER, B. (2001). Catastrophic shifts in ecosystems. *Nature* **413**, 591–596.

SCHNEIDER, F.D. & KÉFI, S., 2016. Spatially heterogeneous pressure raises risk of catastrophic shifts. *Theoretical Ecology* **9**, 207–217.

SINCLAIR, B.J., MARSHALL, K.E., SEWELL, M.A., LEVESQUE, D.L., WILLETT, C.S., SLOTSBO, S., DONG, Y., HARLEY, C.D., MARSHALL, D.J., HELMUTH, B.S. & HUEY, R.B. (2016). Can we predict ectotherm responses to climate change using thermal performance curves and body temperatures?. *Ecology Letters* **19**, 1372–1385.

SONDEREGGER, D.L., WANG, H., CLEMENTS, W.H. & NOON, B.R. (2009). Using SiZer to detect thresholds in ecological data. *Frontiers in Ecology and the Environment* **7,** 190–195.

TAUD, H. & MAS, J.F., (2018). Multilayer perceptron (MLP). *Geomatic approaches for modelling land change scenarios* 451–455.

THOMPSON, L.M., LYNCH, A.J., BEEVER, E.A., ENGMAN, A.C., FALKE, J.A., JACKSON, S.T., KRABBENHOFT, T.J., LAWRENCE, D.J., LIMPINSEL, D., MAGILL, R.T. AND MELVIN, T.A. (2021). Responding to ecosystem transformation: Resist, accept, or direct? *Fisheries* **46**, 8–21.

TILMAN, D., MAY, R.M., LEHMAN, C.L. & NOWAK, M.A. (1994). Habitat destruction and the extinction debt. *Nature* **371**, 65–66.

TOMS, J.D. & LESPERANCE, M.L. (2003). Piecewise regression: a tool for identifying ecological thresholds. *Ecology* **84**, 2034–2041.

TONNO, I., TALAS, L., FREIBERG, R., KISAND, A., BELLE, S., STIVRINS, N., ALLIKSAAR, T., HEINSALU, A., VESKI, S. & KISAND, V. (2021). Environmental drivers and abrupt changes of phytoplankton community in temperate lake Lielais Svētiņu, Eastern Latvia, over the last Post-Glacial period from 14.5 kyr. *Quaternary Science Reviews* **263**, p.107006.

VAN DER MAATEN, L., POSTMA, E.O. & VAN DEN HERIK, H.J. (2009). Dimensionality reduction: A comparative review. *Journal of Machine Learning Research* **10**, p.13.

VAN NES, E.H. & SCHEFFER, M. (2004). Large species shifts triggered by small forces. *The American Naturalist*, 164 255–266.

VASILAKOPOULOS, P. & MARSHALL, C.T. (2015). Resilience and tipping points of an exploited fish population over six decades. *Global change biology 21*, 1834–1847.

VASILAKOPOULOS, P., RAITSOS, D.E., TZANATOS, E. & MARAVELIAS, C.D. (2017). Resilience and regime shifts in a marine biodiversity hotspot. *Scientific reports* **7**, p.13647.

WARD, N.K., LYNCH, A.J., BEEVER, E.A., BOOKER, J., BOUSKA, K.L., EMBKE, H., HOUSER, J.N., KOCIK, J.F., KOCIK, J., LAWRENCE, D.J. & LEMON, M.G. (2023). Reimagining large river management using the Resist–Accept–Direct (RAD) framework in the Upper Mississippi River. *Ecological Processes* 1**2**, 48.

WATSON, J.E.M. & VENTER, O. (2019). Mapping the continuum of humanity's footprint on land. *One Earth* **1**, 175–180.

XUAN, L., SHENG, Z., LU, J., QIU, Q., CHEN, J. & XIONG, J. (2019). Bacterioplankton community responses and the potential ecological thresholds along disturbance gradients. *Science of the Total Environment* **696**, p.134015.

YADAV, B., WENNERBERG, K., AITTOKALLIO, T., & TANG, J. (2015). Searching for drug synergy in complex dose–response landscapes using an interaction potency model. *Computational and structural biotechnology journal* **13**, 504–513.

YING, X. (2019). An overview of overfitting and its solutions. In *Journal of physics: Conference series* **1168**, 022022. IOP Publishing.

ZHAO, Q., VAN DEN BRINK, P.J., XU, C., WANG, S., CLARK, A.T., KARAKOÇ, C., SUGIHARA, G., WIDDICOMBE, C.E., ATKINSON, A., MATSUZAKI, S.I.S. & SHINOHARA, R. (2023). Relationships of temperature and biodiversity with stability of natural aquatic food webs. *Nature Communications* **14**, p.3507.

*ACEVEDO, M. F. & WALLER, W. T. (2000). Modelling and control of a simple trophic aquatic system. *Ecological Modelling* **131**(2-3), 269-284.

*AGATHOKLEOUS, E., ANAV, A., ARAMINIENE, V., DE MARCO, A., DOMINGOS, M., KITAO, M., KOIKE, T., MANNING, W. J., PAOLETTI, E. & SAITANIS, C. J. (2019). Commentary: EPA's proposed expansion of dose-response analysis is a positive step towards improving its ecological risk assessment. *Environmental Pollution* **246**, 566-570.

*ALCARAZ, M., FELIPE, J., GROTE, U., ARASHKEVICH, E. & NIKISHINA, A. (2014). Life in a warming ocean: thermal thresholds and metabolic balance of arctic zooplankton. *Journal of plankton research* **36**, 3-10.

*ALEXANDRA, J. (2012). Australia's landscapes in a changing climate—caution, hope, inspiration, and transformation. *Crop and Pasture Science* **63**, 215-231.

*ALI, G., TETZLAFF, D., MCDONNELL, J. J., SOULSBY, C., CAREY, S., LAUDON, H., MCGUIRE, K., BUTTLE, J., SEIBERT, J. & SHANLEY, J. (2015). Comparison of threshold hydrologic response across northern catchments. *Hydrological Processes* **29**, 3575-3591.

*ALLEN, C. D. (2007). Interactions across spatial scales among forest dieback, fire, and erosion in northern New Mexico landscapes. *Ecosystems* **10**, 797-808.

*ALLEN-DIAZ, B. & BARTOLOME, J. W. (1998). Sagebrush–grass vegetation dynamics: comparing classical and state-transition models. *Ecological Applications* **8**, 795-804.

*AN, H., BASKIN, C. C. & MA, M. (2022). Nonlinear response of the soil seed bank and its role in plant community regeneration with increased grazing disturbance. *Journal of Applied Ecology* **59**, 2593-2603.

*ANDERSON, L. J., MAHERALI, H., JOHNSON, H. B., POLLEY, H. W. & JACKSON, R. B. (2001). Gas exchange and photosynthetic acclimation over subambient to elevated CO2 in a C3–C4 grassland. *Global Change Biology* **7**, 693-707.

*ANDRE, J., HADDON, M. & PECL, G. T. (2010). Modelling climate-change-induced nonlinear thresholds in cephalopod population dynamics. *Global Change Biology* **16**, 2866-2875.

*ANG, T. K., SAFUAN, H. M., SIDHU, H. S., JOVANOSKI, Z. & TOWERS, I. N. (2019). Impact of harvesting on a bioeconomic predator–prey fishery model subject to environmental toxicant. *Bulletin of mathematical biology* **81**, 2748-2767.

*ANUNCIAÇÃO, P. R., BARROS, F. M., RIBEIRO, M. C., DE CARVALHO, L. M. T. & ERNST, R. (2021). Taxonomic and functional threshold responses of vertebrate communities in the Atlantic Forest Hotspot. *Biological Conservation* **257**, 109137.

*ARBIEU, U., GRÜNEWALD, C., SCHLEUNING, M. & BÖHNING-GAESE, K. (2017). The importance of vegetation density for tourists' wildlife viewing experience and satisfaction in African savannah ecosystems. *PloS one* **12**, e0185793.

*ARMITAGE, A. & FOURQUREAN, J. W. (2016). Carbon storage in seagrass soils: long-term nutrient history exceeds the effects of near-term nutrient enrichment. *Biogeosciences* **13**, 313-321.

*ARUMUGAM, R., CHANDRASEKAR, V. & SENTHILKUMAR, D. (2021). Rate-induced tipping and regime shifts in a spatial ecological system. *The European Physical Journal Special Topics* **230**, 3221-3227.

*AYDIN, C., ESEN, Ö. & AYDIN, R. (2019). Is the ecological footprint related to the Kuznets curve a real process or rationalizing the ecological consequences of the affluence? Evidence from PSTR approach. *Ecological indicators* **98**, 543-555.

*AYDIN, C., ESEN, Ö. & AYDIN, R. (2022). Analyzing the economic development-driven ecological deficit in the EU-15 countries: new evidence from PSTR approach. *Environmental Science and Pollution Research*, 1-17.

*BACHELET, D., NEILSON, R. P., LENIHAN, J. M. & DRAPEK, R. J. (2001). Climate change effects on vegetation distribution and carbon budget in the United States. *Ecosystems* **4**, 164-185.

*BAKER, M. E. & KING, R. S. (2013). Of TITAN and straw men: an appeal for greater understanding of community data. *Freshwater Science* **32**, 489-506.

*BARTON, H. A. (2003). Endocrine active substances and dose response for individuals and populations. *Pure and applied chemistry* **75**(11-12), 2159-2166.

*BARTUSZEVIGE, A. M., PAVLACKY, D. C., BURRIS, L. & HERBENER, K. (2012). Inundation of playa wetlands in the western Great Plains relative to landcover context. *Wetlands* **32**, 1103-1113.

*BASCOMPTE, J. & SOLÉ, R. V. (1998). Effects of habitat destruction in a prey–predator metapopulation model. *Journal of theoretical biology* **195**, 383-393.

*Batllori, E., De Cáceres, M., Brotons, L., Ackerly, D. D., Moritz, M. A. & Lloret, F. (2019). Compound fire-drought regimes promote ecosystem transitions in Mediterranean ecosystems. *Journal of Ecology* **107**, 1187-1198.

*Batool, Z., Ali, S. & Rehman, A. (2022). Environmental impact of ICT on disaggregated energy consumption in China: A threshold regression analysis. *Sustainability* **14**, 15600.

*Béal, D., Brasseur, P., Brankart, J.-M., Ourmieres, Y. & Verron, J. (2010). Characterization of mixing errors in a coupled physical biogeochemical model of the North Atlantic: implications for nonlinear estimation using Gaussian anamorphosis. *Ocean Science* **6**, 247-262.

*Beauchesne, D., Jaeger, J. A. & St-Laurent, M.-H. (2014). Thresholds in the capacity of boreal caribou to cope with cumulative disturbances: Evidence from space use patterns. *Biological Conservation* **172**, 190-199.

*Beaugrand, G. (2009). Decadal changes in climate and ecosystems in the North Atlantic Ocean and adjacent seas. *Deep Sea Research Part II: Topical Studies in Oceanography* **56**(8-10), 656-673.

*Beaugrand, G., Mackas, D. & Goberville, E. (2013). Applying the concept of the ecological niche and a macroecological approach to understand how climate influences zooplankton: advantages, assumptions, limitations and requirements. *Progress in oceanography* **111**, 75-90.

*Bedia, J., Herrera, S., Gutiérrez, J. M., Benali, A., Brands, S., Mota, B. & Moreno, J. M. (2015). Global patterns in the sensitivity of burned area to fire-weather: Implications for climate change. *Agricultural and Forest Meteorology* **214**, 369-379.

*BELLINGERI, M. & VINCENZI, S. (2013). Robustness of empirical food webs with varying consumer's sensitivities to loss of resources. *Journal of theoretical biology* **333**, 18-26.

*BENITO, X., LUETHJE, M., SCHNEIDER, T., FRITZ, S. C., BAKER, P. A., PEDERSEN, E. J., GAÜZÈRE, P., DE NOVAES NASCIMENTO, M., BUSH, M. & RUHI, A. (2022). Ecological resilience in tropical Andean lakes: A paleolimnological perspective. *Limnology and Oceanography* **67**, S23-S37.

*BERDUGO, M., VIDIELLA, B., SOLÉ, R. V. & MAESTRE, F. T. (2022). Ecological mechanisms underlying aridity thresholds in global drylands. *Functional Ecology* **36**, 4-23.

*BERL, J. L., EDWARDS, J. W. & BOLSINGER, J. S. (2015). Scale-dependent and multi-metric nest habitat thresholds for Red-headed Woodpeckers at the northern periphery of their range. *The Condor: Ornithological Applications* **117**, 203-216.

*BETTS, M. G., FORBES, G. J., DIAMOND, A. W. & TAYLOR, P. D. (2006). Independent effects of fragmentation on forest songbirds: an organism-based approach. *Ecological Applications* **16**, 1076-1089.

*BEYER, H. L., UNG, R., MURRAY, D. L. & FORTIN, M. J. (2013). Functional responses, seasonal variation and thresholds in behavioural responses of moose to road density. *Journal of Applied Ecology* **50**, 286-294.

*BHATTACHARYYA, J., PIIROINEN, P. T. & BANERJEE, S. (2021). Dynamics of a Filippov predator-prey system with stage-specific intermittent harvesting. *Nonlinear Dynamics* **105**, 1019-1043.

*BILDIRICI, M. & ERSIN, Ö. (2018). Markov-switching vector autoregressive neural networks and sensitivity analysis of environment, economic growth and petrol prices. *Environmental Science and Pollution Research* **25**, 31630-31655.

*BISWAS, S., TIWARI, P. K. & PAL, S. (2021). Delay-induced chaos and its possible control in a seasonally forced eco-epidemiological model with fear effect and predator switching. *Nonlinear Dynamics* **104**, 2901-2930.

*BITTICK, S. J., SUTULA, M. & FONG, P. (2018). A tale of two algal blooms: Negative and predictable effects of two common bloom-forming macroalgae on seagrass and epiphytes. *Marine environmental research* **140**, 1-9.

*BJØRNSTAD, O. N., FINKENSTÄDT, B. F. & GRENFELL, B. T. (2002). Dynamics of measles epidemics: estimating scaling of transmission rates using a time series SIR model. *Ecological monographs* **72**, 169-184.

*BLECKER, S., STILLINGS, L., AMACHER, M., IPPOLITO, J. & DECRAPPEO, N. (2012). Development of vegetation based soil quality indices for mineralized terrane in arid and semi-arid regions. *Ecological indicators* **20**, 65-74.

*BOCHET, E., MOLINA, M. J., MONLEÓN, V., ESPIGARES, T., NICOLAU, J. M., DE LAS HERAS, M. M. & GARCÍA-FAYOS, P. (2021). Interactions of past human disturbance and aridity trigger abrupt shifts in the functional state of Mediterranean holm oak woodlands. *Catena* **206**, 105514.

*BOERS, N., MARWAN, N., BARBOSA, H. M. & KURTHS, J. (2017). A deforestation-induced tipping point for the South American monsoon system. *Scientific reports* **7**, 41489.

*BOHRER, G., KATUL, G. G., NATHAN, R., WALKO, R. L. & AVISSAR, R. (2008). Effects of canopy heterogeneity, seed abscission and inertia on wind-driven dispersal kernels of tree seeds. *Journal of Ecology* **96**, 569-580.

*BOLDT, J. L., HAZEN, E. L., HUNSICKER, M. E., FU, C., PERRY, R. I. & SHAN, X. (2021). Quantifying ecosystem responses to environmental and human pressures in the marine ecosystem off the west coast of Vancouver Island. *Ecological indicators* **132**, 108232.

*BONFIM, F. C. G., DODONOV, P., GUIMARÃES JR, P. R. & CAZETTA, E. (2023). Habitat loss shapes the structure and species roles in tropical plant–frugivore networks. *Oikos* **2023**, e09399.

*BOUGHTON, E. H., QUINTANA-ASCENCIO, P. F., JENKINS, D. G., BOHLEN, P. J., FAUTH, J. E., ENGEL, A., SHUKLA, S., KIKER, G., HENDRICKS, G. & SWAIN, H. M. (2019). Trade-offs and synergies in a payment-for-ecosystem services program on ranchlands in the Everglades headwaters. *Ecosphere* **10**, e02728.

*BOULTON, C. A., RITCHIE, P. D. & LENTON, T. M. (2020). Abrupt changes in Great Britain vegetation carbon projected under climate change. *Global Change Biology* **26**, 4436-4448.

*BOZZEDA, F., ZANGRILLI, M. P. & DEFEO, O. (2016). Assessing sandy beach macrofaunal patterns along large-scale environmental gradients: A Fuzzy Naïve Bayes approach. *Estuarine, Coastal and Shelf Science* **175**, 70-78.

*BRENNAN, R., SCOTT, J., SHARPLEY, A., LALLY, H., JARVIE, H., BOWES, M., HAGGARD, B. & GBUR, E. (2017). Linking soil erosion to instream dissolved phosphorus cycling and periphyton growth. *JAWRA Journal of the American Water Resources Association* **53**, 809-821.

*BRINDIS-BADILLO, D. A., ARROYO-RODRÍGUEZ, V., MENDOZA, E., WIES, G. & MARTÍNEZ-RAMOS, M. (2022). Conserving dominant trees in human-modified landscapes at the Lacandon tropical rainforest. *Biological Conservation* **270**, 109548.

*BRODIE, J. F. & POST, E. (2010). Nonlinear responses of wolverine populations to declining winter snowpack. *Population Ecology* **52**, 279-287.

*BROWN, B. J., MCLAUGHLIN, B. C., BLAKEY, R. V. & MORUETA-HOLME, N. (2018). Future vulnerability mapping based on response to extreme climate events: dieback thresholds in an endemic California oak. *Diversity and Distributions* **24**, 1186-1198.

*BRUEL, R., MARCHETTO, A., BERNARD, A., LAMI, A., SABATIER, P., FROSSARD, V. & PERGA, M. E. (2018). Seeking alternative stable states in a deep lake. *Freshwater Biology* **63**, 553-568.

*BU, W. & REN, S. (2023). Does economic growth target constraint put pressure on green energy efficiency? Evidence from China. *Environmental Science and Pollution Research* **30**, 31171-31187.

*BUECHLING, A., MARTIN, P. H. & CANHAM, C. D. (2017). Climate and competition effects on tree growth in Rocky Mountain forests. *Journal of Ecology* **105**, 1636-1647. *

*BUHLE, E. R., FEIST, B. E. & HILBORN, R. (2012). Population dynamics and control of invasive Spartina alterniflora: inference and forecasting under uncertainty. *Ecological Applications* **22**, 880-893.

*BURDON, F. J., MCINTOSH, A. R. & HARDING, J. S. (2013). Habitat loss drives threshold response of benthic invertebrate communities to deposited sediment in agricultural streams. *Ecological Applications* **23**, 1036-1047.

*BURKETT, V. R., WILCOX, D. A., STOTTLEMYER, R., BARROW, W., FAGRE, D., BARON, J., PRICE, J., NIELSEN, J. L., ALLEN, C. D. & PETERSON, D. L. (2005). Nonlinear dynamics in ecosystem response to climatic change: case studies and policy implications. *Ecological complexity* **2**, 357-394.

*BURTHE, S. J., HENRYS, P. A., MACKAY, E. B., SPEARS, B. M., CAMPBELL, R., CARVALHO, L., DUDLEY, B., GUNN, I. D., JOHNS, D. G. & MABERLY, S. C. (2016). Do early warning indicators consistently predict nonlinear change in long-term ecological data? *Journal of Applied Ecology* **53**, 666-676.

*BURTHE, S., DAUNT, F., BUTLER, A., ELSTON, D. A., FREDERIKSEN, M., JOHNS, D., NEWELL, M., THACKERAY, S. J. & WANLESS, S. (2012). Phenological trends and trophic mismatch across multiple levels of a North Sea pelagic food web. *Marine Ecology Progress Series* **454**, 119-133.

*BUSKIRK, J. V. & OSTFELD, R. S. (1998). Habitat heterogeneity, dispersal, and local risk of exposure to Lyme disease. *Ecological Applications* **8**, 365-378.

*CAMARERO, J. J., GAZOL, A., SANGÜESA-BARREDA, G., OLIVA, J. & VICENTE-SERRANO, S. M. (2015). To die or not to die: early warnings of tree dieback in response to a severe drought. *Journal of Ecology* **103**, 44-57.

*CAMPBELL, B., STAFFORD SMITH, D. & MCKEON, G. (1997). Elevated CO 2and water supply interactions in grasslands: a pastures and rangelands management perspective. *Global Change Biology* **3**, 177-187.

*CAMPOS, C. A., TONIN, A. M., KENNARD, M. J. & GONÇALVES JÚNIOR, J. F. (2022). Setting thresholds of ecosystem structure and function to protect streams of the Brazilian savanna. *Frontiers in Environmental Science* **10**, 867905.

*CANTARELLO, E., NEWTON, A. C., MARTIN, P. A., EVANS, P. M., GOSAL, A. & LUCASH, M. S. (2017). Quantifying resilience of multiple ecosystem services and biodiversity in a temperate forest landscape. *Ecology and Evolution* **7**, 9661-9675.

*CAO, X., CHAI, L., JIANG, D., WANG, J., LIU, Y. & HUANG, Y. (2018). Loss of biodiversity alters ecosystem function in freshwater streams: potential evidence from benthic macroinvertebrates. *Ecosphere* **9**, e02445.

*CAO, X., WANG, J., LIAO, J., GAO, Z., JIANG, D., SUN, J., ZHAO, L., HUANG, Y. & LUAN, S. (2017). Bacterioplankton community responses to key environmental variables in plateau freshwater lake ecosystems: A structural equation modeling and change point analysis. *Science of the Total Environment* **580**, 457-467. *

*CAPON, S. J., LYNCH, A. J. J., BOND, N., CHESSMAN, B. C., DAVIS, J., DAVIDSON, N., FINLAYSON, M., GELL, P. A., HOHNBERG, D. & HUMPHREY, C. (2015). Regime shifts, thresholds and multiple stable states in freshwater ecosystems; a critical appraisal of the evidence. *Science of the Total Environment* **534**, 122-130.

*CARDOSO, A. W., ARCHIBALD, S., BOND, W. J., COETSEE, C., FORREST, M., GOVENDER, N., LEHMANN, D., MAKAGA, L., MPANZA, N. & NDONG, J. E. (2022). Quantifying the environmental limits to fire spread in grassy ecosystems. *Proceedings of the National Academy of Sciences* **119**, e2110364119.

*CARMACK, E., MCLAUGHLIN, F., WHITEMAN, G. & HOMER-DIXON, T. (2012). Detecting and coping with disruptive shocks in Arctic marine systems: A resilience approach to place and people. *Ambio* **41**, 56-65.

*CARPENTER, S. R., LUDWIG, D. & BROCK, W. A. (1999). Management of eutrophication for lakes subject to potentially irreversible change. *Ecological Applications* **9**, 751-771.

*CARR, J., MARIOTTI, G., FAHGERAZZI, S., MCGLATHERY, K. & WIBERG, P. (2018). Exploring the impacts of seagrass on coupled marsh-tidal flat morphodynamics. *Frontiers in Environmental Science* **6**, 92.

*CARVALHO, L., MCDONALD, C., DE HOYOS, C., MISCHKE, U., PHILLIPS, G., BORICS, G., POIKANE, S., SKJELBRED, B., SOLHEIM, A. L. & VAN WICHELEN, J. (2013). Sustaining recreational quality of European lakes: minimizing the health risks from algal blooms through phosphorus control. *Journal of Applied Ecology* **50**, 315-323.

*CASTOLDI, N. & BECHINI, L. (2010). Integrated sustainability assessment of cropping systems with agro-ecological and economic indicators in northern Italy. *European journal of agronomy* **32**, 59-72.

*CAVIN, L., MOUNTFORD, E. P., PETERKEN, G. F. & JUMP, A. S. (2013). Extreme drought alters competitive dominance within and between tree species in a mixed forest stand. *Functional Ecology* **27**, 1424-1435.

*CHA, Y., SHIN, J., GO, B., LEE, D.-S., KIM, Y., KIM, T. & PARK, Y.-S. (2021). An interpretable machine learning method for supporting ecosystem management: Application to species distribution models of freshwater macroinvertebrates. *Journal of Environmental Management* **291**, 112719.

*CHADWICK, O. A., GAVENDA, R. T., KELLY, E. F., ZIEGLER, K., OLSON, C. G., ELLIOTT, W. C. & HENDRICKS, D. M. (2003). The impact of climate on the biogeochemical functioning of volcanic soils. *Chemical Geology* **202**(3-4), 195-223.

*CHAPMAN, M., WILTSHIRE, S., BAUR, P., BOWLES, T., CARLISLE, L., CASTILLO, F., ESQUIVEL, K., GENNET, S., ILES, A. & KARP, D. (2022). Social-ecological feedbacks drive tipping points in farming system diversification. *One Earth* **5**, 283-292.

*CHAPMAN, P. M. (2017). Assessing and managing stressors in a changing marine environment. *Marine Pollution Bulletin* **124**, 587-590.

*CHEN, K. & OLDEN, J. D. (2020). Threshold responses of riverine fish communities to land use conversion across regions of the world. *Global Change Biology* **26**, 4952-4965.

*CHEN, N., ZHANG, Y., ZHU, J., CONG, N., ZHAO, G., ZU, J., WANG, Z., HUANG, K., WANG, L. & LIU, Y. (2021). Multiple-scale negative impacts of warming on ecosystem carbon use efficiency across the Tibetan Plateau grasslands. *Global Ecology and Biogeography* **30**, 398-413.

*CHEN, T., WANG, Q., WANG, Y. & PENG, L. (2022). Processes and mechanisms of vegetation ecosystem responding to climate and ecological restoration in China. *Frontiers in plant science* **13**, 1062691.

*CHEN, W., ZHU, D., CIAIS, P., HUANG, C., VIOVY, N. & KAGEYAMA, M. (2019). Response of vegetation cover to CO2 and climate changes between Last Glacial Maximum and pre-industrial period in a dynamic global vegetation model. *Quaternary Science Reviews* **218**, 293-305.

*CHENG, S., FANG, H. & YU, G. (2018). Threshold responses of soil organic carbon concentration and composition to multi-level nitrogen addition in a temperate needle-broadleaved forest. *Biogeochemistry* **137**, 219-233.

*CHENG, X., HU, M., ZHOU, Y., WANG, F., LIU, L., WANG, Y., HUANG, H. & ZHANG, J. (2022). The divergence of micrometeorology sensitivity leads to changes in GPP/SIF between cork oak and poplar. *Agricultural and Forest Meteorology* **326**, 109189.

*CHRISTIANEN, M. J., SMULDERS, F. O., VONK, J. A., BECKING, L. E., BOUMA, T. J., ENGEL, S. M., JAMES, R. K., NAVA, M. I., DE SMIT, J. C. & VAN DER ZEE, J. P. (2023). Seagrass ecosystem multifunctionality under the rise of a flagship marine megaherbivore. *Global Change Biology* **29**, 215-230.

*CODEÇO, C. T., LELE, S., PASCUAL, M., BOUMA, M. & KO, A. I. (2008). A stochastic model for ecological systems with strong nonlinear response to environmental drivers: application to two water-borne diseases. *Journal of the Royal Society Interface* **5**, 247-252.

*COLDEN, A. M., LATOUR, R. J. & LIPCIUS, R. N. (2017). Reef height drives threshold dynamics of restored oyster reefs. *Marine Ecology Progress Series* **582**, 1-13.

*COLLIE, J. S., RICHARDSON, K. & STEELE, J. H. (2004). Regime shifts: can ecological theory illuminate the mechanisms? *Progress in oceanography* **60**(2-4), 281-302.

*COMER-WARNER, S. A., ROMEIJN, P., GOODDY, D. C., ULLAH, S., KETTRIDGE, N., MARCHANT, B., HANNAH, D. M. & KRAUSE, S. (2018). Thermal sensitivity of CO2 and CH4 emissions varies with streambed sediment properties. *Nature Communications* **9**, 2803.

*CONDO, T. K. & REINHARDT, K. (2019). Large variation in branch and branch-tip hydraulic functional traits in Douglas-fir (Pseudotsuga menziesii) approaching lower treeline. *Tree Physiology* **39**, 1461-1472.

*CONTE, A., OTU-LARBI, F., ALIVERNINI, A., HOSHIKA, Y., PAOLETTI, E., ASHWORTH, K. & FARES, S. (2021). Exploring new strategies for ozone-risk assessment: A dynamic-threshold case study. *Environmental Pollution* **287**, 117620.

*COOKE, B. J. & CARROLL, A. L. (2017). Predicting the risk of mountain pine beetle spread to eastern pine forests: Considering uncertainty in uncertain times. *Forest Ecology and Management* **396**, 11-25.

*CORDELL, J. R., TEAR, L. M. & BOLLENS, S. M. (2010). Modelling physico-chemical factors affecting occurrences of a non-indigenous planktonic copepod in northeast Pacific estuaries. *Biological Invasions* **12**, 1427-1445.

*D'AMARIO, S. C., REARICK, D. C., FASCHING, C., KEMBEL, S. W., PORTER-GOFF, E., SPOONER, D. E., WILLIAMS, C. J., WILSON, H. F. & XENOPOULOS, M. A. (2019). The prevalence of nonlinearity and detection of ecological breakpoints across a land use gradient in streams. *Scientific reports* **9**, 3878.

*D'AMARIO, S. C., WILSON, H. F. & XENOPOULOS, M. A. (2021). Concentration-discharge relationships derived from a larger regional dataset as a tool for watershed management. *Ecological Applications* **31**, e02447.

*DAVIS, J., SIM, L. & CHAMBERS, J. (2010). Multiple stressors and regime shifts in shallow aquatic ecosystems in antipodean landscapes. *Freshwater Biology* **55**, 5-18.

*DAW, T. M., HICKS, C. C., BROWN, K., CHAIGNEAU, T., JANUCHOWSKI-HARTLEY, F. A., CHEUNG, W. W., ROSENDO, S., CRONA, B., COULTHARD, S. & SANDBROOK, C. (2016). Elasticity in ecosystem services: exploring the variable relationship between ecosystems and human well-being. *Ecology and Society* **21**.

*DEBOER, J. A., THOMS, M. C., CASPER, A. F. & DELONG, M. D. (2019). The response of fish diversity in a highly modified large river system to multiple anthropogenic stressors. *Journal of Geophysical Research: Biogeosciences* **124**, 384-404.

*DEE, L. E., OKAMTOTO, D., GÅRDMARK, A., MONTOYA, J. M. & MILLER, S. J. (2020). Temperature variability alters the stability and thresholds for collapse of interacting species. *Philosophical Transactions of the Royal Society B* **375**(1814), 20190457.

*DENT, C. L., CUMMING, G. S. & CARPENTER, S. R. (2002). Multiple states in river and lake ecosystems. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* **357**(1421), 635-645.

*DIEHL, R. M., WILCOX, A. C., MERRITT, D. M., PERKINS, D. W. & SCOTT, J. A. (2018). Development of an eco-geomorphic modeling framework to evaluate riparian ecosystem response to flow-regime changes. *Ecological engineering* **123**, 112-126.

*DIGIOVINAZZO, P., FICETOLA, G. F., BOTTONI, L., ANDREIS, C. & PADOA-SCHIOPPA, E. (2010). Ecological thresholds in herb communities for the management of suburban fragmented forests. *Forest Ecology and Management* **259**, 343-349.

*DIMITRIOU, P. D., SANTI, I., MORAITIS, M. L., TSIKOPOULOU, I., PITTA, P. & KARAKASSIS, I. (2022). Benthic–Pelagic Coupling in the Oligotrophic Eastern Mediterranean: A Synthesis of the HYPOXIA Project Results. *Frontiers in Marine Science* **9**, 886335.

*DÖBERT, T. F., WEBBER, B. L., SUGAU, J. B., DICKINSON, K. J. & DIDHAM, R. K. (2017). Logging increases the functional and phylogenetic dispersion of understorey plant communities in tropical lowland rain forest. *Journal of Ecology* **105**, 1235-1245.

*DODD, N. L., SCHWEINSBURG, R. E. & BOE, S. (2006). Landscape-scale forest habitat relationships to tassel-eared squirrel populations: implications for ponderosa pine forest restoration. *Restoration Ecology* **14**, 537-547.

*DODDS, W. K., CLEMENTS, W. H., GIDO, K., HILDERBRAND, R. H. & KING, R. S. (2010). Thresholds, breakpoints, and nonlinearity in freshwaters as related to management. *Journal of the North American Benthological Society* **29**, 988-997.

*DOHERTY, K. E., EVANS, J. S., COATES, P. S., JULIUSSON, L. M. & FEDY, B. C. (2016). Importance of regional variation in conservation planning: a rangewide example of the greater sage-grouse. *Ecosphere* **7**, e01462.

*DONG, Q., MCCORMICK, P. V., SKLAR, F. H. & DEANGELIS, D. L. (2002). Structural instability, multiple stable states, and hysteresis in periphyton driven by phosphorus enrichment in the Everglades. *Theoretical Population Biology* **61**, 1-13.

*DROBYSHEV, I., SIMARD, M., BERGERON, Y. & HOFGAARD, A. (2010). Does soil organic layer thickness affect climate–growth relationships in the black spruce boreal ecosystem? *Ecosystems* **13**, 556-574.

*DUAN, W., MADASI, J. D. & KHURSHID, A. (2023). How network structure and exchange rate volatility drive the industrial ecosystem towards collapse: a global perspective. *Economic research-Ekonomska istraživanja* **36**, 1688-1707.

*DUARTE, C. M. (2009). Coastal eutrophication research: a new awareness. *Hydrobiologia* **629**, 263-269.

*DUCHARDT, C. J., AUGUSTINE, D. J. & BECK, J. L. (2019). Threshold responses of grassland and sagebrush birds to patterns of disturbance created by an ecosystem engineer. *Landscape Ecology* **34**, 895-909.

*DUNIC, J. C. & CÔTÉ, I. M. (2023). Management thresholds shift under the influence of multiple stressors: Eelgrass meadows as a case study. *Conservation Letters* **16**, e12938.

*DURANT, J. M., ONO, K. & LANGANGEN, Ø. (2022). Empirical evidence of nonlinearity in bottom up effect in a marine predator–prey system. *Biology Letters* **18**, 20220309.

*EKROOS, J., HELIÖLÄ, J. & KUUSSAARI, M. (2010). Homogenization of lepidopteran communities in intensively cultivated agricultural landscapes. *Journal of Applied Ecology* **47**, 459-467.

*ELGERSMA, K. J. & EHRENFELD, J. G. (2011). Linear and non-linear impacts of a non-native plant invasion on soil microbial community structure and function. *Biological Invasions* **13**, 757-768.

*EL-ZEIN, A. & TONMOY, F. N. (2017). Nonlinearity, fuzziness and incommensurability in indicator-based assessments of vulnerability to climate change: A new mathematical framework. *Ecological indicators* **82**, 82-93.

*ERVIN, D. E., WELSH, R., BATIE, S. S. & CARPENTIER, C. L. (2003). Towards an ecological systems approach in public research for environmental regulation of transgenic crops. *Agriculture, ecosystems & environment* **99**(1-3), 1-14.

*ESEN, Ö., YILDIRIM, D. Ç. & YILDIRIM, S. (2021). Pollute less or tax more? Asymmetries in the EU environmental taxes–Ecological balance nexus. *Environmental impact assessment review* **91**, 106662.

*ESSELMAN, P. C., STEVENSON, R. J., LUPI, F., RISENG, C. M. & WILEY, M. J. (2015). Landscape prediction and mapping of game fish biomass, an ecosystem service of Michigan rivers. *North American Journal of Fisheries Management* **35**, 302-320.

*ETZOLD, S., FERRETTI, M., REINDS, G. J., SOLBERG, S., GESSLER, A., WALDNER, P., SCHAUB, M., SIMPSON, D., BENHAM, S. & HANSEN, K. (2020). Nitrogen deposition is the most important environmental driver of growth of pure, even-aged and managed European forests. *Forest Ecology and Management* **458**, 117762.

*FALK-PETERSEN, J. & ARMSTRONG, C. W. (2013). To have one's cake and eat it too managing the alien invasive red king crab. *Marine Resource Economics* **28**, 65-81.

*FANG, Y. (2012). Investment threshold and management reflection for industrial system cleaning: a case for China. *Environmental Science and Pollution Research* **19**, 666-676.

*FEIJÓ-LIMA, R., MCLEAY, S. M., SILVA-JUNIOR, E. F., TROMBONI, F., MOULTON, T. P., ZANDONÀ, E. & THOMAS, S. A. (2018). Quantitatively describing the downstream effects of an abrupt land cover transition: buffering effects of a forest remnant on a stream impacted by cattle grazing. *Inland Waters* **8**, 294-311.

*FEKI-SAHNOUN, W., NJAH, H., HAMZA, A., BARRAJ, N., MAHFOUDI, M., REBAI, A. & HASSEN, M. B. (2018). Using general linear model, Bayesian Networks and Naive Bayes classifier for prediction of Karenia selliformis occurrences and blooms. *Ecological Informatics* **43**, 12-23.

*FITZPATRICK, M. C. & KELLER, S. R. (2015). Ecological genomics meets community-level modelling of biodiversity: Mapping the genomic landscape of current and future environmental adaptation. *Ecology letters* **18**, 1-16.

*FLETCHER, C. S. & HILBERT, D. W. (2007). Resilience in landscape exploitation systems. *Ecological Modelling* **201**(3-4), 440-452.

*FOLEY, M. M., MARTONE, R. G., FOX, M. D., KAPPEL, C. V., MEASE, L. A., ERICKSON, A. L., HALPERN, B. S., SELKOE, K. A., TAYLOR, P. & SCARBOROUGH, C. (2015). Using ecological thresholds to inform resource management: current options and future possibilities. *Frontiers in Marine Science* **2**, 95.

*FOSTEN, J., MORLEY, B. & TAYLOR, T. (2012). Dynamic misspecification in the environmental Kuznets curve: evidence from CO2 and SO2 emissions in the United Kingdom. *Ecological Economics* **76**, 25-33.

*FOX, J. T. & MAGOULICK, D. D. (2022). Hydrologic and environmental thresholds in stream fish assemblage structure across flow regimes. *Ecological indicators* **144**, 109500.

*FRAIR, J. L., MERRILL, E. H., BEYER, H. L. & MORALES, J. M. (2008). Thresholds in landscape connectivity and mortality risks in response to growing road networks. *Journal of Applied Ecology* **45**, 1504-1513.

*FU, C., XU, Y., GRÜSS, A., BUNDY, A., SHANNON, L., HEYMANS, J. J., HALOUANI, G., AKOGLU, E., LYNAM, C. P. & COLL, M. (2020). Responses of ecological indicators to fishing pressure under environmental change: exploring non-linearity and thresholds. *ICES Journal of Marine Science* **77**, 1516-1531.

*FUCHSMAN, P. C., SFERRA, J. C. & BARBER, T. R. (2000). Three lines of evidence in a sediment toxicity evaluation for hexachlorobutadiene. *Environmental Toxicology and Chemistry: An International Journal* **19**, 2328-2337.

*FUNG, T., SEYMOUR, R. M. & JOHNSON, C. R. (2013). Warning signals of regime shifts as intrinsic properties of endogenous dynamics. *The American Naturalist* **182**, 208-222.

*GABLER, C. A., OSLAND, M. J., GRACE, J. B., STAGG, C. L., DAY, R. H., HARTLEY, S. B., ENWRIGHT, N. M., FROM, A. S., MCCOY, M. L. & MCLEOD, J. L. (2017). Macroclimatic change expected to transform coastal wetland ecosystems this century. *Nature Climate Change* **7**, 142-147.

*GALIC, N., HAWKINS, T. & FORBES, V. E. (2019). Adverse impacts of hypoxia on aquatic invertebrates: A meta-analysis. *Science of the Total Environment* **652**, 736-743.

*GAUDRY, W., GAILLARD, J.-M., SAÏD, S., MÅRELL, A., BALTZINGER, C., ROCQUENCOURT, A. & BONENFANT, C. (2022). Population density and plant availability interplay to shape browsing intensity by roe deer in a deciduous forest. *Forest Ecology and Management* **515**, 120153.

*GÉNIN, A., MAJUMDER, S., SANKARAN, S., DANET, A., GUTTAL, V., SCHNEIDER, F. D. & KÉFI, S. (2018). Monitoring ecosystem degradation using spatial data and the R package spatialwarnings. *Methods in Ecology and Evolution* **9**, 2067-2075.

*GERA, A., PAGES, J. F., ARTHUR, R., FARINA, S., ROCA, G., ROMERO, J. & ALCOVERRO, T. (2014). The effect of a centenary storm on the long-lived seagrass Posidonia oceanica. *Limnology and Oceanography* **59**, 1910-1918.

*GERBER, S., JOOS, F. & PRENTICE, I. C. (2004). Sensitivity of a dynamic global vegetation model to climate and atmospheric CO2. *Global Change Biology* **10**, 1223-1239.

*GÖKÇE, A., YAZAR, S. & SEKERCI, Y. (2020). Delay induced nonlinear dynamics of oxygen-plankton interactions. *Chaos, Solitons & Fractals* **141**, 110327.

*GOLDIN, S. R. & HUTCHINSON, M. F. (2015). Thermal refugia in cleared temperate Australian woodlands: coarse woody debris moderate extreme surface soil temperatures. *Agricultural and Forest Meteorology* **214**, 39-47.

*GOMEZ-CASANOVAS, N., HUDIBURG, T. W., BERNACCHI, C. J., PARTON, W. J. & DELUCIA, E. H. (2016). Nitrogen deposition and greenhouse gas emissions from grasslands: uncertainties and future directions. *Global Change Biology* **22**, 1348-1360.

*GOODEN, B., FRENCH, K., TURNER, P. J. & DOWNEY, P. O. (2009). Impact threshold for an alien plant invader, Lantana camara L., on native plant communities. *Biological Conservation* **142**, 2631-2641.

*GORDON, L. J., PETERSON, G. D. & BENNETT, E. M. (2008). Agricultural modifications of hydrological flows create ecological surprises. *Trends in Ecology & Evolution* **23**, 211-219.

*GOVE, J. M., WILLIAMS, G. J., MCMANUS, M. A., CLARK, S. J., EHSES, J. S. & WEDDING, L. M. (2015). Coral reef benthic regimes exhibit non-linear threshold responses to natural physical drivers. *Marine Ecology Progress Series* **522**, 33-48.

*GRENFELL, B. T., BJØRNSTAD, O. N. & FINKENSTÄDT, B. F. (2002). Dynamics of measles epidemics: scaling noise, determinism, and predictability with the TSIR model. *Ecological monographs* **72**, 185-202.

*GRIFFEN, B. D., BELGRAD, B. A., CANNIZZO, Z. J., KNOTTS, E. R. & HANCOCK, E. R. (2016). Rethinking our approach to multiple stressor studies in marine environments. *Marine Ecology Progress Series* **543**, 273-281.

*GROFFMAN, P. M., BARON, J. S., BLETT, T., GOLD, A. J., GOODMAN, I., GUNDERSON, L. H., LEVINSON, B. M., PALMER, M. A., PAERL, H. W. & PETERSON, G. D. (2006). Ecological thresholds: the key to successful environmental management or an important concept with no practical application? *Ecosystems* **9**, 1-13.

*GU, H., WANG, J., MA, L., SHANG, Z. & ZHANG, Q. (2019). Insights into the BRT (Boosted Regression Trees) method in the study of the climate-growth relationship of Masson pine in subtropical China. *Forests* **10**, 228.

*GUILLEMETTE, M., GRÉGOIRE, F., BOUILLET, D., RAIL, J.-F., BOLDUC, F., CARON, A. & PELLETIER, D. (2018). Breeding failure of seabirds in relation to fish depletion: Is there one universal threshold of food abundance? *Marine Ecology Progress Series* **587**, 235-245.

*GUO, P., JIA, J., HAN, T., XIE, J., WU, P., DU, Y. & QU, K. (2017). Nonlinear responses of forest soil microbial communities and activities after short-and long-term gradient nitrogen additions. *Applied Soil Ecology* **121**, 60-64.

*GUO, Y., CAO, X., LIU, B. & PENG, K. (2020). El Niño index prediction using deep learning with ensemble empirical mode decomposition. *Symmetry* **12**, 893.

*HAAKONSSON, S., RODRÍGUEZ, M. A., CARBALLO, C., DEL CARMEN PEREZ, M., AROCENA, R. & BONILLA, S. (2020). Predicting cyanobacterial biovolume from water temperature and conductivity using a Bayesian compound Poisson-Gamma model. *Water Research* **176**, 115710.

*HAGERTHEY, S. E., NEWMAN, S., RUTCHEY, K., SMITH, E. P. & GODIN, J. (2008). Multiple regime shifts in a subtropical peatland: community-specific thresholds to eutrophication. *Ecological monographs* **78**, 547-565.

*HALE, J. D., FAIRBRASS, A. J., MATTHEWS, T. J. & SADLER, J. P. (2012). Habitat composition and connectivity predicts bat presence and activity at foraging sites in a large UK conurbation. *PloS one* **7**, e33300.

*HAN, D., LI, T., FENG, S. & SHI, Z. (2020). Does renewable energy consumption successfully promote the green transformation of China's industry? *Energies* **13**, 229.

*HANSEN, A. J., KNIGHT, R. L., MARZLUFF, J. M., POWELL, S., BROWN, K., GUDE, P. H. & JONES, K. (2005). Effects of exurban development on biodiversity: patterns, mechanisms, and research needs. *Ecological Applications* **15**, 1893-1905.

*HAO, G., HU, Z., GUO, Q., DI, K. & LI, S. (2019). Median to strong rainfall intensity favors carbon sink in a temperate grassland ecosystem in China. *Sustainability* **11**, 6376.

*HAO, J., LI, C., YUAN, R., AHMED, M., KHAN, M. A. & OLÁH, J. (2020). The influence of the knowledge-based network structure hole on enterprise innovation performance: The threshold effect of R&D investment intensity. *Sustainability* **12**, 6155.

*HAO, Y., HUANG, J., GUO, Y., WU, H. & REN, S. (2022). Does the legacy of state planning put pressure on ecological efficiency? Evidence from China. *Business strategy and the environment* **31**, 3100-3121.

*HARLEY, C. D., CONNELL, S. D., DOUBLEDAY, Z. A., KELAHER, B., RUSSELL, B. D., SARÀ, G. & HELMUTH, B. (2017). Conceptualizing ecosystem tipping points within a physiological framework. *Ecology and Evolution* **7**, 6035-6045.

*HART, S. J. & LAROQUE, C. P. (2013). Searching for thresholds in climate–radial growth relationships of Engelmann spruce and subalpine fir, Jasper National Park, Alberta, Canada. *Dendrochronologia* **31**, 9-15.

*HAYAJNEH, W. A., AL-AZZAM, S., YUSEF, D., LATTYAK, W. J., LATTYAK, E. A., GOULD, I., LÓPEZ-LOZANO, J.-M., CONWAY, B. R., CONLON-BINGHAM, G. & ALDEYAB, M. A. (2021). Identification of thresholds in relationships between specific antibiotic use and carbapenem-resistant Acinetobacter baumannii (CRAb) incidence rates in hospitalized patients in Jordan. *Journal of Antimicrobial Chemotherapy* **76**, 524-530.

*HE, Q., SILLIMAN, B. R. & CUI, B. (2017). Incorporating thresholds into understanding salinity tolerance: A study using salt-tolerant plants in salt marshes. *Ecology and Evolution* **7**, 6326-6333.

*HE, W., XIE, X., MEI, Y., WAN, S. & ZHAO, S. (2021). Decreasing predictability as a precursor indicator for abrupt climate change. *Climate Dynamics* **56**, 3899-3908.

*HENDERSON, K. A., BAUCH, C. T. & ANAND, M. (2016). Alternative stable states and the sustainability of forests, grasslands, and agriculture. *Proceedings of the National Academy of Sciences* **113**, 14552-14559.

*HERBERT, E. R., BOON, P., BURGIN, A. J., NEUBAUER, S. C., FRANKLIN, R. B., ARDÓN, M., HOPFENSPERGER, K. N., LAMERS, L. P. & GELL, P. (2015). A global perspective on wetland salinization: ecological consequences of a growing threat to freshwater wetlands. *Ecosphere* **6**, 1-43.

*HERNANDEZ, K., REECE, B. & MCINTYRE, N. (2006). Effects of anthropogenic land use on Odonata in playas of the Southern High Plains. *Western North American Naturalist* **66**, 273-278.

*HEWITT, J. E., ELLIS, J. I. & THRUSH, S. F. (2016). Multiple stressors, nonlinear effects and the implications of climate change impacts on marine coastal ecosystems. *Global Change Biology* **22**, 2665-2675.

*HILDERBRAND, R. H., UTZ, R. M., STRANKO, S. A. & RAESLY, R. L. (2010). Applying thresholds to forecast potential biodiversity loss from human development. *Journal of the North American Benthological Society* **29**, 1009-1016.

*HOLON, F., MARRE, G., PARRAVICINI, V., MOUQUET, N., BOCKEL, T., DESCAMP, P., TRIBOT, A.-S., BOISSERY, P. & DETER, J. (2018). A predictive model based on multiple coastal anthropogenic pressures explains the degradation status of a marine ecosystem: Implications for management and conservation. *Biological Conservation* **222**, 125-135.

*HU, S., ZENG, G., CAO, X., YUAN, H. & CHEN, B. (2021). Does technological innovation promote green development? A case study of the Yangtze River Economic Belt in China. *International Journal of Environmental Research and Public Health* **18**, 6111.

*HUA, L., SHAO, G. & ZHAO, J. (2017). A concise review of ecological risk assessment for urban ecosystem application associated with rapid urbanization processes. *International Journal of Sustainable Development & World Ecology* **24**, 248-261.

*HUANG, F., ZHANG, D. & CHEN, X. (2019). Vegetation response to groundwater variation in arid environments: Visualization of research evolution, synthesis of response types, and estimation of groundwater threshold. *International Journal of Environmental Research and Public Health* **16**, 1849.

*HUANG, Q. & LIU, M. (2022). Trade openness and green total factor productivity: Testing the role of environment regulation based on dynamic panel threshold model. *Environment, Development and Sustainability* **24**, 9304-9329.

*HUNSICKER, M. E., KAPPEL, C. V., SELKOE, K. A., HALPERN, B. S., SCARBOROUGH, C., MEASE, L. & AMRHEIN, A. (2016). Characterizing driver–response relationships in marine pelagic ecosystems for improved ocean management. *Ecological Applications* **26**, 651-663.

*HUNTINGFORD, C. (2014). Complexity and determining dangerous levels of climate impacts. *Environmental Research Letters* **9**, 011001.

*IKEMOTO, T. & KIRITANI, K. (2019). Novel method of specifying low and high threshold temperatures using thermodynamic SSI model of insect development. *Environmental Entomology* **48**, 479-488.

*INGRISCH, J., UMLAUF, N. & BAHN, M. (2023). Functional thresholds alter the relationship of plant resistance and recovery to drought. *Ecology* **104**, e3907.

*ITTER, M. S., FINLEY, A. O., D'AMATO, A. W., FOSTER, J. R. & BRADFORD, J. B. (2017). Variable effects of climate on forest growth in relation to climate extremes, disturbance, and forest dynamics. *Ecological Applications* **27**, 1082-1095.

*IVANY, L. C. (1996). Coordinated stasis or coordinated turnover? Exploring intrinsic vs. extrinsic controls on pattern. *Palaeogeography, Palaeoclimatology, Palaeoecology* **127**(1-4), 239-256.

*JACKSON, B. K. & SULLIVAN, S. M. P. (2018). Ecosystem size and flooding drive trophic dynamics of riparian spiders in a fire-prone Sierra Nevada river system. *Canadian Journal of Fisheries and Aquatic Sciences* **75**, 308-318.

*JACKSON, M. C., RUIZ-NAVARRO, A. & BRITTON, J. R. (2015). Population density modifies the ecological impacts of invasive species. *Oikos* **124**, 880-887.

*JAGER, H. I., KING, A. W., GANGRADE, S., HAINES, A., DEROLPH, C., NAZ, B. S. & ASHFAQ, M. (2018). Will future climate change increase the risk of violating minimum flow and maximum temperature thresholds below dams in the Pacific Northwest? *Climate risk management* **21**, 69-84.

*JANSSEN, A. B., DE JAGER, V. C., JANSE, J. H., KONG, X., LIU, S., YE, Q. & MOOIJ, W. M. (2017). Spatial identification of critical nutrient loads of large shallow lakes: Implications for Lake Taihu (China). *Water Research* **119**, 276-287.

*JARVIE, H. P., SHARPLEY, A. N., WITHERS, P. J., SCOTT, J. T., HAGGARD, B. E. & NEAL, C. (2013). Phosphorus mitigation to control river eutrophication: murky waters, inconvenient truths, and "postnormal" science. *Journal of environmental quality* **42**, 295-304.

*JASSEY, V. E., RECZUGA, M. K., ZIELIŃSKA, M., SŁOWIŃSKA, S., ROBROEK, B. J., MARIOTTE, P., SEPPEY, C. V., LARA, E., BARABACH, J. & SŁOWIŃSKI, M. (2018). Tipping point in plant–fungal interactions under severe drought causes abrupt rise in peatland ecosystem respiration. *Global Change Biology* **24**, 972-986.

*JATAV, K. S. & DHAR, J. (2014). Hybrid approach for pest control with impulsive releasing of natural enemies and chemical pesticides: A plant–pest–natural enemy model. *Nonlinear Analysis: Hybrid Systems* **12**, 79-92.

*JENERETTE, G. D., BARRON-GAFFORD, G. A., GUSWA, A. J., MCDONNELL, J. J. & VILLEGAS, J. C. (2012). Organization of complexity in water limited ecohydrology. *Ecohydrology* **5**, 184-199.

*JI, F., SUN, Y. & YANG, Q. (2023). Early warning of red tides using bacterial and eukaryotic communities in nearshore waters. *Environmental Research* **216**, 114711.

*JIANG, C., WANG, X., ZHANG, H., LABZOVSKII, L., WANG, J. & LIU, T. (2019). Re-orienting ecological restoration in degraded drylands for a more sustainable soil–water relationship: Non-linear boundary of limited water resources in combating soil loss. *Journal of Arid Environments* **167**, 87-100.

*JIANYING, X., JIXING, C. & YANXU, L. (2020). Partitioned responses of ecosystem services and their tradeoffs to human activities in the Belt and Road region. *Journal of Cleaner Production* **276**, 123205.

*JIN, L., ZHOU, T., FANG, S., ZHOU, X., HAN, B. & BAI, Y. (2022). The short-term effects of air pollutants on pneumonia hospital admissions in Lanzhou, China, 2014–2019:

evidence of ecological time-series study. *Air Quality, Atmosphere & Health* **15**, 2199-2213.

*JIN, Y., LI, J., LIU, C., LIU, Y., ZHANG, Y., SONG, Q., SHA, L., BALASUBRAMANIAN, D., CHEN, A. & YANG, D. (2019). Precipitation reduction alters herbaceous community structure and composition in a savanna. *Journal of Vegetation Science* **30**, 821-831.

*JOHNSON, C. J. (2013). Identifying ecological thresholds for regulating human activity: Effective conservation or wishful thinking? *Biological Conservation* **168**, 57-65.

*JONES, R. T., MARSHALL, J. D., CROWLEY, S. F., BEDFORD, A., RICHARDSON, N., BLOEMENDAL, J. & OLDFIELD, F. (2002). A high resolution, multiproxy Late-glacial record of climate change and intrasystem responses in northwest England. *Journal of Quaternary Science: Published for the Quaternary Research Association* **17**, 329-340.

*JOSEFSON, A. B., BLOMQVIST, M., HANSEN, J. L., ROSENBERG, R. & RYGG, B. (2009). Assessment of marine benthic quality change in gradients of disturbance: comparison of different Scandinavian multi-metric indices. *Marine Pollution Bulletin* **58**, 1263-1277.

*KAEMINGK, M. A., CHIZINSKI, C. J., ALLEN, C. R. & POPE, K. L. (2019). Ecosystem size predicts social-ecological dynamics. *Ecology and Society* **24**.

*KARR, K. A., FUJITA, R., HALPERN, B. S., KAPPEL, C. V., CROWDER, L., SELKOE, K. A., ALCOLADO, P. M. & RADER, D. (2015). Thresholds in Caribbean coral reefs: implications for ecosystem-based fishery management. *Journal of Applied Ecology*, 402-412.

*KATH, J., REARDON-SMITH, K., LE BROCQUE, A., DYER, F., DAFNY, E., FRITZ, L. & BATTERHAM, M. (2014). Groundwater decline and tree change in floodplain landscapes: Identifying non-linear threshold responses in canopy condition. *Global Ecology and Conservation* **2**, 148-160.

*KATO, S. & AHERN, J. (2011). The concept of threshold and its potential application to landscape planning. *Landscape and Ecological Engineering* **7**, 275-282.

*KEITH, D. A., BENSON, D. H., BAIRD, I. R., WATTS, L., SIMPSON, C. C., KROGH, M., GORISSEN, S., FERRER-PARIS, J. R. & MASON, T. J. (2023). Effects of interactions between anthropogenic stressors and recurring perturbations on ecosystem resilience and collapse. *Conservation Biology* **37**, e13995.

*KHASKHELI, A., JIANG, Y., RAZA, S. A., KHAN, K. A. & QURESHI, M. A. (2021). Financial development, international trade, and environmental degradation: a nonlinear threshold model based on panel smooth transition regression. *Environmental Science and Pollution Research* **28**, 26449-26460.

*KIDWELL, D. M., DIETRICH, J. C., HAGEN, S. C. & MEDEIROS, S. C. (2017). An Earth's Future Special Collection: Impacts of the coastal dynamics of sea level rise on low-gradient coastal landscapes. *Earth's Future* **5**, 2-9.

*KING, R. S. & RICHARDSON, C. J. (2007). Subsidy–stress response of macroinvertebrate community biomass to a phosphorus gradient in an oligotrophic wetland ecosystem. *Journal of the North American Benthological Society* **26**, 491-508.

*KITTEL, T. G., BAKER, B. B., HIGGINS, J. V. & HANEY, J. C. (2011). Climate vulnerability of ecosystems and landscapes on Alaska's North Slope. *Regional Environmental Change* **11**, 249-264.

*KLEMMER, A. J., WISSINGER, S. A., GREIG, H. S. & OSTROFSKY, M. L. (2012). Nonlinear effects of consumer density on multiple ecosystem processes. *Journal of Animal Ecology* **81**, 770-780.

*KLEPPER, O. & BEDAUX, J. J. (1997). Nonlinear parameter estimation for toxicological threshold models. *Ecological Modelling* **102**(2-3), 315-324.

*KLINSHOV, V. V., KIRILLOV, S., KURTHS, J. & NEKORKIN, V. I. (2018). Interval stability for complex systems. *New Journal of Physics* **20**, 043040.

*KLOSE, A. K., KARLE, V., WINKELMANN, R. & DONGES, J. F. (2020). Emergence of cascading dynamics in interacting tipping elements of ecology and climate. *Royal Society Open Science* **7**, 200599.

*KLUEVER, B. M., SMITH, T. N. & GESE, E. M. (2019). Group effects of a non-native plant invasion on rodent abundance. *Ecosphere* **10**, e02544.

*KOVALENKO, K. E., REAVIE, E. D., ALLAN, J. D., CAI, M., SMITH, S. D. & JOHNSON, L. B. (2017). Pelagic phytoplankton community change-points across nutrient gradients and in response to invasive mussels. *Freshwater Biology* **62**, 366-381.

*KOVALENKO, K. E., REAVIE, E. D., BARBIERO, R. P., BURLAKOVA, L. E., KARATAYEV, A. Y., RUDSTAM, L. G. & WATKINS, J. M. (2018). Patterns of long-term dynamics of aquatic communities and water quality parameters in the Great Lakes: Are they synchronized? *Journal of Great Lakes Research* **44**, 660-669.

*KOVALENKO, K., BRADY, V., BROWN, T., CIBOROWSKI, J. J., DANZ, N., GATHMAN, J., HOST, G., HOWE, R., JOHNSON, L. & NIEMI, G. (2014). Congruence of community thresholds in response to anthropogenic stress in Great Lakes coastal wetlands. *Freshwater Science* **33**, 958-971.

*KROEKER, K. J., KORDAS, R. L. & HARLEY, C. D. (2017). Embracing interactions in ocean acidification research: confronting multiple stressor scenarios and context dependence. *Biology Letters* **13**, 20160802.

*LADINO, L. M., SABOGAL, E. I. & VALVERDE, J. C. (2015). General functional response and recruitment in a predator–prey system with capture on both species. *Mathematical Methods in the Applied Sciences* **38**, 2876-2887.

*LANGER, M., WESTERMANN, S., WALTER ANTHONY, K., WISCHNEWSKI, K. & BOIKE, J. (2015). Frozen ponds: production and storage of methane during the Arctic winter in a lowland tundra landscape in northern Siberia, Lena River delta. *Biogeosciences* **12**, 977-990.

*LANGFORD, T., SHAW, P., FERGUSON, A. & HOWARD, S. (2009). Long-term recovery of macroinvertebrate biota in grossly polluted streams: re-colonisation as a constraint to ecological quality. *Ecological indicators* **9**, 1064-1077.

*LARGE, S. I., FAY, G., FRIEDLAND, K. D. & LINK, J. S. (2013). Defining trends and thresholds in responses of ecological indicators to fishing and environmental pressures. *ICES Journal of Marine Science* **70**, 755-767.

*LARGE, S. I., FAY, G., FRIEDLAND, K. D. & LINK, J. S. (2015). Quantifying patterns of change in marine ecosystem response to multiple pressures. *PloS one* **10**, e0119922.

*LARSEN, S. & ALP, M. (2015). Ecological thresholds and riparian wetlands: an overview for environmental managers. *Limnology* **16**, 1-9.

*LASPOUMADERES, C., MEUNIER, C. L., MAGNIN, A., BERLINGHOF, J., ELSER, J. J., BALSEIRO, E., TORRES, G., MODENUTTI, B., TREMBLAY, N. & BOERSMA, M. (2022). A common temperature dependence of nutritional demands in ectotherms. *Ecology letters* **25**, 2189-2202.

*LATIF, M. & KEENLYSIDE, N. S. (2009). El Niño/Southern Oscillation response to global warming. *Proceedings of the National Academy of Sciences* **106**, 20578-20583.

*LATTY, T. & DAKOS, V. (2019). The risk of threshold responses, tipping points, and cascading failures in pollination systems. *Biodiversity and Conservation* **28**, 3389-3406.

*LEHNHERR, I. (2014). Methylmercury biogeochemistry: a review with special reference to Arctic aquatic ecosystems. *Environmental Reviews* **22**, 229-243.

*LEI, Y., WANG, Y., QIN, F., LIU, J., FENG, P., LUO, L., JORDAN, R. W. & JIANG, S. (2021). Diatom assemblage shift driven by nutrient dynamics in a large, subtropical reservoir in southern China. *Journal of Cleaner Production* **317**, 128435.

*LENTON, T. M. (2011). Early warning of climate tipping points. *Nature Climate Change* **1**, 201-209.

*LEONARD, R., MOORE, P., KRAUSE, S., DEVITO, K., PETRONE, G., MENDOZA, C., WADDINGTON, J. & KETTRIDGE, N. (2021). The influence of system heterogeneity on peat-surface temperature dynamics. *Environmental Research Letters* **16**, 024002.

*LI YANGFAN, L. Y., LI YI, L. Y. & WU WEI, W. W. (2015). Threshold and resilience management of coupled urbanization and water environmental system in the rapidly changing coastal region.

*LI, D., CAO, W., DOU, Y., WU, S., LIU, J. & LI, S. (2022a). Non-linear effects of natural and anthropogenic drivers on ecosystem services: Integrating thresholds into conservation planning. *Journal of Environmental Management* **321**, 116047.

*LI, F. & ZHAO, W. (2017). Hydrologic thresholds and changes in ANPP of artificial sand-fixing vegetation in a desert-oasis ecotone in Northwest China. *Journal of Arid Environments* **146**, 44-52.

*LI, F., ZHAO, W. & LIU, H. (2015). Productivity responses of desert vegetation to precipitation patterns across a rainfall gradient. *Journal of Plant Research* **128**, 283-294.

*LI, J. & LIN, Q. (2023). Threshold effects of green technology application on sustainable grain production: Evidence from China. *Frontiers in plant science* **14**, 1107970.

*LI, J., HAN, G., WANG, G., LIU, X., ZHANG, Q., CHEN, Y., SONG, W., QU, W., CHU, X. & LI, P. (2022b). Imbalanced nitrogen–phosphorus input alters soil organic carbon storage and mineralisation in a salt marsh. *Catena* **208**, 105720.

*LI, K., COE, M., RAMANKUTTY, N. & DE JONG, R. (2007). Modeling the hydrological impact of land-use change in West Africa. *Journal of Hydrology* **337**(3-4), 258-268.

*LI, L., KANG, X., BIEDERMAN, J. A., WANG, W., QIAN, R., ZHENG, Z., ZHANG, B., RAN, Q., XU, C. & LIU, W. (2021a). Nonlinear carbon cycling responses to precipitation variability in a semiarid grassland. *Science of the Total Environment* **781**, 147062.

*LI, P. & GUO, T. (2024). Is digital finance a powerful means for Chinese cities to reduce environmental pollution in the fourth industrial revolution? *Technology Analysis & Strategic Management* **36**, 2332-2351.

*LI, R., GUO, J. & WANG, Q. (2022c). Impact of foreign aid on the ecological sustainability of sub-Saharan African countries. *Environmental impact assessment review* **95**, 106779.

*LI, R., WANG, X. & WANG, Q. (2022d). Does renewable energy reduce ecological footprint at the expense of economic growth? An empirical analysis of 120 countries. *Journal of Cleaner Production* **346**, 131207.

*LI, R., YU, D., ZHANG, Y., HAN, J., ZHANG, W., YANG, Q., GESSLER, A., LI, M.-H., XU, M. & GUAN, X. (2022e). Investment of needle nitrogen to photosynthesis controls the nonlinear productivity response of young Chinese fir trees to nitrogen deposition. *Science of the Total Environment* **840**, 156537.

*LI, T., SHI, Z. & HAN, D. (2021b). Does renewable energy consumption contribute to the development of low-carbon economy? Evidence from China. *Environmental Science and Pollution Research* **28**, 54891-54908.

*LI, Z., PENG, L. & WU, F. (2021c). The impacts of impervious surface on water quality in the urban agglomerations of middle and lower reaches of the Yangtze River economic belt from remotely sensed data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **14**, 8398-8406.

*LIAN, X., JIAO, L. & LIU, Z. (2023). Saturation response of enhanced vegetation productivity attributes to intricate interactions. *Global Change Biology* **29**, 1080-1095.

*LIAO, C. M. & CHOU, B. Y. H. (2005). Predictive risk thresholds for survival protection of farmed abalone, Haliotis diversicolor supertexta, exposed to waterborne zinc. *Environmental Toxicology: An International Journal* **20**, 202-211.

*LILJEDAHL, A., HINZMAN, L., HARAZONO, Y., ZONA, D., TWEEDIE, C., HOLLISTER, R. D., ENGSTROM, R. & OECHEL, W. (2011). Nonlinear controls on evapotranspiration in arctic coastal wetlands. *Biogeosciences* **8**, 3375-3389.

*LIMA, M. M. & MARIANO-NETO, E. (2014). Extinction thresholds for Sapotaceae due to forest cover in Atlantic Forest landscapes. *Forest Ecology and Management* **312**, 260-270.

*LIN, Z., WU, T., XIAO, Y., RAO, E., SHI, X. & OUYANG, Z. (2022). Protecting biodiversity to support ecosystem services: An analysis of trade-offs and synergies in southwestern China. *Journal of Applied Ecology* **59**, 2440-2451.

*LINDEGREN, M., CHECKLEY JR, D. M., KOSLOW, J. A., GOERICKE, R. & OHMAN, M. D. (2018). Climate-mediated changes in marine ecosystem regulation during El Niño. *Global Change Biology* **24**, 796-809.

*LIU, D., OGAYA, R., BARBETA, A., YANG, X. & PEÑUELAS, J. (2015). Contrasting impacts of continuous moderate drought and episodic severe droughts on the aboveground-biomass increment and litterfall of three coexisting M editerranean woody species. *Global Change Biology* **21**, 4196-4209.

*LIU, H., JIAO, F., YIN, J., LI, T., GONG, H., WANG, Z. & LIN, Z. (2020a). Nonlinear relationship of vegetation greening with nature and human factors and its forecast–a case study of Southwest China. *Ecological indicators* **111**, 106009.

*LIU, J., XU, J., ZHANG, X., LIANG, Z. & RAO, K. (2021). Nonlinearity and threshold effects of landscape pattern on water quality in a rapidly urbanized headwater watershed in China. *Ecological indicators* **124**, 107389.

*LIU, W., JIANG, L., YANG, S., WANG, Z., TIAN, R., PENG, Z., CHEN, Y., ZHANG, X., KUANG, J. & LING, N. (2020b). Critical transition of soil bacterial diversity and composition triggered by nitrogen enrichment. *Ecology* **101**, e03053.

*LIU, Z., WANG, Y., LI, Z. & PENG, J. (2013). Impervious surface impact on water quality in the process of rapid urbanization in Shenzhen, China. *Environmental earth sciences* **68**, 2365-2373.

*LIU, Z., ZHOU, Y., YANG, H. & LIU, Z. (2023). Urban green infrastructure affects bird biodiversity in the coastal megalopolis region of Shenzhen city. *Applied Geography* **151**, 102860.

*LLOYD, A. H., DUFFY, P. A. & MANN, D. H. (2013). Nonlinear responses of white spruce growth to climate variability in interior Alaska. *Canadian Journal of Forest Research* **43**(999), 331-343.

*LO CASCIO, M., MORILLAS, L., OCHOA-HUESO, R., MUNZI, S., ROALES, J., HASSELQUIST, N. J., MANRIQUE, E., SPANO, D., JAOUDÉ, R. A. & MEREU, S. (2017). Contrasting effects of nitrogen addition on soil respiration in two Mediterranean ecosystems. *Environmental Science and Pollution Research* **24**, 26160-26171.

*LOH, W. Y. (2014). Fifty years of classification and regression trees. *International Statistical Revie,* **82**, 329–348.

*LOHMANN, D., TIETJEN, B., BLAUM, N., JOUBERT, D. F. & JELTSCH, F. (2012). Shifting thresholds and changing degradation patterns: climate change effects on the simulated long-term response of a semi-arid savanna to grazing. *Journal of Applied Ecology* **49**, 814-823.

*LONG, Y., JIANG, J., HU, X., HU, J., REN, C. & ZHOU, S. (2021). The response of microbial community structure and sediment properties to anthropogenic activities in Caohai wetland sediments. *Ecotoxicology and Environmental Safety* **211**, 111936.

*LÓPEZ, D. R. & CAVALLERO, L. (2017). The role of nurse functional types in seedling recruitment dynamics of alternative states in rangelands. *Acta Oecologica* **79**, 70-80.

*LOUGHEED, V. L., PARKER, C. A. & STEVENSON, R. J. (2007). Using non-linear responses of multiple taxonomic groups to establish criteria indicative of wetland biological condition. *Wetlands* **27**, 96-109.

*LOUZAO, M., GALLAGHER, R., GARCÍA-BARÓN, I., CHUST, G., INTXAUSTI, I., ALBISU, J., BRERETON, T. & FONTÁN, A. (2019). Threshold responses in bird mortality driven by extreme wind events. *Ecological indicators* **99**, 183-192.

*LU, D., XU, X., TIAN, H., MORAN, E., ZHAO, M. & RUNNING, S. (2010). The effects of urbanization on net primary productivity in southeastern China. *Environmental Management* **46**, 404-410.

*LU, M., CHENG, S., FANG, H., XU, M., YANG, Y., LI, Y., ZHANG, J. & MÜLLER, C. (2021). Organic nitrogen addition causes decoupling of microbial nitrogen cycles by stimulating gross nitrogen transformation in a temperate forest soil. *Geoderma* **385**, 114886.

*LU, M., CHENG, S., FANG, H., YANG, Y., GUO, Y., LI, Y. & ZHOU, Y. (2022). Contrasting response of soil N2O release to ammonium, nitrate, and urea addition rates is determined by substrate availability and microbial community abundance and composition. *European Journal of Soil Biology* **109**, 103393.

*LUO, T., PAN, Y., OUYANG, H., SHI, P., LUO, J., YU, Z. & LU, Q. (2004). Leaf area index and net primary productivity along subtropical to alpine gradients in the Tibetan Plateau. *Global Ecology and Biogeography* **13**, 345-358.

*LYNHAM, J., HALPERN, B., BLENCKNER, T., ESSINGTON, T., ESTES, J., HUNSICKER, M., KAPPEL, C., SALOMON, A., SCARBOROUGH, C. & SELKOE, K. (2017). Costly stakeholder participation creates inertia in marine ecosystems. *Marine Policy* **76**, 122-129.

*MA, S., LIU, D., TIAN, Y., FU, C., LI, J., JU, P., SUN, P., YE, Z., LIU, Y. & WATANABE, Y. (2021a). Critical transitions and ecological resilience of large marine ecosystems in the Northwestern Pacific in response to global warming. *Global Change Biology* **27**, 5310-5328.

*MA, S., REN, J., WU, C., CHENG, F., WANG, X., LI, B. & HE, Q. (2022a). Hydrological control of threshold transitions in vegetation over early-period wetland development. *Journal of Hydrology* **610**, 127931.

*MA, S., WANG, L.-J., JIANG, J., CHU, L. & ZHANG, J.-C. (2021b). Threshold effect of ecosystem services in response to climate change and vegetation coverage change in the Qinghai-Tibet Plateau ecological shelter. *Journal of Cleaner Production* **318**, 128592.

*MA, Z., ZENG, C. & ZHENG, B. (2022b). Relaxation time as an indicator of critical transition to a eutrophic lake state: The role of stochastic resonance. *Europhysics Letters* **137**, 42001.

*MAC NALLY, R., ALBANO, C. & FLEISHMAN, E. (2014). A scrutiny of the evidence for pressure-induced state shifts in estuarine and nearshore ecosystems. *Austral Ecology* **39**, 898-906.

*MADRIGAL-GONZÁLEZ, J., HERRERO, A., RUIZ-BENITO, P. & ZAVALA, M. A. (2017). Resilience to drought in a dry forest: insights from demographic rates. *Forest Ecology and Management* **389**, 167-175.

*MAHDIYASA, A. W., LARGE, D. J., MULJADI, B. P. & ICARDI, M. (2023). Modelling the influence of mechanical-ecohydrological feedback on the nonlinear dynamics of peatlands. *Ecological Modelling* **478**, 110299.

*MANGAN, S., LOHRER, A. M., THRUSH, S. F., ELLIS, J. I. & PILDITCH, C. A. (2022). The Effects of Long-Term Nitrogen Enrichment on Estuarine Benthic-Pelagic Coupling. *Journal of Marine Science and Engineering* **10**, 1955.

*MARIANO-NETO, E. & SANTOS, R. A. (2023). Changes in the functional diversity of birds due to habitat loss in the Brazil Atlantic Forest. *Frontiers in Forests and Global Change* **6**, 1041268.

*MARÍN-GUIRAO, L., BERNARDEAU-ESTELLER, J., BELANDO, M. D., GARCÍA-MUÑOZ, R., RAMOS-SEGURA, A., ALCOVERRO, T., MINGUITO-FRUTOS, M. & RUIZ, J. (2022). Photo-acclimatory thresholds anticipate sudden shifts in seagrass ecosystem state under reduced light conditions. *Marine environmental research* **177**, 105636.

*MARTIN, P., NEWTON, A. C., CANTARELLO, E. & EVANS, P. M. (2017). Analysis of ecological thresholds in a temperate forest undergoing dieback. *PloS one* **12**, e0189578.

*MARTINS, R. T., BRITO, J., DIAS-SILVA, K., LEAL, C. G., LEITAO, R. P., OLIVEIRA, V. C., OLIVEIRA-JÚNIOR, J. M., FERRAZ, S. F., DE PAULA, F. R. & ROQUE, F. O. (2021). Low forest-loss thresholds threaten Amazonian fish and macroinvertebrate assemblage integrity. *Ecological indicators* **127**, 107773.

*MATHUR, M. (2018). Magnitude of temporal fluctuations of composite environment and their impacts on Cassia auriculata (Linn.) performance. *Trop Ecol* **59**, 1-12.

*MATISONS, R., ELFERTS, D., KRIŠĀNS, O., SCHNECK, V., GÄRTNER, H., BAST, A., WOJDA, T., KOWALCZYK, J. & JANSONS, Ā. (2021). Non-linear regional weather-growth relationships indicate limited adaptability of the eastern Baltic Scots pine. *Forest Ecology and Management* **479**, 118600.

*MAYER, A. L. & RIETKERK, M. (2004). The dynamic regime concept for ecosystem management and restoration. *BioScience* **54**, 1013-1020.

*MAYOR, A. G., BAUTISTA, S., RODRIGUEZ, F. & KÉFI, S. (2019). Connectivity-mediated ecohydrological feedbacks and regime shifts in drylands. *Ecosystems* **22**, 1497-1511.

*MAYOR, A. G., KÉFI, S., BAUTISTA, S., RODRÍGUEZ, F., CARTENÍ, F. & RIETKERK, M. (2013). Feedbacks between vegetation pattern and resource loss dramatically decrease ecosystem resilience and restoration potential in a simple dryland model. *Landscape Ecology* **28**, 931-942.

*MCGLATHERY, K. J., REIDENBACH, M. A., D'ODORICO, P., FAGHERAZZI, S., PACE, M. L. & PORTER, J. H. (2013). Nonlinear dynamics and alternative stable states in shallow coastal systems. *Oceanography* **26**, 220-231.

*MCGLATHERY, K. J., REYNOLDS, L. K., COLE, L. W., ORTH, R. J., MARION, S. R. & SCHWARZSCHILD, A. (2012). Recovery trajectories during state change from bare sediment to eelgrass dominance. *Marine Ecology Progress Series* **448**, 209-221.

*MCNEALL, D., HALLORAN, P. R., GOOD, P. & BETTS, R. A. (2011). Analyzing abrupt and nonlinear climate changes and their impacts. *Wiley Interdisciplinary Reviews: Climate Change* **2**, 663-686.

*MCNICOL, G. & SILVER, W. L. (2015). Non-linear response of carbon dioxide and methane emissions to oxygen availability in a drained histosol. *Biogeochemistry* **123**, 299-306.

*MCSWINEY, C. P. & ROBERTSON, G. P. (2005). Nonlinear response of N2O flux to incremental fertilizer addition in a continuous maize (Zea mays L.) cropping system. *Global Change Biology* **11**, 1712-1719.

*MEDEIROS, E. S., CALDAS, I. L., BAPTISTA, M. S. & FEUDEL, U. (2017). Trapping phenomenon attenuates the consequences of tipping points for limit cycles. *Scientific reports* **7**, 42351.

*MERKEL, C., HEETHOFF, M. & BRÜCKNER, A. (2020). Temperature affects chemical defense in a mite-beetle predator-prey system. *Journal of chemical ecology* **46**, 947-955.

*MILKOREIT, M., HODBOD, J., BAGGIO, J., BENESSAIAH, K., CALDERÓN-CONTRERAS, R., DONGES, J. F., MATHIAS, J.-D., ROCHA, J. C., SCHOON, M. & WERNERS, S. E. (2018). Defining tipping points for social-ecological systems scholarship—an interdisciplinary literature review. *Environmental Research Letters* **13**, 033005.

*MILLER, J. E., DAMSCHEN, E. I., RATAJCZAK, Z. & ÖZDOĞAN, M. (2017). Holding the line: three decades of prescribed fires halt but do not reverse woody encroachment in grasslands. *Landscape Ecology* **32**, 2297-2310.

*MILLIE, D. F., WECKMAN, G. R., YOUNG II, W. A., IVEY, J. E., CARRICK, H. J. & FAHNENSTIEL, G. L. (2012). Modeling microalgal abundance with artificial neural networks: Demonstration of a heuristic 'Grey-Box'to deconvolve and quantify environmental influences. *Environmental Modelling & Software* **38**, 27-39.

*MITCHELL, M. G., BENNETT, E. M. & GONZALEZ, A. (2015). Strong and nonlinear effects of fragmentation on ecosystem service provision at multiple scales. *Environmental Research Letters* **10**, 094014.

*MOLLICA, N. R., COHEN, A. L., ALPERT, A. E., BARKLEY, H. C., BRAINARD, R. E., CARILLI, J. E., DECARLO, T. M., DRENKARD, E. J., LOHMANN, P. & MANGUBHAI, S. (2019). Skeletal records of bleaching reveal different thermal thresholds of Pacific coral reef assemblages. *Coral Reefs* **38**, 743-757.

*MONACO, C. J. & HELMUTH, B. (2011). Tipping points, thresholds and the keystone role of physiology in marine climate change research. In *Advances in marine biology*. (Volume 60, pp. 123-160. Elsevier.

*MOORE, J. L. & REMAIS, J. V. (2014). Developmental models for estimating ecological responses to environmental variability: structural, parametric, and experimental issues. *Acta biotheoretica* **62**, 69-90.

*MOORE, J. W., SCHINDLER, D. E. & RUFF, C. P. (2008). Habitat saturation drives thresholds in stream subsidies. *Ecology* **89**, 306-312.

*MORANTE-FILHO, J. C., FARIA, D., MARIANO-NETO, E. & RHODES, J. (2015). Birds in anthropogenic landscapes: the responses of ecological groups to forest loss in the Brazilian Atlantic Forest. *PloS one* **10**, e0128923.

*MORENO-DE LAS HERAS, M., SACO, P. M., WILLGOOSE, G. R. & TONGWAY, D. J. (2012). Variations in hydrological connectivity of Australian semiarid landscapes indicate abrupt changes in rainfall-use efficiency of vegetation. *Journal of Geophysical Research: Biogeosciences* **117**(G3).

*MORISON, M. Q., MACRAE, M. L., PETRONE, R. M. & FISHBACK, L. (2017). Seasonal dynamics in shallow freshwater pond-peatland hydrochemical interactions in a subarctic permafrost environment. *Hydrological Processes* **31**, 462-475.

*MOROŃ, D., SKÓRKA, P., LENDA, M., KAJZER-BONK, J., MIELCZAREK, Ł., ROŻEJ-PABIJAN, E. & WANTUCH, M. (2019). Linear and non-linear effects of goldenrod invasions on native pollinator and plant populations. *Biological Invasions* **21**, 947-960.

*MÜLLER, D., SUN, Z., VONGVISOUK, T., PFLUGMACHER, D., XU, J. & MERTZ, O. (2014). Regime shifts limit the predictability of land-system change. *Global Environmental Change* **28**, 75-83.

*MUNSCH, S. H., ANDREWS, K. S., CROZIER, L. G., FONNER, R., GOSSELIN, J. L., GREENE, C. M., HARVEY, C. J., LUNDIN, J. I., PESS, G. R. & SAMHOURI, J. F. (2020). Potential for ecological nonlinearities and thresholds to inform Pacific salmon management. *Ecosphere* **11**, e03302.

*MURPHY JR, R., ESTABROOKS, A., GAUVIN, J., GRAY, S., KROSKA, A. C., WOLF, N. & HARRIS, B. P. (2021). Using mental models to quantify linear and non-linear relationships in complex fishery systems. *Marine Policy* **132**, 104695.

*MUYLAERT, R. L., STEVENS, R. D. & RIBEIRO, M. C. (2016). Threshold effect of habitat loss on bat richness in cerrado-forest landscapes. *Ecological Applications* **26**, 1854-1867.

*N'GUESSAN, M. & HARTNETT, D. C. (2011). Differential responses to defoliation frequency in little bluestem (Schizachyrium scoparium) in tallgrass prairie: implications for herbivory tolerance and avoidance. *Plant Ecology* **212**, 1275-1285.

*NAMAN, S. M., ROSENFELD, J. S., KIFFNEY, P. M. & RICHARDSON, J. S. (2018). The energetic consequences of habitat structure for forest stream salmonids. *Journal of Animal Ecology* **87**, 1383-1394.

*NASTIS, S. A. & PAGONI, E. G. (2019). Gamification of global climate change: an experimental analysis. *Studies in Nonlinear Dynamics & Econometrics* **23**, 20170105.

*NEWBOLD, T., TITTENSOR, D. P., HARFOOT, M. B., SCHARLEMANN, J. P. & PURVES, D. W. (2020). Non-linear changes in modelled terrestrial ecosystems subjected to perturbations. *Scientific reports* **10**, 14051.

*NGUYEN, K. L., NGUYEN, H. A., RICHTER, O. & PHAM, M. T. (2017). Ecophysiological responses of young mangrove species Rhizophora apiculata (Blume) to different chromium contaminated environments. *Science of the Total Environment* **574**, 369-380.

*NIE, Y., LI, Q., WANG, E. & ZHANG, T. (2019). Study of the nonlinear relations between economic growth and carbon dioxide emissions in the Eastern, Central and Western regions of China. *Journal of Cleaner Production* **219**, 713-722.

*NIJP, J. J., TEMME, A. J., VAN VOORN, G. A., KOOISTRA, L., HENGEVELD, G. M., SOONS, M. B., TEULING, A. J. & WALLINGA, J. (2019). Spatial early warning signals for impending regime shifts: A practical framework for application in real-world landscapes. *Global Change Biology* **25**, 1905-1921.

*NIYOGI, D. K., KOREN, M., ARBUCKLE, C. J. & TOWNSEND, C. R. (2007). Stream communities along a catchment land-use gradient: subsidy-stress responses to pastoral development. *Environmental Management* **39**, 213-225.

*NORBURY, G., WILSON, D. J., CLARKE, D., HAYMAN, E., SMITH, J. & HOWARD, S. (2023). Density-impact functions for invasive house mouse (Mus musculus) effects on indigenous lizards and invertebrates. *Biological Invasions* **25**, 801-815.

*NORKKO, J., NORKKO, A., THRUSH, S. F., VALANKO, S. & SUURKUUKKA, H. (2010). Conditional responses to increasing scales of disturbance, and potential implications for threshold dynamics in soft-sediment communities. *Marine Ecology Progress Series* **413**, 253-266.

*NOYCE, G. L., KIRWAN, M. L., RICH, R. L. & MEGONIGAL, J. P. (2019). Asynchronous nitrogen supply and demand produce nonlinear plant allocation responses to warming and elevated CO2. *Proceedings of the National Academy of Sciences* **116**, 21623-21628.

*O'LOUGHLIN, L. S., PANETTA, F. D. & GOODEN, B. (2021). Identifying thresholds in the impacts of an invasive groundcover on native vegetation. *Scientific reports* **11**, 20512.

*O'MEARA, T. A., HILLMAN, J. R. & THRUSH, S. F. (2017). Rising tides, cumulative impacts and cascading changes to estuarine ecosystem functions. *Scientific reports* **7**, 10218.

*OGANJAN, K. & LAURINGSON, V. (2014). Grazing rate of zebra mussel in a shallow eutrophicated bay of the Baltic Sea. *Marine environmental research* **102**, 43-50.

*OGLE, K. & REYNOLDS, J. F. (2004). Plant responses to precipitation in desert ecosystems: integrating functional types, pulses, thresholds, and delays. *Oecologia* **141**, 282-294.

*OGRIS, N., FERLAN, M., HAUPTMAN, T., PAVLIN, R., KAVČIČ, A., JURC, M. & DE GROOT, M. (2019). RITY–A phenology model of Ips typographus as a tool for optimization of its monitoring. *Ecological Modelling* **410**, 108775.

*O'HARA, P. A. (2009). Political economy of climate change, ecological destruction and uneven development. *Ecological Economics* **69**, 223-234.

*OSLAND, M. J., ENWRIGHT, N. & STAGG, C. L. (2014). Freshwater availability and coastal wetland foundation species: ecological transitions along a rainfall gradient. *Ecology* **95**, 2789-2802.

*OSWALD, C. J., RICHARDSON, M. C. & BRANFIREUN, B. A. (2011). Water storage dynamics and runoff response of a boreal Shield headwater catchment. *Hydrological Processes* **25**, 3042-3060.

*OTTO, S. A., KORNILOVS, G., LLOPE, M. & MÖLLMANN, C. (2014). Interactions among density, climate, and food web effects determine long-term life cycle dynamics of a key copepod. *Marine Ecology Progress Series* **498**, 73-84.

*PANETTA, F. D. & GOODEN, B. (2017). Managing for biodiversity: impact and action thresholds for invasive plants in natural ecosystems.

*PANT, H. K. (2007). Nonlinear effects of climate change on phosphorus stability in wetlands: concept and estimation.

*PAPPAS, M., KARAGIORGOU, E., PAPAIOANNOU, G., KOVEOS, D. & BROUFAS, G. (2013). Developmental temperature responses of Chrysoperla agilis (Neuroptera:

Chrysopidae), a member of the European carnea cryptic species group. *Biological Control* **64**, 291-298.

*PARUELO, J., PÜTZ, S., WEBER, G., BERTILLER, M., GOLLUSCIO, R., AGUIAR, M. R. & WIEGAND, T. (2008). Long-term dynamics of a semiarid grass steppe under stochastic climate and different grazing regimes: A simulation analysis. *Journal of Arid Environments* **72**, 2211-2231.

*PAUSAS, J. G. & KEELEY, J. E. (2014). Abrupt climate-independent fire regime changes. *Ecosystems* **17**, 1109-1120.

*PAWSON, S., BROCKERHOFF, E., NORTON, D. & DIDHAM, R. (2006). Clear-fell harvest impacts on biodiversity: past research and the search for harvest size thresholds. *Canadian Journal of Forest Research* **36**, 1035-1046.

*PEARSON, D. E., ORTEGA, Y. K., EREN, Ö. & HIERRO, J. L. (2016). Quantifying "apparent" impact and distinguishing impact from invasiveness in multispecies plant invasions. *Ecological Applications* **26**, 162-173.

*PELAK, N. F., PAROLARI, A. J. & PORPORATO, A. (2016). Bistable plant–soil dynamics and biogenic controls on the soil production function. *Earth Surface Processes and Landforms* **41**, 1011-1017.

*PENG, J., TIAN, L., LIU, Y., ZHAO, M. & WU, J. (2017a). Ecosystem services response to urbanization in metropolitan areas: Thresholds identification. *Science of the Total Environment* **607**, 706-714.

*PENG, Y., CHEN, H. Y. & YANG, Y. (2020). Global pattern and drivers of nitrogen saturation threshold of grassland productivity. *Functional Ecology* **34**, 1979-1990.

*PENG, Y., LI, F., ZHOU, G., FANG, K., ZHANG, D., LI, C., YANG, G., WANG, G., WANG, J. & MOHAMMAT, A. (2017b). Nonlinear response of soil respiration to increasing nitrogen additions in a Tibetan alpine steppe. *Environmental Research Letters* **12**, 024018.

*PENNING, W. E., MJELDE, M., DUDLEY, B., HELLSTEN, S., HANGANU, J., KOLADA, A., VAN DEN BERG, M., POIKANE, S., PHILLIPS, G. & WILLBY, N. (2008). Classifying aquatic macrophytes as indicators of eutrophication in European lakes. *Aquatic ecology* **42**, 237-251.

*PERAKIS, S. S., MAGUIRE, D. A., BULLEN, T. D., CROMACK, K., WARING, R. H. & BOYLE, J. R. (2006). Coupled nitrogen and calcium cycles in forests of the Oregon Coast Range. *Ecosystems* **9**, 63-74.

*PERETYATKO, A., TEISSIER, S., DE BACKER, S. & TRIEST, L. (2012). Classification trees as a tool for predicting cyanobacterial blooms. *Hydrobiologia* **689**, 131-146.

*PETERS, D. P., BESTELMEYER, B. T. & TURNER, M. G. (2007). Cross–scale interactions and changing pattern–process relationships: consequences for system dynamics. *Ecosystems* **10**, 790-796.

*PETERS, D. P., BESTELMEYER, B. T., HERRICK, J. E., FREDRICKSON, E. L., MONGER, H. C. & HAVSTAD, K. M. (2006). Disentangling complex landscapes: new insights into arid and semiarid system dynamics. *BioScience* **56**, 491-501.

*PFLÜGER, F. J., SIGNER, J. & BALKENHOL, N. (2019). Habitat loss causes non-linear genetic erosion in specialist species. *Global Ecology and Conservation* **17**, e00507.

*PHAM, C.-V., WANG, H.-C., CHEN, S.-H. & LEE, J.-M. (2023). The Threshold Effect of Overfishing on Global Fishery Outputs: International Evidence from a Sustainable Fishery Perspective. *Fishes* **8**, 71.

*PHELAN, L., TAPLIN, R., HENDERSON-SELLERS, A. & ALBRECHT, G. (2011). Ecological viability or liability? Insurance system responses to climate risk. *Environmental Policy and Governance* **21**, 112-130.

*PIATT, J. F., HARDING, A. M. A., SHULTZ, M., SPECKMAN, S. G., VAN PELT, T. I., DREW, G. S. & KETTLE, A. B. (2007). Seabirds as indicators of marine food supplies: Cairns revisited. *Marine Ecology Progress Series* **352**, 221-234.

*PIRANI, M., PANTON, A., PURDIE, D. A. & SAHU, S. K. (2016). Modelling macronutrient dynamics in the Hampshire Avon river: A Bayesian approach to estimate seasonal variability and total flux. *Science of the Total Environment* **572**, 1449-1460.

*PLAGÁNYI, É. E., ELLIS, N., BLAMEY, L. K., MORELLO, E. B., NORMAN-LOPEZ, A., ROBINSON, W., SPORCIC, M. & SWEATMAN, H. (2014). Ecosystem modelling provides clues to understanding ecological tipping points. *Marine Ecology Progress Series* **512**, 99-113.

*PLAGÁNYIA, É. & HOBDAYB, A. Using marine models to road-test climate-smart management responses and strategies and prepare decision makers.

*POWELL, L. A., BROWN, M. B., SMITH, J. A., HARRISON, J. O. & WHALEN, C. E. (2017). Modeling the spatial effects of disturbance: A constructive critique to provide evidence of ecological thresholds. *Wildlife Biology* **2017**, 1-10.

*PRATA, D. N., RODRIGUES, W. & BERMEJO, P. H. (2020). Temperature significantly changes COVID-19 transmission in (sub) tropical cities of Brazil. *Science of the Total Environment* **729**, 138862.

*PUPPE, D., EHRMANN, O., KACZOREK, D., WANNER, M. & SOMMER, M. (2015). The protozoic Si pool in temperate forest ecosystems—Quantification, abiotic controls and interactions with earthworms. *Geoderma* **243**, 196-204.

*QI, J., HOLYOAK, M., NING, Y. & JIANG, G. (2020). Ecological thresholds and large carnivores conservation: Implications for the Amur tiger and leopard in China. *Global Ecology and Conservation* **21**, e00837.

*RADFORD, J. Q., BENNETT, A. F. & CHEERS, G. J. (2005). Landscape-level thresholds of habitat cover for woodland-dependent birds. *Biological Conservation* **124**, 317-337.

*RATAJCZAK, Z., NIPPERT, J. B. & OCHELTREE, T. W. (2014). Abrupt transition of mesic grassland to shrubland: evidence for thresholds, alternative attractors, and regime shifts. *Ecology* **95**, 2633-2645.

*RAW, S., TIWARI, B. & MISHRA, P. (2020). Analysis of a plankton–fish model with external toxicity and nonlinear harvesting. *Ricerche di Matematica* **69**, 653-681.

*RAZAVI, B. S., BLAGODATSKAYA, E. & KUZYAKOV, Y. (2015). Nonlinear temperature sensitivity of enzyme kinetics explains canceling effect—a case study on loamy haplic Luvisol. *Frontiers in Microbiology* **6**, 1126.

*REID, J. M. (2022). Intrinsic emergence and modulation of sex-specific dominance reversals in threshold traits. *Evolution* **76**, 1924-1941.

*RENZI, J. J. & SILLIMAN, B. R. (2021). Increasing grazer density leads to linear decreases in Spartina alterniflora biomass and exponential increases in grazing pressure across a barrier island. *Marine Ecology Progress Series* **659**, 49-58.

*REYERS, B., FOLKE, C., MOORE, M.-L., BIGGS, R. & GALAZ, V. (2018). Social-ecological systems insights for navigating the dynamics of the Anthropocene. *Annual Review of Environment and Resources* **43**, 267-289.

*ROCHA-SANTOS, L., PESSOA, M. S., CASSANO, C. R., TALORA, D. C., ORIHUELA, R. L., MARIANO-NETO, E., MORANTE-FILHO, J. C., FARIA, D. & CAZETTA, E. (2016). The shrinkage of a forest: Landscape-scale deforestation leading to overall changes in local forest structure. *Biological Conservation* **196**, 1-9.

*ROCKSTRÖM, J., STEFFEN, W., NOONE, K., PERSSON, Å., CHAPIN III, F. S., LAMBIN, E., LENTON, T. M., SCHEFFER, M., FOLKE, C. & SCHELLNHUBER, H. J. (2009). Planetary boundaries: exploring the safe operating space for humanity. *Ecology and Society* **14**.

*RODRIGUES, M. E., DE OLIVEIRA ROQUE, F., QUINTERO, J. M. O., DE CASTRO PENA, J. C., DE SOUSA, D. C. & JUNIOR, P. D. M. (2016). Nonlinear responses in damselfly community along a gradient of habitat loss in a savanna landscape. *Biological Conservation* **194**, 113-120.

*ROFF, G., ZHAO, J. X. & MUMBY, P. J. (2015). Decadal-scale rates of reef erosion following El Niño-related mass coral mortality. *Global Change Biology* **21**, 4415-4424.

*ROGERS, L. A. & DOUGHERTY, A. B. (2019). Effects of climate and demography on reproductive phenology of a harvested marine fish population. *Global Change Biology* **25**, 708-720.

*ROLAND PITCHER, C., LAWTON, P., ELLIS, N., SMITH, S. J., INCZE, L. S., WEI, C. L., GREENLAW, M. E., WOLFF, N. H., SAMEOTO, J. A. & SNELGROVE, P. V. (2012). Exploring the role of environmental variables in shaping patterns of seabed biodiversity composition in regional-scale ecosystems. *Journal of Applied Ecology* **49**, 670-679.

*ROSS, C. A., ALI, G. A., SPENCE, C. & COURCHESNE, F. (2021). Evaluating the ubiquity of thresholds in rainfall-runoff response across contrasting environments. *Water Resources Research* **57**, e2020WR027498.

*ROSSI, S., ANFODILLO, T., ČUFAR, K., CUNY, H. E., DESLAURIERS, A., FONTI, P., FRANK, D., GRIČAR, J., GRUBER, A. & KING, G. M. (2013). A meta-analysis of cambium phenology and growth: linear and non-linear patterns in conifers of the northern hemisphere. *Annals of botany* **112**, 1911-1920.

*ROTHWELL, J. J., FUTTER, M. N. & DISE, N. B. (2008). A classification and regression tree model of controls on dissolved inorganic nitrogen leaching from European forests. *Environmental Pollution* **156**, 544-552.

*ROVAI, A. S., TWILLEY, R. R., CASTAÑEDA-MOYA, E., MIDWAY, S. R., FRIESS, D. A., TRETTIN, C. C., BUKOSKI, J. J., STOVALL, A. E., PAGLIOSA, P. R. & FONSECA, A. L. (2021). Macroecological patterns of forest structure and allometric scaling in mangrove forests. *Global Ecology and Biogeography* **30**, 1000-1013.

*RUESINK, J. L. (1998). Variation in per capita interaction strength: thresholds due to nonlinear dynamics and nonequilibrium conditions. *Proceedings of the National Academy of Sciences* **95**, 6843-6847.

*SAGRARIO, M. D. L. Á. G., MUSAZZI, S., CÓRDOBA, F. E., MENDIOLAR, M. & LAMI, A. (2020). Inferring the occurrence of regime shifts in a shallow lake during the last 250 years based on multiple indicators. *Ecological indicators* **117**, 106536.

*SAHA, B., BHOWMICK, A. R., CHATTOPADHYAY, J. & BHATTACHARYA, S. (2013). On the evidence of an Allee effect in herring populations and consequences for population survival: A model-based study. *Ecological Modelling* **250**, 72-80.

*SAKAMAKI, T. & RICHARDSON, J. S. (2013). Nonlinear variation of stream–forest linkage along a stream-size gradient: an assessment using biogeochemical proxies of in-stream fine particulate organic matter. *Journal of Applied Ecology* **50**, 1019-1027.

*SALGUEIRO, P. A., MIRA, A., RABAÇA, J. E. & SANTOS, S. M. (2018). Identifying critical thresholds to guide management practices in agro-ecosystems: insights from bird community response to an open grassland-to-forest gradient. *Ecological indicators* **88**, 205-213.

*SAMHOURI, J. F., ANDREWS, K. S., FAY, G., HARVEY, C. J., HAZEN, E. L., HENNESSEY, S. M., HOLSMAN, K., HUNSICKER, M. E., LARGE, S. I. & MARSHALL, K. N. (2017). Defining ecosystem thresholds for human activities and environmental pressures in the California Current. *Ecosphere* **8**, e01860.

*ŠAMONIL, P., DANĚK, P., SCHAETZL, R., TEJNECKÝ, V. & DRÁBEK, O. (2018). Converse pathways of soil evolution caused by tree uprooting: A synthesis from three regions with varying soil formation processes. *Catena* **161**, 122-136.

*SAMPAIO, E., RODIL, I. F., VAZ-PINTO, F., FERNÁNDEZ, A. & ARENAS, F. (2017). Interaction strength between different grazers and macroalgae mediated by ocean acidification over warming gradients. *Marine environmental research* **125**, 25-33.

*SASAKI, T., KOYAMA, A. & OKURO, T. (2018). Coupling structural and functional thresholds for vegetation changes on a Mongolian shrubland. *Ecological indicators* **93**, 1264-1275.

*SASAKI, T., OHKURO, T., JAMSRAN, U. & TAKEUCHI, K. (2012). Changes in the herbage nutritive value and yield associated with threshold responses of vegetation to grazing in Mongolian rangelands. *Grass and Forage Science* **67**, 446-455.

*SASAKI, T., OKAYASU, T., JAMSRAN, U. & TAKEUCHI, K. (2008). Threshold changes in vegetation along a grazing gradient in Mongolian rangelands. *Journal of Ecology* **96**, 145-154.

*SASAKI, T., OKUBO, S., OKAYASU, T., JAMSRAN, U., OHKURO, T. & TAKEUCHI, K. (2009). Two-phase functional redundancy in plant communities along a grazing gradient in Mongolian rangelands. *Ecology* **90**, 2598-2608.

*SATTERTHWAITE, W. H., ANDREWS, K. S., BURKE, B. J., GOSSELIN, J. L., GREENE, C. M., HARVEY, C. J., MUNSCH, S. H., O'FARRELL, M. R., SAMHOURI, J. F. & SOBOCINSKI, K. L. (2020). Ecological thresholds in forecast performance for key United States West Coast Chinook salmon stocks. *ICES Journal of Marine Science* **77**, 1503-1515.

*SAUNDERS, M. I., ATKINSON, S., KLEIN, C. J., WEBER, T. & POSSINGHAM, H. P. (2017). Increased sediment loads cause non-linear decreases in seagrass suitable habitat extent. *PloS one* **12**, e0187284.

*SCHARFENBERGER, U., MAHDY, A. & ADRIAN, R. (2013). Threshold-driven shifts in two copepod species: Testing ecological theory with observational data. *Limnology and Oceanography* **58**, 741-752.

*SCHARTAU, A. K., MOE, S. J., SANDIN, L., MCFARLAND, B. & RADDUM, G. G. (2008). Macroinvertebrate indicators of lake acidification: analysis of monitoring data from UK, Norway and Sweden. *Aquatic ecology* **42**, 293-305.

*SCHEFFER, M., HIROTA, M., HOLMGREN, M., VAN NES, E. H. & CHAPIN III, F. S. (2012). Thresholds for boreal biome transitions. *Proceedings of the National Academy of Sciences* **109**, 21384-21389.

*SCHIPPER, A. M., HENDRIKS, H. W., KAUFFMAN, M. J., HENDRIKS, A. J. & HUIJBREGTS, M. A. (2013). Modelling interactions of toxicants and density dependence in wildlife populations. *Journal of Applied Ecology* **50**, 1469-1478.

*SCHLAEPFER, D. R., LAUENROTH, W. K. & BRADFORD, J. B. (2012). Consequences of declining snow accumulation for water balance of mid-latitude dry regions. *Global Change Biology* **18**, 1988-1997.

*SCHOLES, R. J. & KRUGER, J. M. (2011). A framework for deriving and triggering thresholds for management intervention in uncertain, varying and time-lagged systems. *Koedoe: African Protected Area Conservation and Science* **53**, 1-8.

*SEDDON, A. W., FROYD, C. A., LENG, M. J., MILNE, G. A. & WILLIS, K. J. (2011). Ecosystem resilience and threshold response in the Galápagos coastal zone. *PloS one* **6**, e22376.

*SEDDON, A. W., FROYD, C. A., WITKOWSKI, A. & WILLIS, K. J. (2014). A quantitative framework for analysis of regime shifts in a Galápagos coastal lagoon. *Ecology* **95**, 3046-3055.

*SEDDON, A. W., MACIAS-FAURIA, M. & WILLIS, K. J. (2015). Climate and abrupt vegetation change in Northern Europe since the last deglaciation. *The Holocene* **25**, 25-36.

*SENGUPTA, A., FANSLER, S. J., CHU, R. K., DANCZAK, R. E., GARAYBURU-CARUSO, V. A., RENTERIA, L., SONG, H.-S., TOYODA, J., HAGER, J. & STEGEN, J. C. (2021). Disturbance triggers non-linear microbe–environment feedbacks. *Biogeosciences* **18**, 4773-4789.

*SHAO, Q. & SHEN, S. (2017). When reduced working time harms the environment: A panel threshold analysis for EU-15, 1970–2010. *Journal of Cleaner Production* **147**, 319-329.

*SHELDON, K. S. & DILLON, M. E. (2016). Beyond the mean: biological impacts of cryptic temperature change. *Integrative and comparative biology* **56**, 110-119.

*SHELEF, E., GRIFFORE, M., MARK, S., COLEMAN, T., WONDOLOWSKI, N., LASHER, G. E. & ABBOTT, M. (2022). Sensitivity of Erosion-Rate in Permafrost Landscapes to Changing Climatic and Environmental Conditions Based on Lake Sediments From Northwestern Alaska. *Earth's Future* **10**, e2022EF002779.

*SHI, Y., JIN, N., MA, X., WU, B., HE, Q., YUE, C. & YU, Q. (2020). Attribution of climate and human activities to vegetation change in China using machine learning techniques. *Agricultural and Forest Meteorology* **294**, 108146.

*SINGH JATAV, K. & DHAR, J. (2013). Hybrid Approach for Pest Control with Impulsive Releasing of Natural Enemies and Chemical Pesticides: A Plant-Pest-Natural Enemy Model. *Nonlinear Analysis: Hybrid Systems* **9**, 1-14.

*SINGH, R., QUINN, J. D., REED, P. M. & KELLER, K. (2018). Skill (or lack thereof) of data-model fusion techniques to provide an early warning signal for an approaching tipping point. *PloS one* **13**, e0191768.

*SNYDER, C. D. & YOUNG, J. A. (2020). Identification of management thresholds of urban development in support of aquatic biodiversity conservation. *Ecological indicators* **112**, 106124.

*SOFAER, H. R., JARNEVICH, C. S. & PEARSE, I. S. (2018). The relationship between invader abundance and impact. *Ecosphere* **9**, e02415.

*SOLOMON, C. T., HOTCHKISS, E. R., MOSLEMI, J. M., ULSETH, A. J., STANLEY, E. H., HALL JR, R. O. & FLECKER, A. S. (2009). Sediment size and nutrients regulate denitrification in a tropical stream. *Journal of the North American Benthological Society* **28**, 480-490.

*SONG, B., NIU, S., LI, L., ZHANG, L. & YU, G. (2014). Soil carbon fractions in grasslands respond differently to various levels of nitrogen enrichments. *Plant and Soil* **384**, 401-412.

*SPITZ, F. & LEK, S. (1999). Environmental impact prediction using neural network modelling. An example in wildlife damage. *Journal of Applied Ecology* **36**, 317-326.

*SPONSLER, D. B., REQUIER, F., KALLNIK, K., CLASSEN, A., MAIHOFF, F., SIEGER, J. & STEFFAN-DEWENTER, I. (2022). Contrasting patterns of richness, abundance, and turnover in mountain bumble bees and their floral hosts. *Ecology* **103**, e3712.

*STAGG, C. L., OSLAND, M. J., MOON, J. A., HALL, C. T., FEHER, L. C., JONES, W. R., COUVILLION, B. R., HARTLEY, S. B. & VERVAEKE, W. C. (2020). Quantifying hydrologic controls on local-and landscape-scale indicators of coastal wetland loss. *Annals of botany* **125**, 365-376.

*STECKBAUER, A., DUARTE, C. M., CARSTENSEN, J., VAQUER-SUNYER, R. & CONLEY, D. J. (2011). Ecosystem impacts of hypoxia: thresholds of hypoxia and pathways to recovery. *Environmental Research Letters* **6**, 025003.

*STEGEN, J. C., ANDERSON, C. G., BOND-LAMBERTY, B., CRUMP, A. R., CHEN, X. & HESS, N. (2017). Soil respiration across a permafrost transition zone: spatial structure and environmental correlates. *Biogeosciences* **14**, 4341-4354.

*STORCH, D., ŠÍMOVÁ, I., SMYČKA, J., BOHDALKOVÁ, E., TOSZOGYOVA, A. & OKIE, J. G. (2022). Biodiversity dynamics in the Anthropocene: how human activities change equilibria of species richness. *Ecography* **2022**.

*STUART-HAËNTJENS, E. J., CURTIS, P. S., FAHEY, R. T., VOGEL, C. S. & GOUGH, C. M. (2015). Net primary production of a temperate deciduous forest exhibits a threshold response to increasing disturbance severity. *Ecology* **96**, 2478-2487.

*STURTEVANT, B. R. & FORTIN, M.-J. (2021). Understanding and modeling forest disturbance interactions at the landscape level. *Frontiers in Ecology and Evolution* **9**, 653647.

*SUN, G.-Q., CHAKRABORTY, A., LIU, Q.-X., JIN, Z., ANDERSON, K. E. & LI, B.-L. (2014). Influence of time delay and nonlinear diffusion on herbivore outbreak. *Communications in Nonlinear Science and Numerical Simulation* **19**, 1507-1518.

*SWIFT, T. L. & HANNON, S. J. (2010). Critical thresholds associated with habitat loss: a review of the concepts, evidence, and applications. *Biological reviews* **85**, 35-53.

*SWINGEDOUW, D., IFEJIKA SPERANZA, C., BARTSCH, A., DURAND, G., JAMET, C., BEAUGRAND, G. & CONVERSI, A. (2020). Early warning from space for a few key tipping points in physical, biological, and social-ecological systems. *Surveys in Geophysics* **41**, 1237-1284.

*TABENI, S., GARIBOTTI, I. A., PISSOLITO, C. & ARANIBAR, J. N. (2014). Grazing effects on biological soil crusts and their interaction with shrubs and grasses in an arid rangeland. *Journal of Vegetation Science* **25**, 1417-1425.

*TAN, C. O. & BEKLIOGLU, M. (2006). Modeling complex nonlinear responses of shallow lakes to fish and hydrology using artificial neural networks. *Ecological Modelling* **196**(1-2), 183-194.

*TANG, Y., JIANG, J., CHEN, C., CHEN, Y. & WU, X. (2018). Rainfall pulse response of carbon fluxes in a temperate grass ecosystem in the semiarid Loess Plateau. *Ecology and Evolution* **8**, 11179-11189.

*TANGNEY, R., MERRITT, D. J., FONTAINE, J. B. & MILLER, B. P. (2019). Seed moisture content as a primary trait regulating the lethal temperature thresholds of seeds. *Journal of Ecology* **107**, 1093-1105.

*TARANU, Z. E., CARPENTER, S. R., FROSSARD, V., JENNY, J. P., THOMAS, Z., VERMAIRE, J. C. & PERGA, M. E. (2018). Can we detect ecosystem critical transitions and signals of changing resilience from paleo-ecological records? *Ecosphere* **9**, e02438.

*TAVERNIA, B. G. & REED, J. M. (2010). Spatial, temporal, and life history assumptions influence consistency of landscape effects on species distributions. *Landscape Ecology* **25**, 1085-1097.

*TAYLOR, J. M., KING, R. S., PEASE, A. A. & WINEMILLER, K. O. (2014). Nonlinear response of stream ecosystem structure to low-level phosphorus enrichment. *Freshwater Biology* **59**, 969-984.

*TCHABOVSKY, A. V., SAVINETSKAYA, L. E., SURKOVA, E. N., OVCHINNIKOVA, N. L. & KSHNYASEV, I. A. (2016). Delayed threshold response of a rodent population to human-induced landscape change. *Oecologia* **182**, 1075-1082.

*TEICHERT, N., BORJA, A., CHUST, G., URIARTE, A. & LEPAGE, M. (2016). Restoring fish ecological quality in estuaries: implication of interactive and cumulative effects among anthropogenic stressors. *Science of the Total Environment* **542**, 383-393.

*THOMAS, A., MASANTE, D., JACKSON, B., COSBY, B., EMMETT, B. & JONES, L. (2020). Fragmentation and thresholds in hydrological flow-based ecosystem services. *Ecological Applications* **30**, e02046.

*THOMPSON, S. J., HANDEL, C. M., RICHARDSON, R. M. & MCNEW, L. B. (2016). When winners become losers: predicted nonlinear responses of arctic birds to increasing woody vegetation. *PloS one* **11**, e0164755.

*TIAN, D., NIU, S., PAN, Q., REN, T., CHEN, S., BAI, Y. & HAN, X. (2016a). Nonlinear responses of ecosystem carbon fluxes and water-use efficiency to nitrogen addition in Inner Mongolia grassland. *Functional Ecology* **30**, 490-499.

*TIAN, D., WANG, H., SUN, J. & NIU, S. (2016b). Global evidence on nitrogen saturation of terrestrial ecosystem net primary productivity. *Environmental Research Letters* **11**, 024012.

*TIWARI, P. K., SAMANTA, S., BONA, F., VENTURINO, E. & MISRA, A. K. (2019). The time delays influence on the dynamical complexity of algal blooms in the presence of bacteria. *Ecological complexity* **39**, 100769.

*TIWARI, T., BUFFAM, I., SPONSELLER, R. A. & LAUDON, H. (2017). Inferring scale-dependent processes influencing stream water biogeochemistry from headwater to sea. *Limnology and Oceanography* **62**(S1), S58-S70.

*TOMAL, J. H. & CIBOROWSKI, J. J. (2020). Ecological models for estimating breakpoints and prediction intervals. *Ecology and Evolution* **10**, 13500-13517.

*TOOTOONCHI, M., GETTYS, L. A., FERRELL, J. A., ERICKSON, J. E. & BHADHA, J. H. (2023). Salt tolerance assessment of aquatic and wetland plants: increased salinity can reshape aquatic vegetation communities. *Hydrobiologia* **850**, 4575-4587.

*TRAILL, L. W., BRADSHAW, C. J., DELEAN, S. & BROOK, B. W. (2010). Wetland conservation and sustainable use under global change: a tropical Australian case study using magpie geese. *Ecography* **33**, 818-825.

*TREMBLAY, J.-P., HUOT, J. & POTVIN, F. (2006). Divergent nonlinear responses of the boreal forest field layer along an experimental gradient of deer densities. *Oecologia* **150**, 78-88.

*UCHE, E. & EFFIOM, L. (2021). Financial development and environmental sustainability in Nigeria: fresh insights from multiple threshold nonlinear ARDL model. *Environmental Science and Pollution Research* **28**, 39524-39539.

*UCHE, E., DAS, N. & BERA, P. (2023). Re-examining the environmental Kuznets curve (EKC) for India via the multiple threshold NARDL procedure. *Environmental Science and Pollution Research* **30**, 11913-11925.

*ULLAH, A., AHMED, M., RAZA, S. A. & ALI, S. (2021). A threshold approach to sustainable development: nonlinear relationship between renewable energy consumption, natural resource rent, and ecological footprint. *Journal of Environmental Management* **295**, 113073.

*ULUCAK, R., KOÇAK, E., ERDOĞAN, S. & KASSOURI, Y. (2020). Investigating the non-linear effects of globalization on material consumption in the EU countries: Evidence from PSTR estimation. *Resources policy* **67**, 101667.

*URRUTIA-CORDERO, P., EKVALL, M. K., RATCOVICH, J., SOARES, M., WILKEN, S., ZHANG, H. & HANSSON, L. A. (2017). Phytoplankton diversity loss along a gradient of future warming and brownification in freshwater mesocosms. *Freshwater Biology* **62**, 1869-1878.

*USINOWICZ, J., QIU, J. & KAMARAINEN, A. (2017). Flashiness and flooding of two lakes in the Upper Midwest during a century of urbanization and climate change. *Ecosystems* **20**, 601-615.

*VALENTE-NETO, F., MARTÍNEZ, B. T., HUGHES, R. M., FERREIRA, A., SEVERO-NETO, F., DE SOUZA, F. L., SOUZA, R. P., ESCARPINATI, S. C. & DE OLIVEIRA ROQUE, F. (2021). Incorporating costs, thresholds and spatial extents for selecting stream bioindicators in an ecotone between two Brazilian biodiversity hotspots. *Ecological indicators* **127**, 107761.

*VAN DER HEIDE, T., ANGELINI, C., DE FOUW, J. & EKLÖF, J. S. (2021). Facultative mutualisms: A double-edged sword for foundation species in the face of anthropogenic global change. *Ecology and Evolution* **11**, 29-44.

*VEETTIL, A. V. (2020). Multiscale hydrological drought analysis: Role of climate, catchment and morphological variables and associated thresholds. *Journal of Hydrology* **582**, 124533.

*VEGAS-VILARRÚBIA, T., RULL, V., MONTOYA, E. & SAFONT, E. (2011). Quaternary palaeoecology and nature conservation: a general review with examples from the neotropics. *Quaternary Science Reviews* **30**(19-20), 2361-2388.

*VIDAL, M. M., BANKS-LEITE, C., TAMBOSI, L. R., HASUI, É., DEVELEY, P. F., SILVA, W. R., GUIMARÃES JR, P. R. & METZGER, J. P. (2019). Predicting the non-linear collapse of plant–frugivore networks due to habitat loss. *Ecography* **42**, 1765-1776.

*VIDIELLA, B., FONTICH, E., VALVERDE, S. & SARDANYÉS, J. (2021). Habitat loss causes long extinction transients in small trophic chains. *Theoretical Ecology* **14**, 641-661.

*VILLALOBOS, V. I., VALDIVIA, N., FÖRSTERRA, G., BALLYRAM, S., ESPINOZA, J. P., WADHAM, J. L., BURGOS-ANDRADE, K. & HÄUSSERMANN, V. (2021). Depth-dependent diversity patterns of rocky subtidal macrobenthic communities along a temperate fjord in Northern Chilean Patagonia. *Frontiers in Marine Science* **8**, 635855.

*Virta, L. & Teittinen, A. (2022). Threshold effects of climate change on benthic diatom communities: evaluating impacts of salinity and wind disturbance on functional traits and benthic biomass. *Science of the Total Environment* **826**, 154130.

*Wakwella, A., Mumby, P. J. & Roff, G. (2020). Sedimentation and overfishing drive changes in early succession and coral recruitment. *Proceedings of the Royal Society B* **287**(1941), 20202575.

*Wang, B., Zheng, X., Zhang, H., Xiao, F., He, Z. & Yan, Q. (2020a). Keystone taxa of water microbiome respond to environmental quality and predict water contamination. *Environmental Research* **187**, 109666.

*Wang, C., Ren, F., Zhou, X., Ma, W., Liang, C., Wang, J., Cheng, J., Zhou, H. & He, J.-S. (2020b). Variations in the nitrogen saturation threshold of soil respiration in grassland ecosystems. *Biogeochemistry* **148**, 311-324.

*Wang, C., Yao, Z., Zhan, P., Yi, X., Chen, J. & Xiong, J. (2023). Significant tipping points of sediment microeukaryotes forewarn increasing antibiotic pollution. *Journal of Environmental Sciences* **124**, 429-439.

*Wang, J., Chadwick, D. R., Cheng, Y. & Yan, X. (2018). Global analysis of agricultural soil denitrification in response to fertilizer nitrogen. *Science of the Total Environment* **616**, 908-917.

*Wang, J., Liu, Y., Cao, W., Li, W., Wang, X., Zhang, D., Shi, S., Pan, D. & Liu, W. (2020c). Effects of grazing exclusion on soil respiration components in an alpine meadow on the north-eastern Qinghai-Tibet Plateau. *Catena* **194**, 104750.

*Wang, Q., Wang, X. & Li, R. (2022a). Does official development assistance alleviate the environmental pressures during the urbanization of recipient countries? Evidence from the sub-Saharan Africa countries. *Environmental impact assessment review* **95**, 106787.

*WANG, Q., WANG, X. & LI, R. (2022b). Does population aging reduce environmental pressures from urbanization in 156 countries? *Science of the Total Environment* **848**, 157330.

*WANG, Q., WANG, X. & LI, R. (2022c). Does urbanization redefine the environmental Kuznets curve? An empirical analysis of 134 Countries. *Sustainable Cities and Society* **76**, 103382.

*WANG, S., WANG, J. & FAN, F. (2021). The hidden mediating role of innovation efficiency in coordinating development of economy and ecological environment: Evidence from 283 Chinese cities. *Environmental Science and Pollution Research* **28**, 47668-47684.

*WANG, S., YAN, Y., FU, Z. & CHEN, H. (2022d). Rainfall-runoff characteristics and their threshold behaviors on a karst hillslope in a peak-cluster depression region. *Journal of Hydrology* **605**, 127370.

*WANG, T., WANG, P., WU, Z., YU, J., POZDNIAKOV, S. P., GUAN, X., WANG, H., XU, H. & YAN, D. (2022e). Modeling revealed the effect of root dynamics on the water adaptability of phreatophytes. *Agricultural and Forest Meteorology* **320**, 108959.

*WANG, X. & LUO, Y. (2020d). Has technological innovation capability addressed environmental pollution from the dual perspective of FDI quantity and quality? Evidence from China. *Journal of Cleaner Production* **258**, 120941.

*WARD, D. F., WOTHERSPOON, S., MELBOURNE-THOMAS, J., HAAPKYLÄ, J. & JOHNSON, C. R. (2018). Detecting ecological regime shifts from transect data. *Ecological monographs* **88**, 694-715.

*WATSON, S. C., GRANDFIELD, F. G., HERBERT, R. J. & NEWTON, A. C. (2018). Detecting ecological thresholds and tipping points in the natural capital assets of a protected coastal ecosystem. *Estuarine, Coastal and Shelf Science* **215**, 112-123.

*WATSON, S. C., NEWTON, A. C., RIDDING, L. E., EVANS, P. M., BRAND, S., MCCRACKEN, M., GOSAL, A. S. & BULLOCK, J. M. (2021). Does agricultural intensification cause tipping points in ecosystem services? *Landscape Ecology* **36**, 3473-3491.

*WEI, W., WEILE, C. & SHAOPENG, W. (2010). Forest soil respiration and its heterotrophic and autotrophic components: Global patterns and responses to temperature and precipitation. *Soil Biology and Biochemistry* **42**, 1236-1244.

*WEI, Z., YU, Y. & YI, Y. (2022). Spatial distribution of nutrient loads and thresholds in large shallow lakes: The case of Chaohu Lake, China. *Journal of Hydrology* **613**, 128466.

*WEISBERG, P. J., BONAVIA, F. & BUGMANN, H. (2005). Modeling the interacting effects of browsing and shading on mountain forest tree regeneration (Picea abies). *Ecological Modelling* **185**(2-4), 213-230.

*WELEGEDARA, N. P., GRANT, R. F., QUIDEAU, S. A. & DAS GUPTA, S. (2020). Modelling nitrogen mineralization and plant nitrogen uptake as affected by reclamation cover depth in reclaimed upland forestlands of Northern Alberta. *Biogeochemistry* **149**, 293-315.

*WHITTEN, S. M., HERTZLER, G. & STRUNZ, S. (2012). How real options and ecological resilience thinking can assist in environmental risk management. *Journal of Risk Research* **15**, 331-346.

*WIENS, J. A., SCHOOLEY, R. L. & WEEKS JR, R. D. (1997). Patchy landscapes and animal movements: do beetles percolate? *Oikos*, 257-264.

*WILLIAMS, J. W., BLOIS, J. L. & SHUMAN, B. N. (2011). Extrinsic and intrinsic forcing of abrupt ecological change: case studies from the late Quaternary. *Journal of Ecology* **99**, 664-677.

*WILLIAMS, J. W., ORDONEZ, A. & SVENNING, J.-C. (2021). A unifying framework for studying and managing climate-driven rates of ecological change. *Nature ecology & evolution* **5**, 17-26.

*WILLIAMS, J. W., SHUMAN, B. & BARTLEIN, P. J. (2009). Rapid responses of the prairie-forest ecotone to early Holocene aridity in mid-continental North America. *Global and Planetary Change* **66**(3-4), 195-207.

*WRONA, F. J., PROWSE, T. D., REIST, J. D., HOBBIE, J. E., LÉVESQUE, L. M. & VINCENT, W. F. (2006). Climate impacts on Arctic freshwater ecosystems and fisheries: background, rationale and approach of the Arctic Climate Impact Assessment (ACIA). *AMBIO: A Journal of the Human Environment* **35**, 326-329.

*WU, J., LIU, Z. & QIAN, J. (2013). Non-linear effect of habitat fragmentation on plant diversity: evidence from a sand dune field in a desertified grassland in northeastern China. *Ecological engineering* **54**, 90-96.

*WU, X., GUO, W., LIU, H., LI, X., PENG, C., ALLEN, C. D., ZHANG, C., WANG, P., PEI, T. & MA, Y. (2019). Exposures to temperature beyond threshold disproportionately reduce vegetation growth in the northern hemisphere. *National Science Review* **6**, 786-795.

*WUNDERLING, N., STAAL, A., SAKSCHEWSKI, B., HIROTA, M., TUINENBURG, O. A., DONGES, J. F., BARBOSA, H. M. & WINKELMANN, R. (2022). Recurrent droughts increase risk of cascading tipping events by outpacing adaptive capacities in the Amazon rainforest. *Proceedings of the National Academy of Sciences* **119**, e2120777119.

*XU, C., YANG, Z., QIAN, W., CHEN, S., LIU, X., LIN, W., XIONG, D., JIANG, M., CHANG, C. T. & HUANG, J. C. (2019). Runoff and soil erosion responses to rainfall and vegetation cover under various afforestation management regimes in subtropical montane forest. *Land Degradation & Development* **30**, 1711-1724.

*XU, T., YANG, T. & XIONG, M. (2020). Time scales of external loading and spatial heterogeneity in nutrients-chlorophyll a response: Implication on eutrophication control in a large shallow lake. *Ecological engineering* **142**, 105636.

*XUAN, L., SHENG, Z., LU, J., QIU, Q., CHEN, J. & XIONG, J. (2019). Bacterioplankton community responses and the potential ecological thresholds along disturbance gradients. *Science of the Total Environment* **696**, 134015.

*YAN, J., CUI, B., ZHENG, J., XIE, T., WANG, Q. & LI, S. (2015). Quantification of intensive hybrid coastal reclamation for revealing its impacts on macrozoobenthos. *Environmental Research Letters* **10**, 014004.

*YANG, X., SCUDERI, L. A., WANG, X., SCUDERI, L. J., ZHANG, D., LI, H., FORMAN, S., XU, Q., WANG, R. & HUANG, W. (2015). Groundwater sapping as the cause of irreversible desertification of Hunshandake Sandy Lands, Inner Mongolia, northern China. *Proceedings of the National Academy of Sciences* **112**, 702-706.

*YANG, X.-G., CHEN, L., WANG, L., WANG, X., GU, J.-L., QU, W.-J. & SONG, N.-P. (2019). Dynamic rainfall-partitioning relationships among throughfall, stemflow, and interception loss by Caragana intermedia. *Journal of Hydrology* **574**, 980-989.

*YOUNG, A. M., HIGUERA, P. E., DUFFY, P. A. & HU, F. S. (2017). Climatic thresholds shape northern high-latitude fire regimes and imply vulnerability to future climate change. *Ecography* **40**, 606-617.

*YU, W., WEINTRAUB, S. R. & HALL, S. J. (2021). Climatic and geochemical controls on soil carbon at the continental scale: interactions and thresholds. *Global Biogeochemical Cycles* **35**, e2020GB006781.

*YUAN, X., NIU, D., GHERARDI, L. A., LIU, Y., WANG, Y., ELSER, J. J. & FU, H. (2019). Linkages of stoichiometric imbalances to soil microbial respiration with increasing nitrogen addition: Evidence from a long-term grassland experiment. *Soil Biology and Biochemistry* **138**, 107580.

*YUHUI, W., JIQUAN, C., GUANGSHENG, Z., CHANGLIANG, S., JUN, C., YU, W. & JIANMIN, S. (2018). Predominance of precipitation event controls ecosystem CO2 exchange in an Inner Mongolian desert grassland, China. *Journal of Cleaner Production* **197**, 781-793.

*ZAMPELLA, R. A., BUNNELL, J. F., LAIDIG, K. J. & PROCOPIO III, N. A. (2010). Aquatic degradation in shallow coastal plain lakes: Gradients or thresholds? *Ecological indicators* **10**, 303-310.

*ZHANG, A., ZHENG, S., DIDHAM, R. K., HOLT, R. D. & YU, M. (2021). Nonlinear thresholds in the effects of island area on functional diversity in woody plant communities. *Journal of Ecology* **109**, 2177-2189.

*ZHANG, H., DUAN, Z., WANG, Z., ZHONG, M., TIAN, W., WANG, H. & HUANG, H. (2019). Freshwater lake ecosystem health assessment and its response to pollution stresses based on planktonic index of biotic integrity. *Environmental Science and Pollution Research* **26**, 35240-35252.

*ZHANG, H., SUN, T. & YANG, W. (2020). Short-term environmental flow assessment of a functional estuarine tidal flat ecosystem: A nonlinear ecological response to flow alteration. *Water Resources Research* **56**, e2020WR027084.

*ZHANG, X., ZHAO, J., DING, L., LI, Y., LIU, H., ZHAO, Y. & FU, G. (2022a). Eutrophication evolution trajectory influenced by human activities and climate in the shallow Lake Gehu, China. *Ecological indicators* **138**, 108821.

*ZHANG, Y., HUO, S., XI, B., MA, C. & HE, Z. (2016). Establishing Nutrient Criteria in Nine Typical Lakes in China: A Conceptual Model. *CLEAN–Soil, Air, Water* **44**, 1335-1344.

*ZHANG, Z., HUANG, J., DUAN, S., HUANG, Y., CAI, J. & BIAN, J. (2022b). Use of interpretable machine learning to identify the factors influencing the nonlinear linkage between land use and river water quality in the Chesapeake Bay watershed. *Ecological indicators* **140**, 108977.

*ZHENG, Y., YANG, T., WANG, N., WAN, X., HU, C., SUN, L. & YAN, X. (2023). Quantifying hydrological-ecological response relationships based on zooplankton index of biotic integrity and comprehensive habitat quality index-A case study of typical rivers in Xi'an, China. *Science of the Total Environment* **858**, 159925.

*ZHOU, Q., ZHANG, Y., TAO, J., YE, L., WANG, H., SHAN, K., JEPPESEN, E. & SONG, L. (2022). Water depth and land-use intensity indirectly determine phytoplankton functional diversity and further regulate resource use efficiency at a multi-lake scale. *Science of the Total Environment* **834**, 155303.

*ZHOU, X., WENG, E. & LUO, Y. (2008). Modeling patterns of nonlinearity in ecosystem responses to temperature, CO2, and precipitation changes. *Ecological Applications* **18**, 453-466.

*ZHU, Q., GUO, X., GUO, J., WU, J., YE, Y., CAI, W. & LIU, S. (2022). The quality attribute of watershed ecosystem is more important than the landscape attribute in controlling erosion of red soil in southern China. *International Soil and Water Conservation Research* **10**, 507-517.

*ZWEIG, C. L. & KITCHENS, W. M. (2009). Multi-state succession in wetlands: a novel use of state and transition models. *Ecology* **90**, 1900-1909.

IX. SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

.

Table S1. Classification of included studies based on the number of predictors and response variables used to determine non-linear thresholds in data.

Appendix S1. Identification and screening process through systematic review following the PRISMA method.

Appendix S2. Stepwise workflow to use SiZer for detecting tipping points from data.

Appendix S3. Steps to use gradient forest as a tool to identify thresholds in response data along predictor gradients.

Appendix S4. Steps for combining principal components analyses (PCAs) from multiple response variables and predictors to determine state shift on a temporal scale.

Figure legends

Fig. 1. Example of a simplified coastal ecosystem with three subsystems. Internal feedback loops within all three subsystems and their interaction with external forcing need to be accounted for to achieve an ecosystem-based approach to evaluate state shifts of this ecosystem. The relationships between components of subsystems and the forcing parameters may be linear or non-linear and can change through cumulative effects. The arrows and symbols indicate the type of interaction (positive or negative) between ecosystem components or pressures.

Fig. 2. Model ecosystem response to multi-layered stressors. A shift in stable state, if present, may be delayed due to multiple ecological mechanisms (e.g. population dynamics, community interactions, functional redundancy, evolutionary mechanisms, extinction debt), however, it is often preceded by a loss of resilience and stability. Recovery to a reference state is generally lengthy and complex, requiring careful management of the ecosystem.

Fig. 3. Categorisation of included studies that utilise or describe methods for identifying thresholds ($N = 507$). Studies were classified (as percentage of total number of studies reviewed) based on the dimensionality (one or multiple dimensions) of data used. The remaining 17% were review studies on specific habitats, methods, or management perspectives. The graphs within each category show the percentage contribution of the most common methods used for threshold detection to that category. The green arrows show the percentage of studies within each category that did not integrate intervariable interactions in their analyses, and thus reduced the dimensionality of their data for analysis. N = number of studies in each category. MDS, multidimensional scaling; PCA, principal components analysis.

Fig. 4. Illustration of the segmented (A) and changepoint (B) methods for detection of shifts in empirical data. Segmented regression usually uses a non-linear regression with a 'hinge' function (A: dotted line). Changepoint analysis is generally based on sequential F-tests.

Fig. 5. Using significant zero crossing (SiZer) to detect a threshold from non-linear models by considering different fitting bandwidths (h = tuning parameter). Different bandwidths are used to fit the model to the data (A), after which the SiZer maps of first and second derivatives are produced (B). In (B) the red are represents a negative derivative, blue

represents a positive derivative, and purple is when the derivative is zero. The white lines give a visual representation of the size of the bandwidth. Figures were made by generating a data set specifically to exemplify this method and using the *SiZer* package in R (https://cran.r-project.org/web/packages/SiZer/index.html) to run the analyses. The data and R code used for model testing are available publicly on Zenodo at https://zenodo.org/records/10077651.

Fig. 6. Workflow of the gradient forest method involving measurement of impurity importance from random forest (A), ranking of pressures (B), assessing the effect of each predictor on all response variables (C), determining the effect of each predictor on each variable (D), and identifying the thresholds caused by each predictor based on the cumulative importance of splits (E). See Appendix S3 for further details. Figures were made by generating a random data set and using the *gradientForest* package in R (https://gradientforest.r-forge.r-project.org/) to exemplify the analyses. The data and R code used for model testing are available publicly on Zenodo at https://zenodo.org/records/10077651.

Fig. 7. Representation of the method developed by Vasilakopoulos & Marshall (2015) that utilises principal components analyses (PCAs) of response and predictor variables (A) to determine shifts in ecosystems. The changes in each predictor (Pred) and response variable (RV) over time are visualised with a traffic light plot (B) and the combined PC scores are plotted against time (C). The relationship between response variables and predictor variables PC scores then is analysed with a general additive model (GAM) and threshold-GAM (TGAM), after which a bifurcation graph is plotted, indicating the possible year around

which major shifts have occurred (E). All graphs are from an imaginary scenario. See Appendix S3 for further details.

Fig. 8. Dose–response representation of an ecosystem response in relation to interacting pressures. Circles indicate the model's EC_{50} (half-maximal effective concentration). Additive effects (A) lead to a reduction in the EC_{50} due to addition of the effects of pressure A and B. Antagonistic effects (B) reduce the overall effect of pressures A and B. Finally, synergistic effects (C) are produced through an interaction of pressures that together cause a stronger impact than each individual pressure alone, but not in a simply additive manner.

Figure 1

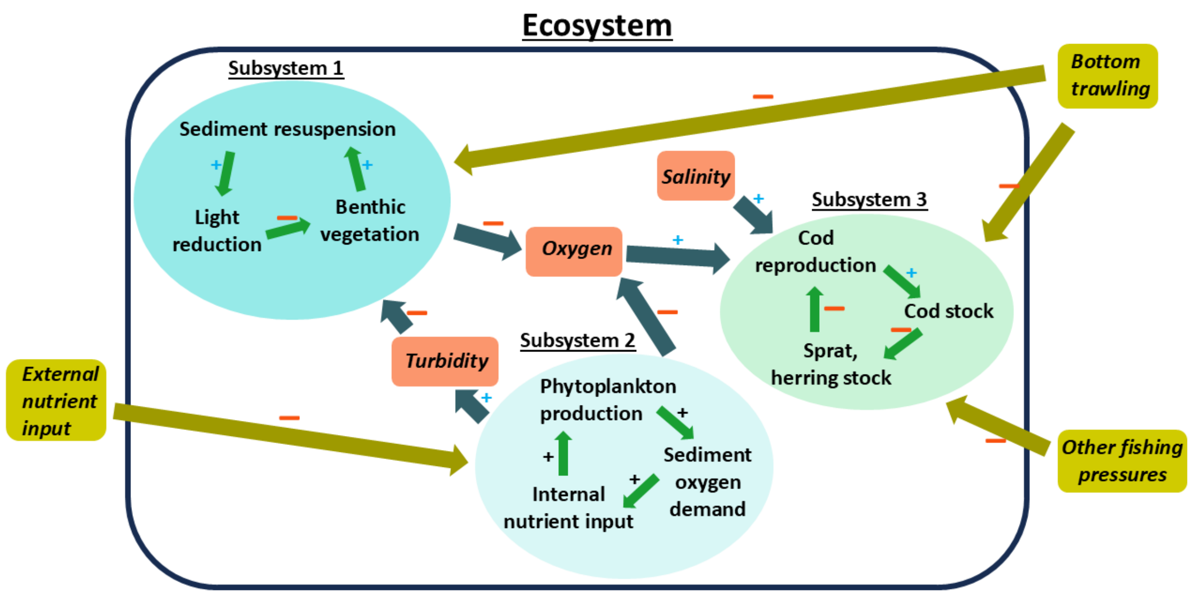
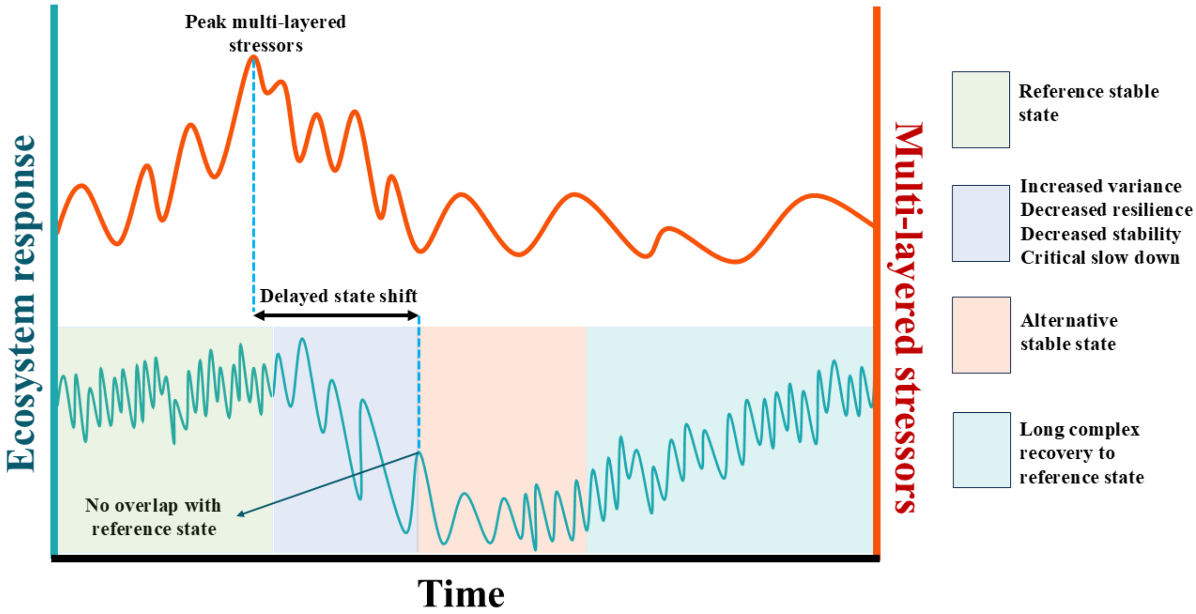
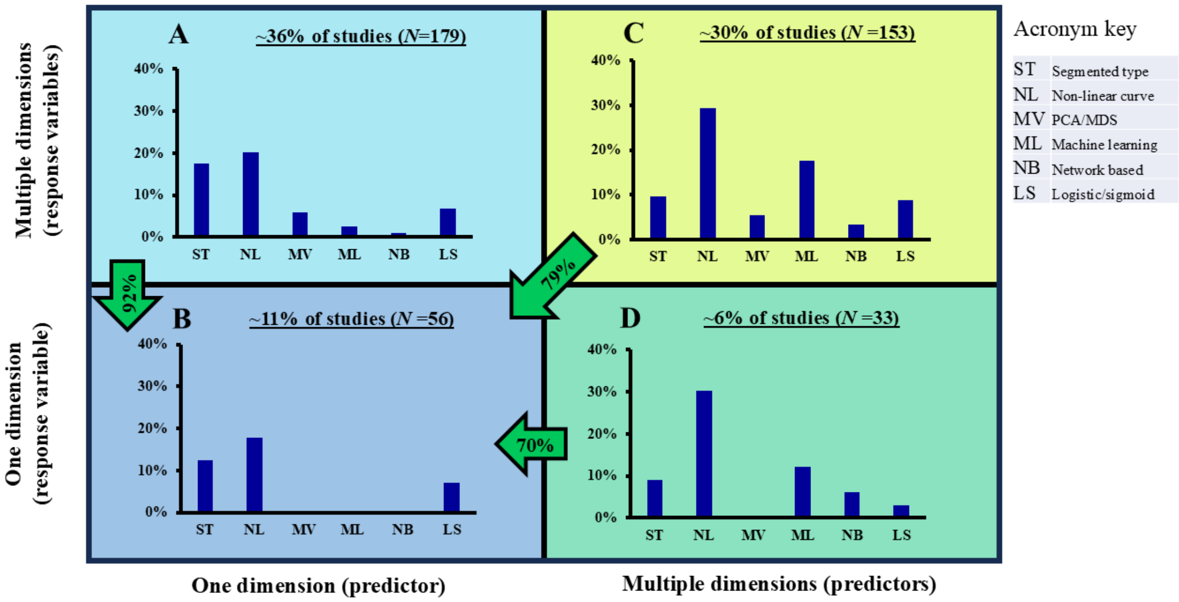


Figure 2



Peak multi-layered stressors
Delayed state shift
No overlap with reference state
Ecosystem response
Multi-layered stressors
Time
Reference stable state
Increased variance
Decreased resilience
Decreased stability
Critical slow down
Alternative stable state
Long complex recovery to reference state

Figure 3



A
~36% of studies (N=179)
C
~30% of studies (N =153)
B
~11% of studies (N =56)
D
~6% of studies (N =33)
40%
30%
20%
10%
0%
ST
NL
MV
ML
NB
LS
92%
79%
70%
Multiple dimensions (response variables)
One dimension (response variable)
One dimension (predictor)
Multiple dimensions (predictors)
Acronym key
ST Segmented type
NL Non-linear curve
MV PCA/MDS
ML Machine learning
NB Network based
LS Logistic/sigmoid

Figure 4

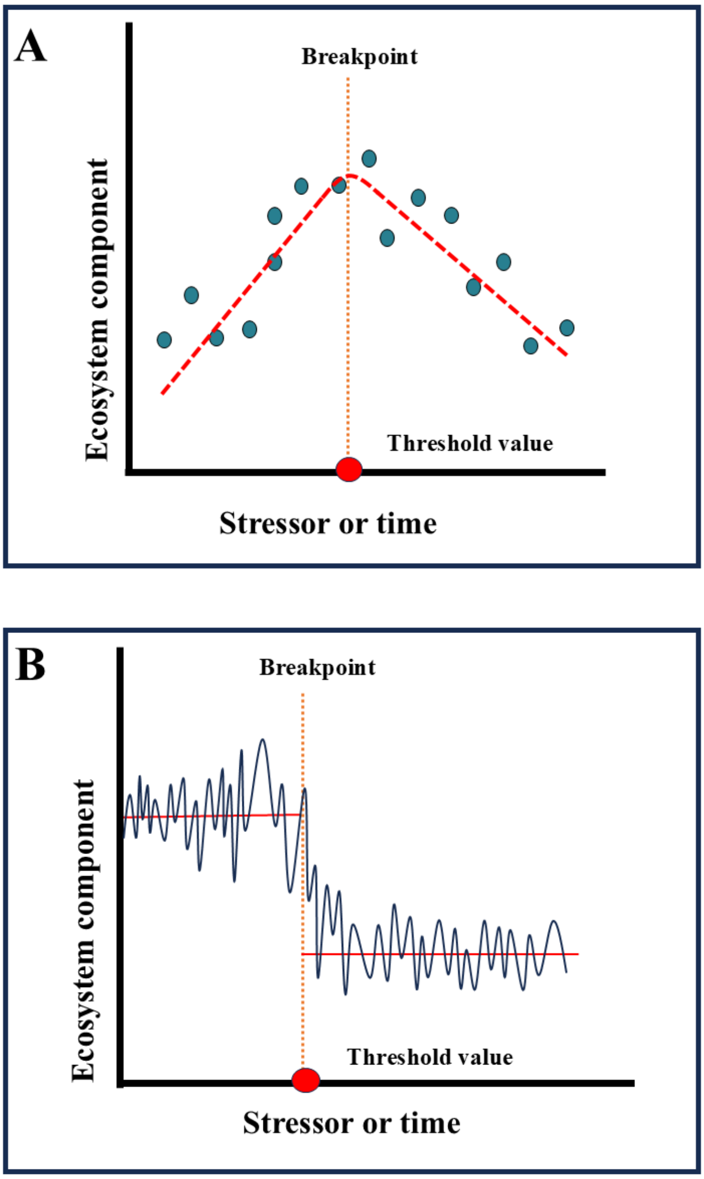

A
Breakpoint
Ecosystem component
Threshold value
Stressor or time
B
Breakpoint
Ecosystem component
Threshold value
Stressor or time

Figure 5

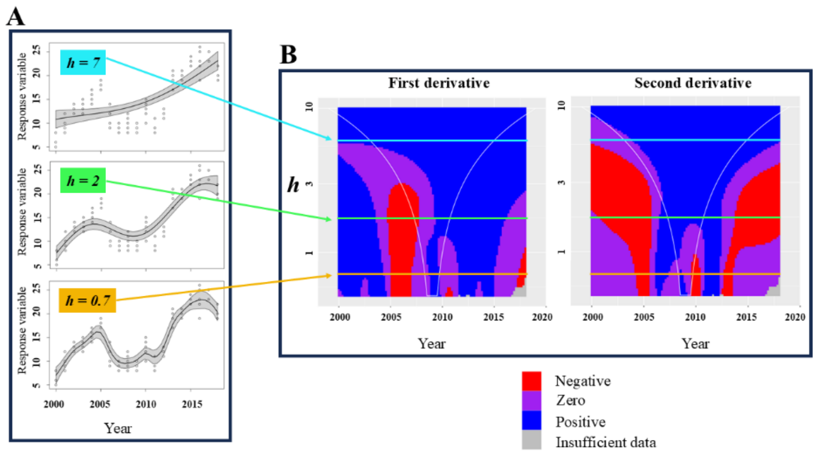
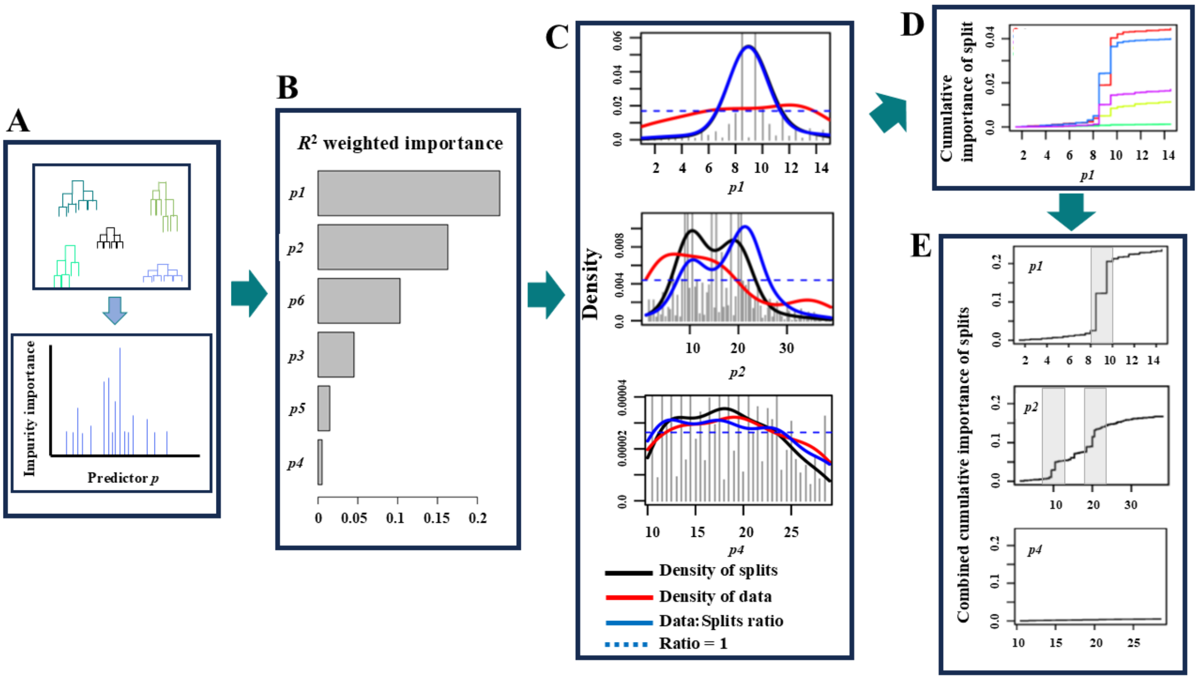

A
h = 7
h = 2
h = 0.7
Response variable
Year
B
First derivative
Second derivative
h
Year
Negative
Zero
Positive
Insufficient data

Figure 6



A
Impurity importance
Predictor p
B
R² weighted importance
p1
p2
p6
p3
p5
p4
0
0.05
0.1
0.15
0.2
C
Density
p1
p2
p4
Density of splits
Density of data
Data:Splits ratio
Ratio = 1
D
Cumulative importance of split
p1
E
Combined cumulative importance of splits
p1
p2
p4

Figure 7

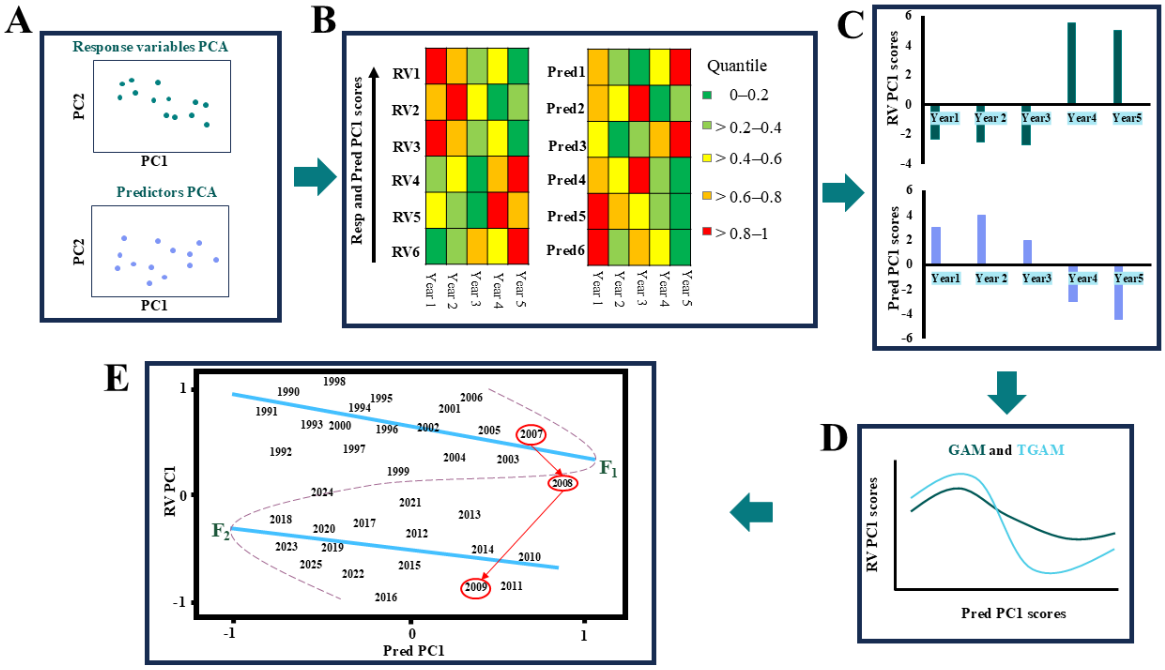
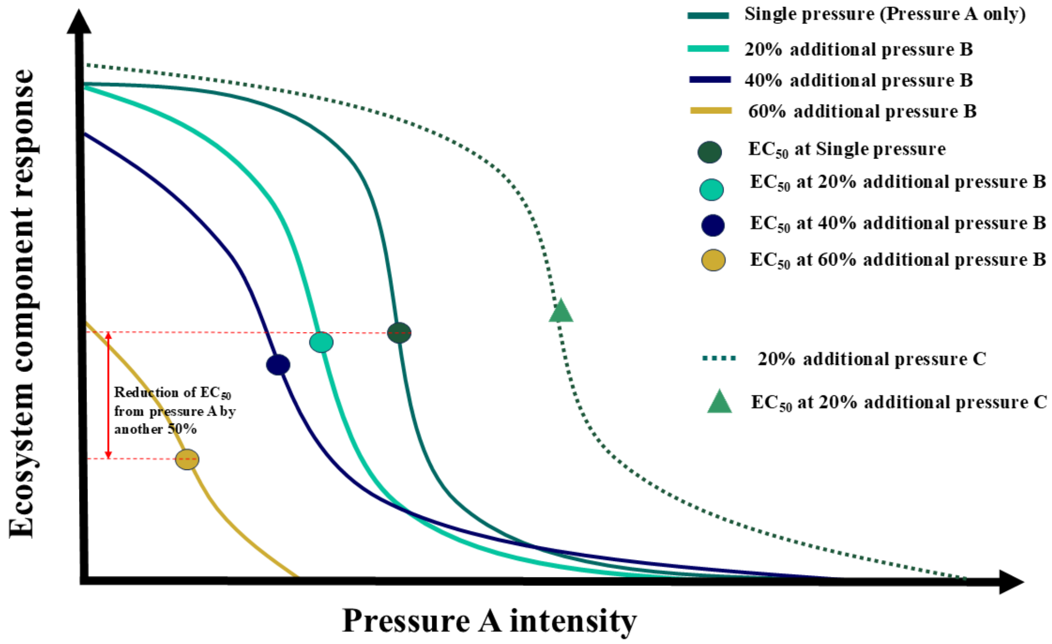

A
Response variables PCA
PC2
PC1
Predictors PCA
PC2
PC1
B
Resp and Pred PC1 scores
RV1
RV2
RV3
RV4
RV5
RV6
Year 1
Year 2
Year 3
Year 4
Year 5
Pred1
Pred2
Pred3
Pred4
Pred5
Pred6
Quantile
0–0.2
> 0.2–0.4
> 0.4–0.6
> 0.6–0.8
> 0.8–1
C
RV PC1 scores
Pred PC1 scores
Year1
Year 2
Year3
Year4
Year5
D
GAM and TGAM
RV PC1 scores
Pred PC1 scores
E
RV PC1
Pred PC1
F1
F2

Figure 8


Single pressure (Pressure A only)
20% additional pressure B
40% additional pressure B
60% additional pressure B
EC_{50} at Single pressure
EC_{50} at 20% additional pressure B
EC_{50} at 40% additional pressure B
EC_{50} at 60% additional pressure B
20% additional pressure C
EC_{50} at 20% additional pressure C
Reduction of EC_{50} from pressure A by another 50%
Ecosystem component response
Pressure A intensity